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Predictive latent states for extreme vortex gust–airfoil interactions

Alberto Solera-Rico^{1,2}, Arnau Miró^{3,4}, Oriol Lehmkuhl⁴ and Carlos Sanmiguel Vila^{1,2}

¹Aerial Platforms Department, Spanish National Institute for Aerospace Technology (INTA), San Martín de la Vega, 28330, Madrid, Spain

²Department of Aerospace Engineering, Universidad Carlos III de Madrid, Leganés, 28911, Madrid, Spain

³TUAREG Research Group, Universitat Politècnica de Catalunya (UPC), Carrer Colom 11, 08222 Terrassa, Spain

⁴Barcelona Supercomputing Centre (BSC), CASE Large-Scale Computational Fluid Dynamics, Plaça d'Eusebi Güell, 1-3, 08034 Barcelona, Spain

Corresponding author: Carlos Sanmiguel Vila, csanvil@inta.es

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[AB-01] Extreme vortex gusts drive a post-stall airfoil through transient wake reorganisations that the gust parameters alone do not determine. We investigate whether a reduced state can retain this wake dynamics, remain predictable and be estimated from sparse wall pressure. Direct numerical simulations of a NACA 0012 airfoil at $\alpha = 14^\circ$ and $Re = 5000$, subjected to Taylor vortices of varying strength, size and offset, compare predictive, reconstruction-trained and proper-orthogonal-decomposition states at matched dimension. A controlled supervision study shows that, within the predictive formulation considered, a lift target is necessary to avoid latent collapse, a near-body force-element target improves peak-load readability, and wake supervision is required to retain the wake enstrophy and circulation. Lift stays linearly accessible at four coefficients over the dimensions tested, whereas the spatial wake structure requires a larger distributed representation. Under a common direct multi-horizon forecaster the predictive states give the most accurate lift forecasts, while recursive prediction is limited by accumulated rollout error. A pressure-only ensemble estimator with eight wall-pressure sensors tracks the lift through impact and relaxation, reaching impact-phase $R^2 = 0.794$ at a root-mean-square lift error of 0.286 with the final two-stage configuration. Where a static single-frame inverse matches it in distribution, the filter's advantage is in the tails, in continuous state tracking, and at the held-out $|G| = 4$ boundary. Wake enstrophy is not reliably recovered by the wall-sensor configurations considered. The results separate the roles of observable supervision, predictive training and sensing geometry in constructing reduced states for extreme aerodynamic transients.

AB-01

KEEP

The paper's contract with the reader, now carrying the in-distribution pair. Every clause maps to a Results exhibit.

Key words: vortex shedding, low-dimensional models, machine learning

Editorial overlay (not part of the manuscript). Every paragraph and caption is colour-coded for its value to the paper and carries a unique identifier; the margin note gives the reason. Switch the overlay off with `\reviewmarksfalse` in `reviewmarks.tex`.

■ **K** keep in the body. ■ **T** could be removed.

■ **A** move to an appendix. ■ **S** move to the supplementary material. ■ **D** delete.

Identifiers are `<unit>-<nn>`: AB abstract, S1 to S6 the numbered sections (S31, S34 and so on for the v4 subfiles), APA to APC the appendices; C in the number marks a caption. The full ledger, with the same identifiers, is `editorial/RELEVANCE_MAP.md`.

1. Introduction

[S1-01] Small uncrewed aerial vehicles are increasingly expected to operate where the incoming flow is neither steady nor weakly disturbed. Urban canyons, mountainous terrain, coastal regions and the wakes of ships and buildings can expose a small wing to gusts whose length scale is comparable to the chord and whose velocity is of the order of the flight speed (Jones *et al.* 2022; Mohamed *et al.* 2023; Fukami & Taira 2023). In this regime a gust is not a small perturbation of an otherwise steady aerodynamic state: the strongest encounters reorganise separation, vortex shedding and wake recovery rather than merely modulate the instantaneous force, and the resulting transient loads bear directly on navigability, control authority and flight safety.

[S1-02] This regime has motivated a growing body of work on extreme aerodynamics (Taira 2026). Its canonical problem is the interaction between a discrete vortex gust and a wing, where the disturbance is strong enough to reorganise the near wake rather than merely modulate the force (Fukami *et al.* 2025), and its central empirical discovery is that these apparently complex encounters admit unexpectedly low-dimensional descriptions. Lift-augmented autoencoders reveal compact manifolds for extreme vortex-gust airfoil interactions (Fukami & Taira 2023), the same reduced description supports phase-amplitude analysis and data-driven transient lift attenuation (Fukami *et al.* 2024), and large-scale vortical structures show similar organisation across laminar and turbulent extreme-gust encounters (Odaka *et al.* 2026). The dynamical organisation of these flows is therefore simpler than the full field dimension suggests.

[S1-03] For deployment, however, compactness alone is not sufficient. A reduced description intended for gust mitigation must serve as a plant state: it must be advanced in time by a low-order forecast model and estimated from the sparse sensors a wing can carry (Loiseau *et al.* 2018). Surface pressure is central here, because it is one of the few high-bandwidth measurements available on an aerodynamic body, and recent gust studies use sparse pressure histories to infer unsteady loads and to bridge spatial pressure correlations with temporal flow memory (Zhong *et al.* 2023; Haughn *et al.* 2024; Chen *et al.* 2026), while Bayesian sequential estimation has begun to close the loop between pressure and low-order aerodynamic states, carrying uncertainty alongside the estimate (Mousavi & Eldredge 2025; Eldredge & Mousavi 2025). These advances sharpen a prior question: what reduced state should be estimated from the wall in the first place?

[S1-04] The two established routes to such a state pull in opposite directions. Nonlinear reconstruction buys compactness but not forecastability. A decoder fixes the latent only up to an arbitrary reparametrisation of the manifold, so the coordinates a reconstruction objective produces need not be the coordinates a time-advancing model can use, and controlled comparisons in wake flows report that autoencoder latents mix high and low frequencies in their coordinates and evolve irregularly, so that autoregressive predictors built on them diverge, whereas linear proper-orthogonal-decomposition predictors forecast more accurately and with a narrower spread precisely because the basis is linear, but pay

S1-01

TRIM?

Generic small-UAV motivation that the extreme-aerodynamics literature already establishes. S1-02 restarts with the same framing, so the section survives without it.

S1-02

KEEP

The empirical premise the paper builds on: these encounters admit low-dimensional descriptions. Names the lineage the comparison is against.

S1-03

KEEP

States the gap: compactness is not a plant state. Motivates wall pressure as the only sensor a wing carries.

S1-04

KEEP

The pivot. Compactness against forecastability, and nonlinearity against wake reach, is the tension the whole paper resolves.

for it in compactness, needing many more coordinates to represent the same flow structures (Solera-Rico *et al.* 2026). The linear basis, however, buys that forecastable dynamics at the price of the nonlinear wake content of the encounter: in the present flow the POD coefficients do not reach the wake enstrophy at any dimension we tested. Compactness trades against forecastability, and nonlinearity against wake reach, and extreme gusts are the arena where the trade cannot be evaded, because there the linear escape route genuinely fails.

[S1-05] This tension motivates a predictive training objective: rather than compressing for reconstruction and hoping the latent is forecastable, one trains the encoder together with a latent predictor so the coordinates are shaped in the space the forward model will use. This is a joint-embedding predictive architecture (LeCun 2022), which learns an encoder and a predictor such that the present latent predicts a future latent rather than reconstructing the field, developed for images and video as non-generative prediction in representation space (Assran *et al.* 2023, 2025). Here it is a diagnostic instrument for the paper’s central question: whether predictive training yields a nonlinear state at once more compact than the linear basis and more forecastable than a reconstruction-trained latent.

[S1-06] Training such a state on a forced wake exposes a structural requirement an autoencoder does not face: without an observable anchor the predictive latent collapses, and it is the scalar lift, not a high-dimensional field descriptor, that prevents it. This turns the observable heads from an assumption into a finding and connects the objective to the observable-augmented manifold learning of the autoencoder lineage (Fukami & Taira 2023, 2025), in which physical observables orient the latent towards physically relevant quantities. A healthy latent and a useful one are different achievements: the lift anchors the state, and a wake target loads it with the wake.

[S1-07] We examine whether a reduced representation of an extreme vortex-gust encounter can satisfy three requirements simultaneously: it must retain the wake variables that distinguish post-impact states, remain usable under forward prediction, and be recoverable from sparse wall pressure. Direct numerical simulations of a post-stall NACA 0012 airfoil at $\alpha = 14^\circ$ and $Re = 5000$, perturbed by Taylor vortex gusts spanning gust ratio, core diameter and offset, are used to compare predictive, reconstruction-trained and proper-orthogonal-decomposition states at matched dimension. Controlled supervision ablations identify which physical targets organise the latent representation; a common direct forecaster separates representation quality from operator choice, the encoder never seeing the gust parameters and the reconstruction baselines carrying the identical observable heads; and a pressure-only sequential estimator tests which parts of the state are observable from the wall. The results support four claims: (i) observable supervision determines the physical content of the state; (ii) multi-step predictive training improves its forward stability; (iii) wall-pressure histories recover the load-bearing directions more reliably than the complete wake state; and (iv) the strongest gusts expose the remaining limitation of a mid-plane representation and an under-dispersed estimator.

[S1-08] The remainder of the paper is organised as follows. §2 describes the flow configuration, data and endpoints; §3 the reduced states, the shared forecaster and the wall-pressure estimator; §4 the results; §5 discusses the mechanisms and their limits, and §6 concludes.

2. Flow configuration, data and endpoints

2.1. Vortex gust-airfoil interaction

[S2-01] We study a NACA 0012 airfoil at angle of attack $\alpha = 14^\circ$ and chord Reynolds number $Re = 5000$, perturbed by a discrete vortex gust modelled as a spanwise-oriented Taylor vortex released upstream of the leading edge. The data analysed here are our own direct

S1-05

KEEP

Introduces the predictive objective as a diagnostic instrument rather than a proposal. This framing is what keeps the paper from reading as an architecture paper.

S1-06

KEEP

Previews the collapse finding, which turns the observable heads from an assumption into a result. Load-bearing for the contribution claim.

S1-07

KEEP

The three requirements, the design of the comparison, and the four claims. This is the contribution list a referee will check the Results against.

S1-08

TRIM?



Pure signposting, and §4’s own subsection openings already announce the order. Removing it costs the reader nothing that the section headings do not supply.

S2-01

KEEP

Configuration, solver provenance and the gust parametrisation. The DNS-not-LES and Fukami-as-reference-not-source distinctions live here.

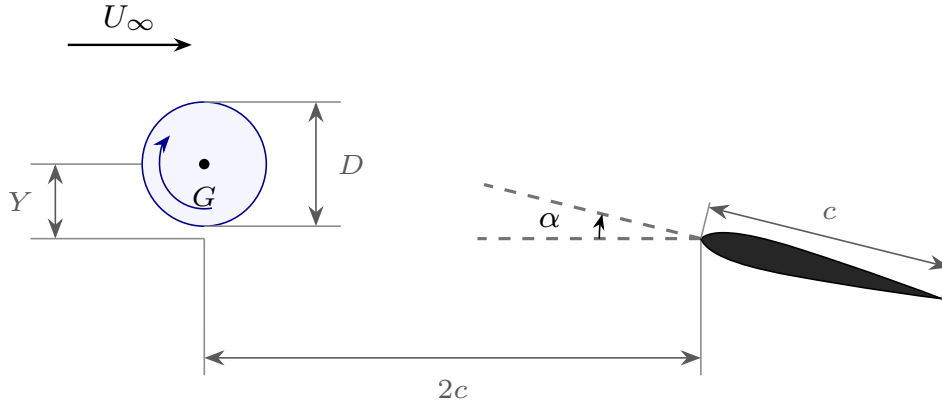


Figure 1. [S2-C1] Case geometry (schematic, not to scale), drawn directly in the free-stream-aligned axes in which (G, D, Y) are defined, following Fukami *et al.* (2025). The vortex core is released a fixed $2c$ upstream of the leading edge (LE), offset by Y normal to the free stream U_∞ , and convects downstream parallel to it, not along the chord. The chord is pitched by the angle of attack $\alpha = 14^\circ$ relative to these axes, trailing edge (TE) below leading edge. The core has diameter $D = 2R/c$ (core radius R) and peak azimuthal velocity $u_{\theta,\max}$ setting the signed gust ratio $G \equiv u_{\theta,\max}/u_\infty$; the arrow shows the clockwise sense of rotation for $G > 0$. Only (G, D, Y) vary across cases.

S2-C1

KEEP

Defines the free-stream-aligned axes in which (G, D, Y) are read. Without it the sign conventions of the envelope result are ambiguous.

numerical simulations of this configuration, computed with the GPU-enabled spectral-element solver SOD2D (Gasparino *et al.* 2024) with no subgrid-scale model, so that all scales of the separated wake are resolved at this Reynolds number. The solver uses an incompressible, constant-density pressure-projection formulation ($M = 0$; table 1). The configuration and its physical characterisation follow Fukami, Smith & Taira (2025), which is the reference for the physics, not the source of the data. The vortex has the azimuthal velocity profile

$$u_\theta(r) = u_{\theta,\max} \frac{r}{R} \exp\left[\frac{1}{2}\left(1 - \frac{r^2}{R^2}\right)\right], \quad (2.1)$$

which peaks at the core radius $r = R$ (Taylor 1918), and is parametrised by three scalars: the signed gust ratio $G \equiv u_{\theta,\max}/u_\infty$, the core diameter $D \equiv 2R/c$, and the offset Y of the vortex centre, measured normal to the free stream (figure 1). The release point is fixed at $2c$ upstream of the leading edge and offset by Y , both measured in axes aligned with the free stream, following Fukami *et al.* (2025) and the chord is pitched by the angle of attack α relative to these axes. The profile and the (G, D, Y) sampling follow the configuration of Fukami *et al.* (2025). The training envelope spans $|G| \in \{0.25, 0.5, 1.0, 1.5, 2.0, 3.0\}$ with $G = 0$ as the undisturbed reference, $D \in \{0.5, 1.0, 1.5\}$ chords, and $Y \in [-0.4, +0.4]$; the value $|G| = 4$ is held out as an extrapolation boundary, for reasons that are physical and that we return to below. Convective time t is measured in chords from the vortex release, $t = 0$, while t_{imp} denotes the instant at which the vortex centre reaches the leading edge, near $t/c = 2$ for this configuration; all phase windows below are referred to t_{imp} .

[S2-02] Without a gust, the $\alpha = 14^\circ$ wake at this Reynolds number is separated and sheds periodically; in dynamical terms the undisturbed flow is a limit cycle. A gust encounter is then a transient excursion from this limit cycle and back, and it is helpful to read it in stages (figure 2) (Smith *et al.* 2024; Fukami *et al.* 2025; Odaka *et al.* 2026). The vortex first induces a strong wall-normal vorticity flux at the leading edge, the airfoil experiences

S2-02

KEEP

The staged reading of the encounter that the five observables summarise. Physics framing the Results repeatedly refer back to.

a sharp lift excursion as a leading-edge vortex (LEV) forms, the LEV rolls up and sheds together with a trailing-edge structure as the load reaches its extremum, and the wake then relaxes back toward the baseline shedding. The sign of G sets whether the first lift excursion is positive or negative, Y sets the side and strength of the impingement, D sets how much of the encounter is dominated by the vortex core, and the impact timing sets the phase of the baseline cycle at which the disturbance arrives. The five observables defined in §2.2 are quantitative summaries of exactly this sequence.

[S2-03] A consequence of the source characterisation (Fukami *et al.* 2025) is central to how we read the extrapolation case. For $|G| \leq 3$ the encounter is primarily two-dimensional, so a mid-plane slice of the spanwise vorticity carries the relevant large-scale dynamics; for $|G| \geq 4$ the interaction becomes genuinely three-dimensional, with fine-scale instabilities developing during impact. We confirm this from the full three-dimensional vorticity fields: the fraction of the spanwise-vorticity (ω_z) enstrophy (the integral of squared vorticity over a region, formalised for the wake window in §2.2, eq. 2.2) not captured by its spanwise mean, the content a mid-plane slice discards, is roughly constant across the training range ($|G| \leq 3$, post-impact wake median ≈ 0.20) and rises about threefold at $|G| = 4$ (≈ 0.56 , computed on the four boundary cases for which the full three-dimensional fields were retained). Because the encoder in this work consumes a single mid-plane vorticity field (§2.2), the held-out $|G| = 4$ case is not merely one parameter step beyond the training range; it is a regime in which the mid-plane representation omits a substantially larger fraction of the three-dimensional vorticity content, so that more of the dynamics is not captured by what the encoder sees. We therefore treat $|G| = 4$ as a physically more demanding extrapolation set in which a mid-plane state is incomplete, rather than as a claim of parametric generalisation.

2.2. Data, fields, and observables

[S2-04] Each simulation case is one long run partitioned into encounters of 120 cached frames at $\Delta t = 0.05 t/c$, one encounter per gust release. The simulations use a spectral-element discretisation of polynomial order $p = 4$, and the fields are cached on a structured grid of $192 \times 96 \times 32$ points spanning $x/c \in [-1.5, 4.5]$ and $y/c \in [-1.5, 1.5]$ with a spanwise extent of approximately one chord; this analysis subdomain matches the data-driven subdomain of the source characterisation (Fukami *et al.* 2025). The lift and drag coefficients are $C_L = F_L / (\frac{1}{2} \rho u_\infty^2 c)$ and $C_D = F_D / (\frac{1}{2} \rho u_\infty^2 c)$, obtained by integrating the pressure and viscous stresses over the airfoil surface. The remaining DNS solver-resolution details required for full reproducibility are listed in table 1; the grid and time-step sensitivity check is still pending from the simulation collaborators. The dataset comprises 102 cases partitioned into training, validation (the held-out last encounters of the training cases; archive name test_a), in-distribution test (test_b) and boundary test (test_c) sets; the archive names in parentheses label the corresponding records in the data release and are not used again in the running text. The in-distribution test set is stratified to span the gust strengths, the three core diameters, and the two source groups (figure 3); the boundary test set is the out-of-distribution $|G| = 4$ extrapolation set, sampled symmetrically in both signs of the gust. These 102 cases are separated in preprocessing into 450 encounter windows of 120 frames each, one per gust release: 268 training, 100 validation, 42 in-distribution test and 40 boundary test encounters. The undisturbed reference case is retained in training. Behind these encounter counts are far fewer distinct cases: the 42 in-distribution test encounters come from only 10 held-out cases (6 interior and 4 boundary tier, the latter the edge-of-training tier of figure 3) and the 40 boundary test encounters from 8 cases, with most cases contributing four to six encounters that share a baseline shedding history. Encounters of the same case are therefore not independent, so encounter-level intervals

S2-03

KEEP

The physical reason $|G| = 4$ is a regime change and not one parameter step. This paragraph is what licenses the boundary reading in §4.6, §5.4 and §6.

S2-04

KEEP

Partition, encounter counts and the non-independence caveat behind the case-clustered statistics. Reproducibility rests on it.

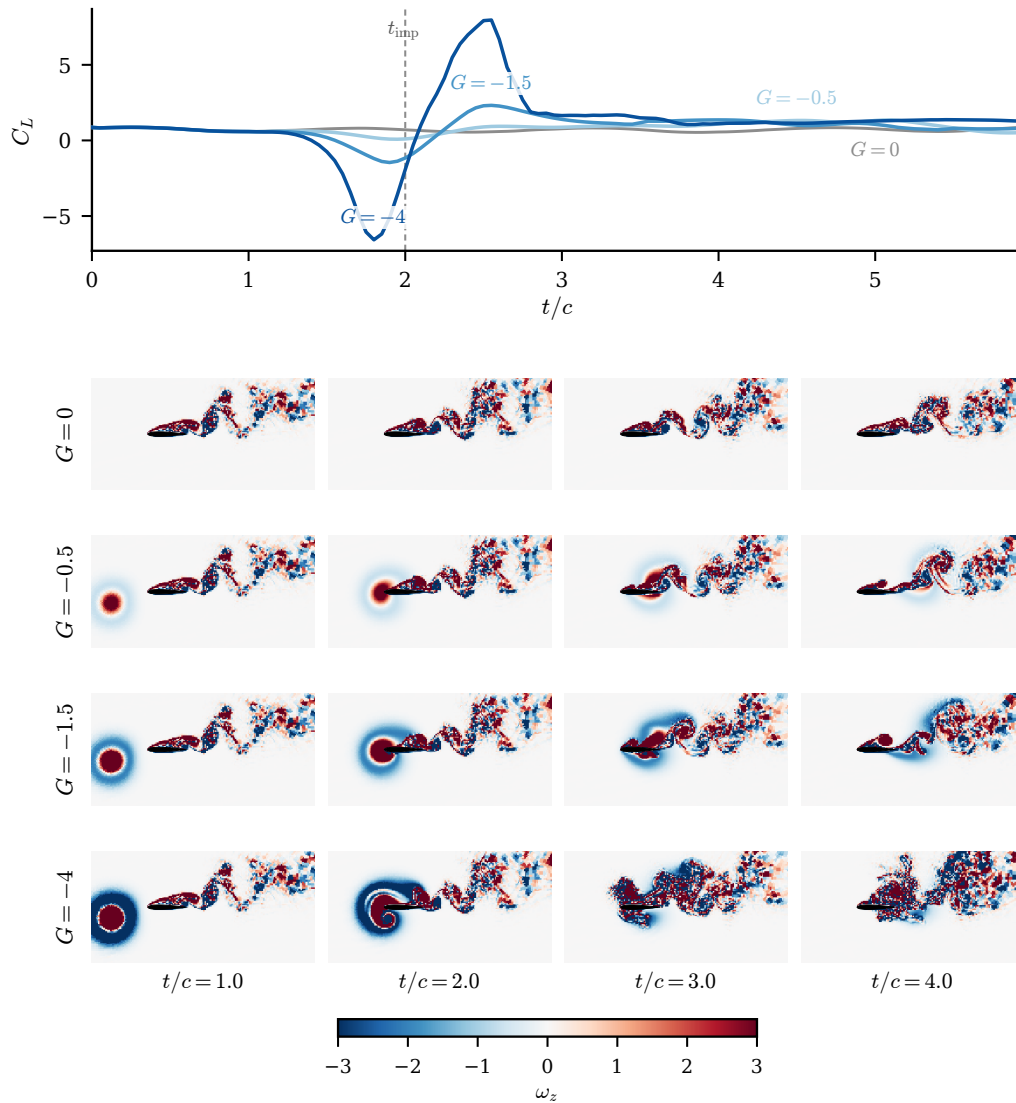


Figure 2. [S2-C2] The vortex gust as a wake-reorganisation event: a gust-strength sweep at fixed core diameter $D = 0.5$ and offset $Y = -0.1$ (first encounter of each case). Top: lift coefficient C_L against convective time, with $t/c = 0$ the release and t_{imp} (dashed) the leading-edge crossing near $t/c = 2$; the load excursion deepens and widens with $|G|$, the baseline ($G = 0$) in grey. Bottom: mid-plane spanwise vorticity ω_z per case (rows $G = 0, -0.5, -1.5, -4$) at four times (columns $t/c = 1, 2, 3, 4$), raw units, colourbar clipped at ± 3 (intense cores saturate), airfoil filled. The strong gust reorganises the wake through impact while the weak cases barely perturb the shedding; $G = -4$ is the $|G| = 4$ extrapolation case, shown only for illustration. The release parameters alone do not fix where the wake sits afterwards, which is why the wake, not only C_L , is the endpoint a reduced state must carry.

S2-C2

KEEP

The gust-strength sweep that shows the wake reorganisation the paper is about. Also states why the wake, not only C_L , is the endpoint.

S2-05

SUPPL.

A data-integrity check on the run4 cases, not a result: the mirror-image lift transients confirm the file metadata. The one load-bearing sentence, that the boundary is sampled at both signs, is already in S2-04 and the fig:paramspace caption.

overstate the effective sample size; the headline comparisons are reported additionally with a case-clustered bootstrap and a case-level paired test (Appendix A). The partition is fixed once and used identically by every model and baseline.

[S2-05] The $|G| = 4$ extrapolation boundary is sampled symmetrically by combining the periodic archive cases with four additionally released cases at the opposite sign of the gust. At matched core diameter and offset the two signs of the $|G| = 4$ gust produce near

Quantity	Value	Why it is reported
Free-stream Mach number	$M = 0$ (constant-density incompressible solver)	Confirms the solver formulation is incompressible at $Re = 5000$.
Computational domain and spanwise extent L_z/c	$x/c \in [-15, 17]$, $y/c \in [-15, 15]$, $L_z/c = 1$	Bounds domain blockage and confirms the span resolves the three-dimensional wake.
Element counts and solution-point counts	39 342 elements ($p = 4$), 2 632 900 solution points	Fixes the spatial resolution of the spectral-element discretisation.
Minimum wall-normal spacing $\Delta n_{\min}/c$ and wall units Δn^+	$\Delta n_{\min}/c \approx 2.2 \times 10^{-3}$; $\Delta n^+ \sim 1-4$ (instantaneous estimate)	Shows the near-wall layer is resolved without a wall model.
Time step $\Delta t u_\infty/c$ and maximum CFL	$\Delta t u_\infty/c = 2.5 \times 10^{-4}$ (fixed), $CFL < 0.5$	Establishes the temporal resolution and stability of the integration.
Gust-release station x_0/c	$x_0/c = -2$	Sets the upstream distance over which the Taylor vortex develops before impact.
Grid and time-step sensitivity check (baseline and a representative gust case)	[PENDING:]	Demonstrates that the reported statistics are converged.

Table 1. [TDNS-C1] DNS solver-resolution details required for full reproducibility (§2.2). The data are direct numerical simulations with no subgrid-scale model, so all scales are resolved and no large-eddy closure is involved. Rows marked pending await the simulation collaborators.

TDNS-C1 **KEEP**
The DNS reproducibility record. Generated from the DNS metadata YAML, so any wording change goes through that file and the renderer, never this one.

S2-06 **KEEP**
What the cache stores and how the vorticity is normalised. Trim internally: the SSIM constants are restated verbatim in S35-01.

S2-07 **KEEP**
Pre-registers the two co-primary endpoints. The multiplicity correction in appendix A is applied against exactly this declaration.

170 mirror-image lift transients, one a positive-first excursion peaking near +8.3 and the other
171 troughing near −7.6, as the sign of the gust ratio sets the sign of the first excursion (§2.1),
172 so the two halves of the symmetric boundary set are physically consistent. Sampling the
173 boundary at both signs is a genuine improvement over a one-sided $|G| = 4$ set: it lets the
174 sign asymmetry be measured rather than assumed (§4.6.2).

175 [S2-06] For each frame the cache stores the mid-plane spanwise vorticity $\omega_z(192, 96)$,
176 the span-averaged wall pressure $p_{\text{wall}}(192)$, and the lift and drag coefficients C_L, C_D .
177 The vorticity is normalised by a fixed pipeline: a spatial mask removes the solid and one
178 adjacent cell layer to suppress the leading-edge finite-difference artefact, a per-encounter
179 high-percentile clip caps spikes, and the field is scaled by three times the training standard
180 deviation with no mean shift, preserving the vorticity antisymmetry. All representation
181 and probe losses are computed in this normalised space; unnormalised values are used
182 only for physical-units metrics and figures. The field-reconstruction similarity reported
183 below uses the structural similarity index of Wang *et al.* (2004) with the standard constants
184 $K_1 = 0.01$, $K_2 = 0.03$ and a dataset-dependent dynamic range $L = 8.487$, set to twice the
185 global 99.9th percentile of the normalised target magnitude over the validation split so that
186 the index is comparable across reruns. Appendix A repeats the primary wake probe under
187 no amplitude clip and under a single training-pooled leak-free threshold, and the family
188 ordering is unchanged, so the per-encounter clip does not drive the result.

189 [S2-07] We evaluate forward closure against five physical observables that together span
190 body-force and spatially distributed wake diagnostics. The lift and drag coefficients C_L
191 and C_D are the body forces. The wake enstrophy and the positive and negative circulation
192 (defined below, eq. 2.2 and eq. 2.3) are spatially distributed wake observables: they integrate
193 (signed) vorticity over the wake region and are the most demanding to forecast, because
194 they are sensitive to the redistribution of vorticity that the LEV roll-up and shedding
195 produce. The choice is deliberate: the wake observables can change sharply while the
196 scalar forces change little, so they are the part of the encounter most likely to separate a
197 representation that has captured the gust-vortex physics from one that has only captured its
198 force signature. On these grounds we fix two co-primary endpoints in advance: the wake
199 enstrophy, which the reduced state must carry beyond the release parameters, and the lift
200 coefficient C_L , which the sequential wall-pressure estimator of §3.3 tracks and in which the
201 operating envelope is stated. The remaining three observables are reported as secondary

and descriptive. This pre-registration is what the family-wide multiplicity correction in Appendix A is applied against.
 [S2-08] The five observables are defined on each frame of the masked vorticity field ω_z described above. The lift and drag coefficients C_L, C_D are as defined earlier in this section. The remaining three are computed on a fixed downstream window $\Omega_w = \{(x, y) : x/c \in [0.5, 4], |y/c| \leq 1\}$, chosen to exclude the boundary layer and leading-edge region ($x/c < 0.5$) and to cover the region into which the shed wake convects. The wake enstrophy is the mean-square vorticity integrated over this window,

$$E_w(t) = \int_{\Omega_w} \omega_z^2 dA, \quad (2.2)$$

a single non-negative scalar that summarises how much rotational activity, of either sign, the wake carries at a given instant; it rises sharply as the LEV rolls up and sheds and falls back as the wake relaxes. Because E_w sums ω_z^2 , it cannot distinguish a wake dominated by one strong vortex from one holding two opposite-signed structures of the same net strength, so the signed wake circulations recover that sign information by integrating the vorticity itself, separately over its positive and negative parts, above a magnitude threshold that keeps only the coherent vortex cores and discards the weak, spatially diffuse vorticity of the ambient shear layer:

$$\Gamma^+(t) = \int_{\Omega_w, \omega_z > \omega_c} \omega_z dA, \quad \Gamma^-(t) = \int_{\Omega_w, \omega_z < -\omega_c} \omega_z dA, \quad (2.3)$$

with threshold $\omega_c = 1$ in the cleaned vorticity units. Γ^+ tracks the strength of positively-signed structures in the wake and Γ^- the negatively-signed ones (for instance the leading- and trailing-edge vortices of an encounter), independently of how the two partially cancel inside E_w . The signed circulations at $\omega_c \in \{0.5, 1, 2\}$ are nearly collinear (pairwise Pearson well above 0.9 across held-out frames) and the predictive latent's circulation closure advantage over the reconstructive baseline is essentially unchanged across them, so the closure does not depend on this threshold. Every case uses the same fixed window and mask, and values are reported in the cache vorticity units integrated over chord-normalised area. The forecast horizon $H = 16$ is the sixteenth cached frame after impact, $t = 16 \Delta t = 0.8 c/u_\infty$.

[S2-09] One definitional point bounds how these observables are read. The wake and flow observables (the enstrophy and the two circulations) are computed on the same mid-plane slice the encoder consumes, so they omit the out-of-plane content quantified in §2.1 (roughly a fifth of the ω_z enstrophy in distribution), whereas C_L and C_D are true three-dimensional surface-integrated forces. The closure therefore compares mid-plane wake diagnostics against three-dimensional forces, and the mid-plane proxy is weakest at the $|G| = 4$ extrapolation boundary, where the out-of-plane fraction rises substantially. This split also matches what a physical rig can measure: the mid-plane vorticity field is the kind of planar slice a particle-image velocimetry (PIV) system would resolve, while C_L and C_D are exactly the integrated quantities a load cell reports, so the five observables are realisable measurements in an experimental setup and not only diagnostics available because the full DNS field happens to exist.

S2-08

KEEP

Defines E_w and $\Gamma^\pm$, the endpoints of the whole comparison. The threshold-collinearity check inside it could move to the data record.

S2-09

KEEP

Bounds how the closure may be read: mid-plane wake diagnostics against three-dimensional forces. The PIV and load-cell realisability sentence is the removable part.

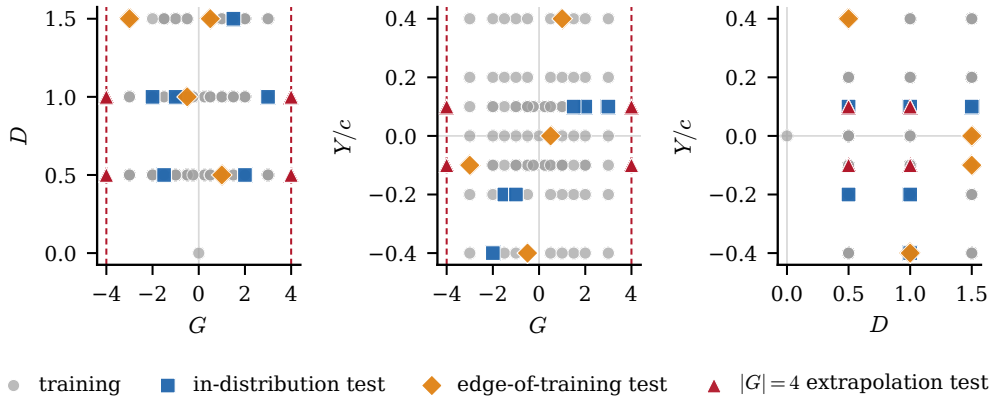


Figure 3. [S2-C3] Parameter-space sampling of the 102 cases in the (G, D, Y) gust envelope, shown as three two-dimensional projections and coloured by split: training (268 encounters), the stratified in-distribution test set (42 encounters from 10 cases, split into in-distribution and edge-of-training tiers), and the $|G| = 4$ extrapolation set (40 encounters from 8 cases). The dashed lines mark the $|G| = 4$ extrapolation boundary, now sampled at both signs of the gust so that the boundary set is symmetric rather than one-sided. Training mass concentrates near $Y/c = 0$, which is why the wall-normal offset is the least well resolved parameter (§4.3).

S2-C3

KEEP

Shows the sampling and the symmetric boundary, and names the Y under-sampling that §5.4 lists as a limitation.

3. Reduced states, forecasting and wall-pressure estimation

3.1. Model and training objective

[S3-01] The model takes no gust parameters: neither the encoder nor the predictor receives (G, D, Y) . The predictive encoder E_θ maps the mid-plane vorticity field $\omega_z(192, 96)$ to a latent $\mathbf{z} \in \mathbb{R}^d$ at the headline dimension $d = 32$. This dimension is a deployment criterion rather than an accuracy-tuned choice, and the criterion is the sensing budget the estimator has to work with. Reading K wall-pressure sensors over a window of m delays gives Km scalar measurements per update, and the delay-embedding bound of §3.3 holds a state of dimension d to be generically reconstructible from them once $Km > 2d$. The production budget of §3.3 is $K = 8$ sensors over a ten-frame window, which admits d up to 40; $d = 32$ sits just inside that ceiling, so the state is as large as the wall can support and no larger. The representation results below confirm it also sits on a readability plateau. The encoder is a hybrid network (figure 4): three convolutional downsampling stages produce a $24 \times 12 \times 256$ feature map, a six-layer vision transformer processes the resulting tokens, and the class-token output is projected to $\mathbf{z}$ through a single linear layer with a batch-normalisation boundary, which the characteristic-function regulariser below requires. Layer counts, widths and parameter counts for the encoder and the predictor are in table 11 (appendix A.3). The encoder is therefore a pure state map, and this is the property that lets the latent itself serve as the observable a controller would have access to, since at deployment the gust parameters are not known a priori (§5).

[S3-02] The predictor P_ϕ advances the pooled state in time, from its own history alone, so the model is unconditioned end to end. Throughout the paper the forecasting task is one map, from a context of L past states to the future latent sequence over a horizon H ,

$$[\hat{\mathbf{z}}_{t+1}, \dots, \hat{\mathbf{z}}_{t+H}] = f(\mathbf{z}_{t-L+1}, \dots, \mathbf{z}_t), \quad (3.1)$$

with no gust parameters among the inputs; what varies between models is how f is realised. The training predictor is a recursive single-step model rather than a sequence-to-sequence one: a six-layer autoregressive transformer with rotary temporal positions and a causal mask maps a short history of 32-vectors to the next 32-vector alone, and the horizon is covered by feeding that prediction back in as the newest history element and repeating, so

S3-01

KEEP

The encoder and the justification of $d = 32$ as a deployment criterion rather than a tuned choice. Referees will look for exactly this.

S3-02

KEEP

Carries equation (1), the one statement of the forecasting problem. The two-network description duplicates APA-01, which is where the detail belongs.

its errors can compound along the horizon. The one-shot alternative, which emits all H future states from one forward pass and so cannot compound, is the direct multi-horizon forecaster of §3.2.1; the two are compared in §4.4. During training it rolls eight steps open loop from the two most recent encoded states, and the loss is the multi-step error between the rolled coefficients and the online encoder outputs at the matching frames, detached. The pipeline is therefore as plain as a linear or autoencoder reduced-order model: 32 numbers in, a dynamics model on the vector, 32 numbers out, with no spatial map anywhere in the loop. The state is the pooled vector at every stage: the anti-collapse regulariser, the observable heads, the probes, and the estimator all read the same 32-vector, and nothing in the model carries more. The estimator of §3.3, an ensemble Kalman filter that advances a cloud of candidate states through a prediction model and corrects them against sparse wall-pressure measurements at each frame, refits a predictor of this same class on the frozen pooled trajectories for its predict step, so the training and estimation dynamics share a function class. Figure 4 shows the encoder and predictor, and figure 6 the deployment predict-correct loop; the matched evaluation protocol used throughout is figure 23 in appendix A, which also draws the contrast between the predictive training signal, a latent-space error against a shared-weight target embedding, and the reconstructive one, a field-space error through a decoder (figure 17). Full configuration, widths and parameter counts for both networks are in table 11 in appendix A.3.

[S3-03] The encoder and the predictor are trained jointly on two terms. The first is a multi-step rollout error. From the two most recent encoded states the predictor is rolled eight steps forward on its own outputs, never being handed the true state at any intermediate step, and the eight predicted latents are compared with the latents the encoder produces from the corresponding true fields. Those targets are computed by the same encoder as it trains, not by a separate frozen copy, but the gradient is blocked from flowing back through them, so the encoder is optimised to make its own trajectory predictable rather than to move the target towards the prediction. The second term exists because that first one has a trivial solution: an encoder that maps every field to the same constant vector makes the future perfectly predictable and drives the rollout error to zero while carrying no information about the flow. This is representational collapse, and it is a genuine attractor of the objective rather than a hypothetical. We suppress it with the characteristic-function regulariser of Balestriero & LeCun (2025); Maes *et al.* (2026), which matches the distribution of latents in a batch to an isotropic Gaussian along a small number of random projections; a constant encoder has zero variance along every projection, so the term penalises exactly the degenerate solution and keeps the latent covariance full rank. An automatic fall-back to a variance-covariance objective (Bardes *et al.* 2022) is armed should the participation ratio of the latent indicate collapse regardless. Its configuration is given in appendix A, and §4.1.1 shows which supervision choices collapse in practice. A variant lineage, referred to below as the direct-forecaster-coupled lineage, replaces the autoregressive transformer in this rollout term with the direct multi-horizon forecaster of §3.2.1 as the training predictor, so that the encoder is shaped against a one-shot rather than a recursive forward model; it is compared at low dimension in §4.2.1.

[S3-04] Three auxiliary heads map the latent state to aerodynamic observables during training. They exist because predictability alone does not fix what the state contains: the rollout term rewards any latent whose own evolution is easy to forecast, and the regulariser only keeps that latent from degenerating, so neither requires the state to carry the aerodynamic quantities a user of the model would want to read out of it. The heads supply that requirement directly, by making the load and the wake structure recoverable from $\mathbf{z}_t$ by a small map, and the conditioning study of §4.1.1 is what establishes that they change the representation rather than merely decorating it. The three are

S3-03**KEEP**

The training objective and the anti-collapse term. Names the direct-forecaster-coupled lineage used at low dimension.

**S3-04****KEEP**

Defines the three supervision heads and the wake descriptor. The five-citation provenance of the descriptor design could move to appendix A.



1. a lift head, mapping $\mathbf{z}_t$ to the instantaneous lift coefficient $C_L(t)$;
2. a wake head, mapping $\mathbf{z}_t$ to an 80-dimensional signed wake spectrum; and
3. a near-body head, described below and constructed in appendix A.1, mapping $\mathbf{z}_t$ to an 80-dimensional lift-element observable of byte-matched capacity.

All three read the encoder output, not the predictor output, so they constrain the state itself and not the forecast. All are trained with small weights jointly with the representation, none is part of the latent-prediction loss, and all are discarded once training ends. The scalar-lift lineage is the observable augmentation of Fukami & Taira (2023, 2025); which head supplies which property of the state is the subject of the conditioning study of §4.1.1. The wake target is a fixed descriptor, chosen by design and never tuned against an outcome. Its 80 numbers are two summaries of the masked wake region, concatenated. The first 64 are patch energies on an 8×4 grid, taken separately for positive and negative ω_z , so they record where the alternating vortices sit and with which sign while discarding pixel-level detail; this is the spatial-pyramid construction of Lazebnik *et al.* (2006). The remaining 16 are a radially-averaged power spectrum, which records the scales at which the wake energy resides and is both the canonical isotropic scale descriptor of a turbulent field (Pope 2001) and an established compact image descriptor in machine learning (Durall *et al.* 2020). Pairing a localised spatial summary with a multiscale spectral one is the spatial-plus-scale representation formalised by the wavelet scattering transform (Bruna & Mallat 2013), which has been applied to turbulent flow fields precisely to recover geometry that the power spectrum alone cannot resolve (Skinner *et al.* 2025). Keeping the descriptor to 80 dimensions constrains the wake structure without over-parameterising the target on the available training encounters. Because the three heads switch on and off independently, their effects are separable rather than confounded: §4.1.1 trains all eight on-off combinations, three seeds each, against matched reconstructive anchors, so each head's contribution is read off that cube rather than inferred from a single configuration.

[S3-05] One asymmetry in that comparison is worth naming, because it is a design choice and not an oversight: the linear basis carries no supervision channel at all. Supervision is a property of trainable encoders, so the proper orthogonal basis serves as the objective-free linear reference, read through the identical frozen probes, and the matched-supervision autoencoder exists precisely so that objective and supervision stay separable. The remaining conditions that keep the head comparison fair are the byte-matched targets and shared head class above, the identical heads carried by the reconstructive anchors, and the fit-on-train, score-on-held-out probe rule; the full set of matched controls is table 12 and §3.4.

[S31-01] The near-body head targets a physically constructed observable rather than a raw field, because not all vorticity contributes equally to the lift. The force-element decomposition of Chang (1992), in the form popularised for vortex-dominated flows by Menon & Mittal (2021), weights the vorticity at each point by an auxiliary potential that measures how strongly motion there projects onto the lift direction, so the resulting lift-element density is large where the flow is actually producing load and small where it is not. We supervise that density inside a feathered band within $0.3c$ of the surface, since the cache window truncates the domain and only the band-restricted field tracks the lift, and we compress it to 80 dimensions by the identical construction used for the wake target above. The two field observables are therefore byte-matched in capacity and share one network class, and differ only in where they look: the wake head at the shed vortex layout, the near-body head at the vorticity that is generating the load. Figure 5 shows the pipeline; the construction, its discretisation and its pre-committed quality gate are given in appendix A.1.

S3-05

KEEP

The one asymmetry in the head comparison that is a design choice rather than an oversight: the linear basis carries no supervision channel. Cut from a five-point enumeration whose other four points are stated where they bind.

S31-01

KEEP

The near-body head at the same depth as the other two: what the observable is, why it is not the wake target, and where the construction lives. Full construction moved to appendix A.

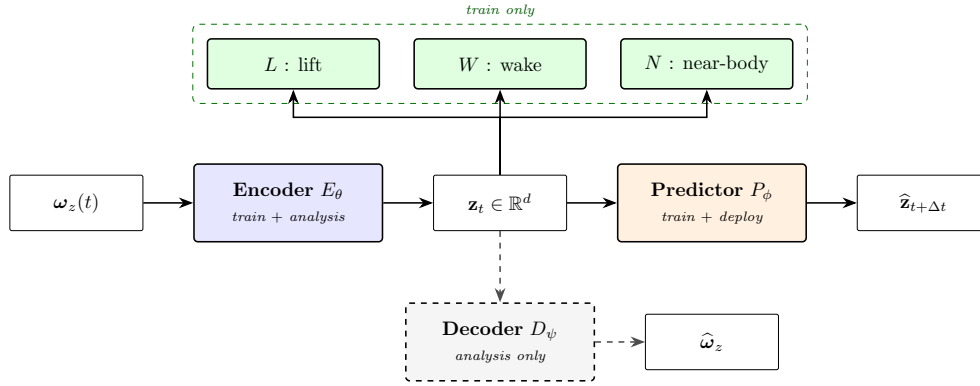


Figure 4. [S3-C1] The model at training time, and which modules survive it. The encoder maps the mid-plane vorticity field $\omega_z \in \mathbb{R}^{192 \times 96}$ to a latent $\mathbf{z}_t$ through a CNN stem, a six-layer ViT and a BatchNorm projection, and the predictor advances that latent from its own history in either of its two variants: the six-layer autoregressive transformer with rotary temporal positions of table 11, or the direct multi-horizon forecaster of §3.2.1. No gust parameters enter either network; both are unconditional. The three heads read $\mathbf{z}_t$ during training under a smooth- L_1 loss at unit weight, mapping it to the scalar lift $C_L(t)$ (L), an 80-dimensional wake observable (W) and an 80-dimensional Chang lift-element observable (N); none is part of the latent-prediction loss. Each module carries the stages in which it is active: the encoder is used in training and in every analysis that reads $\mathbf{z}_t$, the three heads are dropped once training ends, the predictor is the only trained module that also runs at deployment (figure 6), and the visualisation decoder D_ψ , which pairs LapFiLM with a spectral loss, is trained on the frozen encoder and serves analysis alone. The evaluation protocol applied identically to every encoder is figure 23 (appendix A).

S3-C1

KEEP

The training-time architecture, with the stage attribution for every module that no other text states.

3.2. Matched-predictor closure protocol

[S3-06] The comparison rests on a single principle: every baseline is the same pipeline, and only the encoder differs. Each encoder produces a latent trajectory $\mathbf{z}_{1:T}$ from the vorticity fields, a pooled coefficient vector at the same dimension for every family (the vector of proper-orthogonal temporal coefficients for the linear basis, the bottleneck vector for the autoencoders), and an identical probe family maps a latent to a physical observable for evaluation (figure 23, appendix A). The primary forecast-merit comparison of §4.4 realises the map of equation (3.1) with a single shared operator, the direct multi-horizon forecaster of §3.2.1, trained identically on every family's latents, so the family ordering cannot be an artefact of operator choice. A superseded protocol that instead fitted a different operator to each family, and what it changes, is documented in the supplementary material. The wake-supervised predictive state's own co-trained predictor, the co-trained reduced-order model, is reported alongside, and the sequential estimator of §3.3 uses the same matched transformer on the wake-supervised predictive state's frozen trajectories. Every state is compared at the matched headline dimension $d = 32$. This isolation is the only fair way to attribute representation quality to the encoder rather than to incidental differences in the readout. Two choices in that setup deserve stating, because both could otherwise be read as favouring our own model. The reconstructive references are run with ReLU activations and group normalisation, the configuration their sources document most fully, rather than the strict published variant with tanh activations and no normalisation. We trained both and the strict variant probes the gust parameters worse, so reporting it would have understated the baseline; the comparison here is against the stronger of the two. Second, every family's observable heads predict their observable at the current frame, the same form throughout, so no family is helped by a supervision signal shaped differently from the rest.

S3-06

KEEP

The single principle the comparison rests on: same pipeline, only the encoder differs. Also names the shared operator and the retired suited-operator protocol.

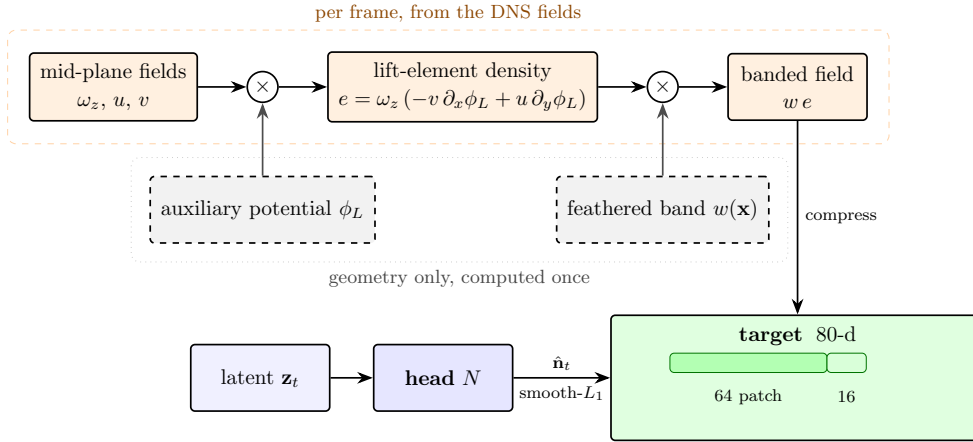


Figure 5. [S31-C1] The near-body supervision head, which supplies the N observable of §3.1. *Top row, per frame.* The mid-plane vorticity and velocity (ω_z, u, v) at frame t are weighted by the auxiliary potential ϕ_L to give the lift-element density $e = \omega_z(-v \partial_x \phi_L + u \partial_y \phi_L)$, which is restricted to a near-body band and compressed to 80 dimensions. This is the force-element decomposition of Chang (1992): ϕ_L weights each point by how strongly motion there projects onto the lift direction, so e is large where the flow is producing load and small where it is not. *Middle branch, grey, geometry only.* ϕ_L solves an exterior Laplace problem set by the airfoil surface and the lift direction alone, and the band is the feathered mask $w(\mathbf{x}) = \max\{0, 1 - \delta(\mathbf{x})/0.3c\}$ with δ the distance to the body. Both are computed once and carry no time or flow state; the band exists because the cache window truncates the domain, so only the band-restricted field tracks the lift. *Bottom row, trained.* The head, a small multilayer perceptron, reads the latent $\mathbf{z}_t$ and is fit to the target under a smooth- L_1 loss at unit weight. The encoder never sees e itself, and this loss is no part of the latent-prediction objective. The target’s 64 + 16 split is the identical construction used for the wake target, which is what makes the two field observables byte-matched in capacity: 64 patch energies, the banded field pooled onto an 8×4 grid with positive and negative parts kept apart, followed by a 16-bin radial power spectrum. The latter is the two-dimensional power spectrum of the same field averaged over rings of constant wavenumber magnitude and collected into 16 bins from the largest scales to the smallest, so each bin reports how much energy sits at that scale irrespective of direction. The patch block therefore records where the structure is and the spectral block at which scales it lives. The discretisation, the solution residual and the pre-committed quality gate are in appendix A.1; ϕ_L and the band are drawn on the cache grid in figure 24.

S31-C1

KEEP

The near-body head pipeline. Makes the geometry-only status of ϕ_L and the shared 64 + 16 construction visible, which the prose can only assert.



3.2.1. The shared direct multi-horizon forecaster

[S33-01] The forecast-merit comparison of §4.4 requires one operator trained identically on every family’s latents; the operator used throughout is the direct multi-horizon forecaster, a direct sequence-to-sequence realisation of equation (3.1) that emits the full horizon in one pass. It is a quantile forecaster whose design transfers the deployable ingredients of the TiRex forecasting line (Auer *et al.* 2025; Podest *et al.* 2026) without its foundation-model machinery. The operator reads a lookback window of the L states preceding the forecast. L is randomised during training, drawn uniformly from 16 to 30 so that the operator is not tuned to one context length, and held fixed at $L = 25$ for every evaluation reported here. Each window is normalised on itself, $\tilde{\mathbf{z}} = \text{arcsinh}((\mathbf{z} - \mu)/\sigma)$ with μ, σ the window mean and standard deviation (floored at 10^{-3}), passed through a two-layer LSTM of width 512, and mapped by a single feed-forward head to the full forecast tensor of 40 horizon steps, d coordinates and nine quantiles $q \in \{0.1, \dots, 0.9\}$ in one shot; the inverse transform restores physical scale. There is no autoregressive feedback, so rollout error cannot compound by construction. That is the reason for the design rather than a side effect of it: forecasts of learned latent coordinates in actuated wake flows degrade with horizon and can diverge

S33-01

KEEP

Defines the one operator under which every family is compared. The forecast result and the filter’s process noise both depend on this specification.



409 catastrophically (Solera-Rico *et al.* 2026), and emitting the whole horizon at once removes
 410 error compounding as one route to that failure. Training minimises the pinball loss of
 411 quantile regression (Koenker & Bassett 1978)

$$412 \quad \mathcal{L} = \frac{1}{|Q|} \sum_{q \in Q} \langle \rho_q(z - \hat{z}_q) \rangle, \quad \rho_q(u) = \max(qu, (q-1)u), \quad (3.2)$$

413 with AdamW. The loss is used because the filter needs a spread and not only a point
 414 estimate: ρ_q is minimised in expectation by the q th quantile of the predictive distribution,
 415 so the nine outputs are calibrated quantiles rather than an arbitrary band, and the width
 416 between them can be read directly as a forecast uncertainty. The median forecast is the
 417 point prediction everywhere; that inter-quantile width supplies the state-dependent process
 418 noise of §3.3.3. The optimiser settings, the one-off backbone and width selection over
 419 two widths and two quantile counts, and the adopted and rejected ingredients of the
 420 TiRex source line, including the oracle-covariate negative of §4.4, are catalogued in the
 421 supplementary material.

422 [S3-07] The headline figure of merit is representational closure: a fixed linear probe
 423 is applied to the latent encoded directly from a held-out field, and we ask whether it
 424 recovers each of the five observables, the wake in particular. This isolates what the encoder
 425 carries, with no rollout. We report it as the held-out coefficient of determination R^2 on
 426 the in-distribution test and boundary test sets, with the per-observable mean absolute
 427 error alongside, and bootstrap 95% confidence intervals over encounters. As a secondary,
 428 confirmatory measure we also report forecast closure: the predictor is rolled recursively
 429 from the pre-impact context to $H = 16$, and the same probe maps the predicted latent
 430 to each observable. A small forecast error means the encoder produces a latent whose
 431 dynamics, propagated by the predictor, retain the information the probe needs; a large error
 432 means either that the encoder discarded that information or that the predictor lost it during
 433 rollout, a distinction the drift diagnostic below resolves. The representational closure is in
 434 table 2, which reports the readability and the matched-predictor forecast merit side by side.
 435 The rollout is reported to show that the latent stays usable, not as a validated forecast.

3.3. Estimating the state from the wall

436
 437 [S34-01] At deployment the vorticity field is unavailable, and the wall pressure at a single
 438 instant does not determine the state: one pressure frame recovers little of the wake through
 439 any of the latents (§4.5). The limitation is dynamical rather than instrumental, and it is
 440 relieved by observing over a window of time rather than by adding sensors. The encounter
 441 evolves on an attractor of low effective dimension, a limit cycle carrying a gust excursion
 442 and a relaxation, into which the encoded latent concentrates its variance (§4.3). For such a
 443 system, delay-embedding theory (Takens 1981; Sauer *et al.* 1991) establishes that a generic
 444 observation function sampled over a window of delays reconstructs the attractor up to a
 445 diffeomorphism, provided the product of the sensor count K and the delay count m exceeds
 446 twice the attractor dimension. Reconstruction up to a diffeomorphism means that the delay
 447 vector and the true state are related by a smooth, smoothly invertible change of coordinates:
 448 the reconstruction preserves the dynamics and every quantity that is a smooth function of
 449 the state, but it does not return that state in its original coordinates, so a learned map from
 450 the delay vector to $\mathbf{z}$ remains necessary. The genericity assumptions of the theorem are
 451 not established for the finite, parametrically forced transient family considered here, so
 452 we treat the result as an organising principle and take the sensor-delay map of §4.5 as the
 453 governing empirical evidence. That map establishes two properties. A small number of
 454 sensors read over a short window suffices, and lengthening the window compensates for

S3-07

KEEP

Defines representational closure, the headline figure of merit, and scopes the rollout as secondary. The whole Results section is read through this definition.



S34-01

KEEP

Why estimation must be sequential, and the delay-embedding principle honestly scoped as an organising idea rather than a guarantee. §4.5 is read against it.



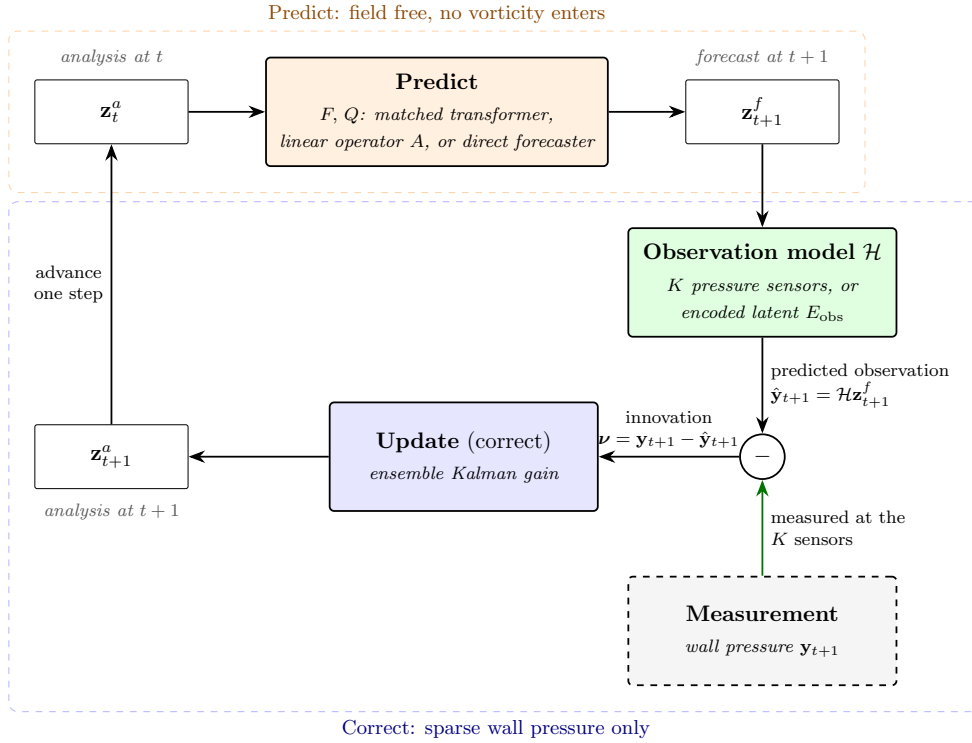


Figure 6. [S34-C1] The predict-correct loop at deployment, and the only route by which information reaches the state once the vorticity field is gone. *Predict*. An ensemble of 64 members, each a copy of the 32-dimensional coefficient state carrying its own perturbation, is advanced one frame through the prediction model without any field input; the forecast $\mathbf{z}_{t+1}^f$ is the result before any measurement is used. *Correct*. The observation model maps that forecast to the observation it implies, $\hat{\mathbf{y}}_{t+1} = \mathcal{H}\mathbf{z}_{t+1}^f$; the wall pressure is then measured at 8 span-averaged sensors, held out from training and placed by the per-model selection of appendix B; and the innovation is the measurement minus that prediction. Only the innovation, never the measurement itself, moves the state. The subtraction is performed in sensor space for the base filter of §3.3.1, so $\mathcal{H}$ is the pressure read-out, and in encoded latent space for the filters of §3.3.2 and §3.3.3, where the sensor history is first lifted to a state estimate by E_{obs} and $\mathcal{H}$ is the identity. The update is a stochastic ensemble Kalman step: the innovation is multiplied by a gain built from the ensemble spread and the calibrated process and observation covariances Q and R , and added to every member, giving the analysis $\mathbf{z}_{t+1}^a$ that the next predict step advances. All four estimator configurations are this one loop with only the two annotated slots swapped, the propagation (F, Q) and the observation path. The production configuration is the two-stage one: the direct forecaster with the encoded observation inside the impact window, the matched transformer with the pressure sensors outside it. Table 13 gives the settings per configuration, including the ensemble inflation applied before each update. No vorticity field enters at any point, the state is the same 32-vector at every step, and the frozen lift and wake probes are excluded from the innovation, which is what makes the reported analysis closure an estimate rather than a quantity the filter was handed.

S34-C1

The deployment loop on its own, with the two slots that distinguish the four estimator configurations. The leakage guarantee sits here because the innovation path is what carries it.

KEEP



455 reducing the sensor count, so temporal resolution is exchangeable for spatial resolution.
 456 The wake circulation, a coordinate to which the instantaneous pressure is insensitive, is
 457 nonetheless recovered from the window, because the circulation is dynamically coupled
 458 to the near-surface flow that the sensors do resolve and therefore leaves a signature in
 459 the pressure history. Estimation must accordingly be sequential, accumulating pressure
 460 innovations through learned dynamics, which is the setting of data assimilation.

[S34-02] Every estimator in this paper solves the same discrete state-estimation problem on the frozen coefficient state. Write $\mathbf{z}_t \in \mathbb{R}^d$ for the encoded state at frame t and $\mathbf{p}_t \in \mathbb{R}^K$ for the wall pressure at the K selected sensors, with successive frames separated by $\Delta t = 0.05 c/u_\infty$, the interval at which the cache stores the simulation. The pair

$$\mathbf{z}_{t+1}^f = F(\mathbf{z}_t^a) + \boldsymbol{\eta}_t, \quad \boldsymbol{\eta}_t \sim \mathcal{N}(0, Q_t), \quad (3.3)$$

$$\mathbf{y}_t = h(\mathbf{z}_t) + \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(0, R), \quad (3.4)$$

defines each member of the suite once its four ingredients (F, h, Q_t, R) are named; the remainder of this subsection does exactly that, and table 15 collects every parameter. The superscripts distinguish the two states of the assimilation cycle: $\mathbf{z}^f$ is the forecast, obtained by advancing the previous estimate through F before the new measurement is used, and $\mathbf{z}^a$ is the analysis, the same state after that measurement has corrected it. The frozen probes that read the lift and the wake enstrophy from the state are excluded from the innovation by construction, so the only measured quantity entering an update is the wall pressure and the analysis closure reported in §4.6.3 is an estimate rather than a quantity supplied to the filter (figure 6).

[S34-03] Two assimilation protocols are used, and each result states which one it follows. Under the envelope protocol, assimilation begins 24 frames before impact and ends 48 frames after it, and the ensemble is initialised at the start of that window without any field information. Under the ladder protocol, assimilation runs over the whole encounter: the ensemble is initialised at $t_0 = 9$, the first frame at which the ten-frame delay window is complete, from the encoded observation there, and scoring is restricted to the same impact window as the envelope protocol so that the two are comparable. Every estimator in either protocol is fitted on the training split alone. Each family senses at its own sensors, selected greedily on the training split so as to best expose that family's own state, with a shared target-blind array used wherever a sensor-budget claim must not depend on a placement tuned to any one model; the selection criterion and the two arrays are given in appendix B.

3.3.1. Base stochastic ensemble filter

[S34-04] The base filter is a stochastic ensemble Kalman filter in the perturbed-observation form of Burgers *et al.* (1998), following Evensen (2003), with $N = 64$ members. It is the reference configuration against which the others are read, not the production estimator of §3.3.3, but its update is the one every configuration in the suite uses. The prediction model F is the matched autoregressive transformer of §3.1, refitted on the frozen pooled training trajectories; each member carries its own buffer of past analysis states (at most 32 frames) and is advanced one step per frame. The process noise is state-independent, $Q_t \equiv Q$, estimated as the sample covariance of one-step teacher-forced predictor residuals over 256 training sub-trajectories of length 32, with a 10^{-6} diagonal floor. The observation map h is a ridge-regularised linear read-out from the state to the K sensors. It is fitted on four fifths of the training encounters and the observation-noise covariance R is the diagonal residual variance of that map on the remaining fifth, with floor 10^{-6} . The split is by whole encounter rather than by frame: frames within one encounter are strongly autocorrelated, so a frame-wise split would place near-copies of the fitted data in the held-out set and return an R far smaller than the error the filter actually meets, which is the one quantity that tells the filter how much to trust its sensors. Both parts of the split lie inside the training split, so no held-out encounter is touched.

[S34-05] Write $\bar{\mathbf{z}}_t^f = N^{-1} \sum_i \mathbf{z}_t^{f,(i)}$ for the ensemble mean of the forecast, $X_t^f = [\mathbf{z}_t^{f,(i)} - \bar{\mathbf{z}}_t^f]_i$ for the $d \times N$ matrix of member deviations from it, and $Y_t^f = [h(\mathbf{z}_t^{f,(i)}) - h(\bar{\mathbf{z}}_t^f)]_i$ for the

S34-02

KEEP



The shared state-space problem, equations (5) and (6), plus the leakage guarantee. Every filter in the suite is defined by naming four ingredients of this pair.

S34-03

KEEP

The two assimilation protocols, split out of S34-02 so the state-space definition and the evaluation setting are separate paragraphs. Every result states which protocol it follows.

S34-04

KEEP



The base filter and its fitted ingredients, including the encounter-grouped split behind R . It is the reference configuration, not the production estimator.

S34-05

KEEP

The ensemble-Kalman gain and update, written in the cross-covariance form the code implements, with the plain-language reading of what the gain does.

508 same deviations carried through the observation map. The gain and the update are

$$509 \quad \Sigma_t^{zy} = \frac{X_t^f (Y_t^f)^\top}{N-1}, \quad \Sigma_t^{yy} = \frac{Y_t^f (Y_t^f)^\top}{N-1} + R, \quad K_t = \Sigma_t^{zy} (\Sigma_t^{yy})^{-1}, \quad (3.5)$$

$$510 \quad \mathbf{z}_t^{a,(i)} = \mathbf{z}_t^{f,(i)} + K_t \left(\mathbf{y}_t + \boldsymbol{\epsilon}_t^{(i)} - h(\mathbf{z}_t^{f,(i)}) \right), \quad \boldsymbol{\epsilon}_t^{(i)} \sim \mathcal{N}(0, R). \quad (3.6)$$

512 where Σ^{zy} measures how each coordinate of the state co-varies with each sensor across
513 the ensemble, and Σ^{yy} is how much the predicted observation varies, from the spread
514 of the ensemble and from sensor noise together. Their ratio K_t is the weight given to
515 being wrong at the wall, so a state coordinate that the sensors see clearly is corrected hard
516 and one they barely see is left near its forecast. Each member is then moved by its own
517 innovation, computed against its own independently perturbed copy of the measurement;
518 perturbing the measurement per member is what stops the ensemble from collapsing to
519 zero spread as it is corrected, and it is what makes the filter stochastic. Written this way
520 the gain is formed from ensemble cross-covariances, exactly as implemented, and no $d \times d$
521 forecast covariance is ever built; for the linear h used here this is algebraically the textbook
522 $P^f H^\top (H P^f H^\top + R)^{-1}$.

523 [S34-06] Multiplicative inflation $\mathbf{z}^{f,(i)} \leftarrow \bar{\mathbf{z}}^f + \rho (\mathbf{z}^{f,(i)} - \bar{\mathbf{z}}^f)$ is applied before the update.
524 It rescales the ensemble about its mean, widening the spread when $\rho > 1$; the standard
525 remedy for the tendency of a small ensemble to become overconfident, which if left alone
526 makes Σ^{yy} too small, the gain too large and the filter progressively deaf to its sensors.
527 Here ρ is selected per family on the validation split only ($\rho = 1.00$ for the predictive and
528 linear families, $\rho = 1.05$ for the reconstructive one). The ensemble is initialised at the
529 window start from a windowed pressure-to-state regressor (a Nyström radial-basis ridge
530 with 300 components on a 12-frame causal pressure window) by sampling its training-
531 residual covariance around the regressed mean, so no field information enters the filter at
532 any time. The frozen observable probes that score the filter never enter the innovation: the
533 only measured quantity in (3.6) is the K -sensor wall pressure.

534 3.3.2. The linear latent filter with encoded observations

535 [S34-07] This configuration replaces both ingredients of the base filter with their simplest
536 stable counterparts, and it exists to separate what the estimator contributes from what
537 the learned dynamics contribute: if a linear propagation driven by a full-state observation
538 matches a nonlinear one, the gain was in the sensing rather than in the model. It follows
539 the latent-space assimilation construction of Tong *et al.* (2026). The dynamics are linear,
540 $F(\mathbf{z}) = A\mathbf{z}$, with A the ridge solution of $\mathbf{z}_{t+1} \approx A\mathbf{z}_t$ over all training transitions, followed by
541 a projection of its singular values onto $[0, 1]$. That projection is what keeps the propagation
542 stable: a singular value above one is a direction the operator amplifies at every step,
543 so a long rollout diverges along it, and clipping the spectrum removes those directions
544 outright rather than penalising them as Tong *et al.* (2026) do. Q is the covariance of the
545 resulting fit residuals. The observation side replaces the sparse sensors by an encoded
546 full-state observation: a delay-embedding operator lifts the normalised sensor history
547 $\mathbf{p}_{t-m+1:t} \in \mathbb{R}^{Km}$ with $m = 10$ (causal, edge-padded), and a ridge map W regresses it onto
548 the state, so that

$$549 \quad \mathbf{y}_t = E_{\text{obs}}(\mathbf{p}_{t-m+1:t}) = W^\top \text{vec}(\mathbf{p}_{t-m+1:t}), \quad h = \text{Id}, \quad R = \Gamma, \quad (3.7)$$

550 with Γ the covariance of the E_{obs} training residuals plus a 10^{-8} jitter. The same E_{obs} ,
551 evaluated framewise with no dynamics at all, is the static delay-embedded inverse that
552 anchors the estimator ladder. Filter consistency failure is declared by one fixed criterion

S34-06

KEEP

Multiplicative inflation, what it corrects, and the field-free initialisation. Small but load-bearing: the inflation values differ by family.

S34-07

KEEP

Carries equation (9), the encoded observation used by three of the four configurations, and the one fixed definition of consistency failure. Load-bearing well beyond its own subsection.

everywhere in the paper: the normalised innovation squared exceeds the $\chi^2_{0.99}$ quantile at the observation dimension for five or more consecutive frames, or any analysis Mahalanobis ratio exceeds 3. Both tests ask whether the filter’s own error bars are honest, that is whether the innovations are as large as the ensemble spread and R together say they should be; persistently larger means the ensemble has become overconfident and the gain is too small for the error it faces. It is therefore distinct from a large-error encounter, for which the analysis lift itself fails (impact-phase R^2 below -1), and a filter can post consistency failures while its mean still tracks the load.

3.3.3. Forecast-derived state-dependent process noise

[S34-08] The forecast-noise filter keeps the encoded observation (3.7) and replaces the member propagation by the direct multi-horizon forecaster of §3.2.1: each member is advanced by the one-step median forecast, and the process noise is read off the forecaster’s own uncertainty band at forecast time,

$$\mathbf{z}_{t+1}^{f,(i)} = \hat{\mathbf{z}}_{t+1}^{q_{0.5}} + c \boldsymbol{\sigma}_{t+1} \odot \boldsymbol{\xi}^{(i)}, \quad \boldsymbol{\sigma}_{t+1} = \frac{\hat{\mathbf{z}}_{t+1}^{q_{0.9}} - \hat{\mathbf{z}}_{t+1}^{q_{0.1}}}{2 \times 1.28155}, \quad \boldsymbol{\xi}^{(i)} \sim \mathcal{N}(0, \text{Id}), \quad (3.8)$$

where $\hat{\mathbf{z}}^q$ is the forecaster’s predicted q th quantile of the next state, $\odot$ is the elementwise product, and $\boldsymbol{\xi}^{(i)}$ is an independent standard normal draw per member. In words: every member is put at the forecaster’s median prediction and then scattered about it by an amount the forecaster itself nominates, so the ensemble is wide exactly where the forecaster expects to be wrong. The divisor turns a quantile band into a standard deviation: for a Gaussian the interval from $q_{0.1}$ to $q_{0.9}$ is $2 \Phi^{-1}(0.9) = 2 \times 1.28155$ standard deviations wide, with Φ the standard normal distribution function, so dividing the predicted band by that number reads off the equivalent $\boldsymbol{\sigma}$. The process noise $\mathbf{Q}_t = c^2 \text{diag}(\boldsymbol{\sigma}_{t+1})^2$ is therefore state-dependent: unlike the fixed $\mathbf{Q}$ of the base filter, it grows and shrinks with the difficulty of the step, which is the property the whole configuration exists to exploit.

[S34-09] The single scalar c is the filter’s only tuned quantity and it carries two independent calibrations, both on validation data alone. The production value $c = 1.77$ is the coverage calibration: it is the scale at which the one-step $q_{0.1}$ to $q_{0.9}$ band actually contains 80 % of held-out training transitions, as an 80 % band should. The second calibration, fixed in advance of seeing any value, tunes c for filter consistency instead. It asks whether the innovations are the size the filter claims to expect, measured by the normalised innovation squared

$$\text{NIS}_t = \frac{1}{d} \mathbf{v}_t^\top \mathbf{S}_t^{-1} \mathbf{v}_t, \quad \mathbf{v}_t = \mathbf{y}_t - \bar{\mathbf{z}}_t^f, \quad \mathbf{S}_t = \mathbf{P}_t^f + \mathbf{\Gamma}, \quad (3.9)$$

in which $\mathbf{v}_t$ is the innovation and $\mathbf{S}_t$ is the covariance the filter predicts for it, the ensemble spread $\mathbf{P}_t^f$ plus the observation noise $\mathbf{\Gamma}$. Dividing by the observation dimension d makes the expected value one when the filter’s uncertainty is honest: below one it is overdispersed, claiming more uncertainty than it has, and above one it is overconfident. We select c by driving the pooled impact-phase NIS toward one over the fixed grid $c \in \{1.0, 1.4, 1.77, 2.5, 3.5, 4.5, 6.0\}$, on the held-out encounters of the training cases, followed by a single frozen run on the test splits. The two calibrations need not agree, and which one wins is itself a result (§4.6.3). The two-stage filter applies the encoded-observation update (3.7) with the median-forecast propagation (3.8) inside the impact window only, and outside it propagates members with the matched transformer and assimilates in sensor space through the linear map of §3.3.1; the schedule boundary is the frozen impact-window mask of the cache and is not tuned. One further member of the suite is a reanalysis rather than a filter: a fixed-lag Rauch–Tung–Striebel smoother on the

S34-08

KEEP

The production route: state-dependent noise, its two independent calibrations, and the two-stage schedule. This is the estimator the headline number comes from.



S34-09

KEEP

The two independent calibrations of the band scale and the NIS statistic they are judged by. Which calibration wins is itself a result.

linear stack of §3.3.2, which revises each estimate once five later frames of pressure have arrived and so buys accuracy at a fixed reporting latency of $0.25 c/u_\infty$. Its recursion is given in appendix C.

3.4. Diagnostics and controls

[S3-08] Two diagnostics turn the closure result into a mechanism. The first is the latent drift, which addresses a discrepancy the closure numbers alone cannot resolve: a predictor with low one-step error can still roll out badly, because each prediction is fed back as the next input and small geometric errors compound. What we need to know is whether a rollout has wandered off the region of latent space the probes were fitted on. The natural measure is the Mahalanobis distance, which scores a point against a distribution in units of that distribution's own spread along each direction, so an excursion counts as large only if it is large relative to how far the training latents themselves range there. A Euclidean distance would not do this: the encoded latents have strongly unequal variance across coordinates, so a fixed radius means something different in every direction. We therefore report the Mahalanobis distance of the rolled-out latent at horizon H against the distribution of simulation-encoded latents for the same encoder, normalised by the Mahalanobis distance of the simulation-encoded latent itself, which removes the encoder's own scale and leaves a ratio comparable across families. A ratio near one means the rollout stays within the natural spread of the training latents; a large ratio means it has left that region, and any probe read there is being queried outside its support.

[S3-09] The second diagnostic is a parameter-only floor, and it decides whether the central closure result means anything at all. Each encounter is generated by three numbers, the gust parameters $\mathbf{c} = (G, D, Y)$, so an observable that those three numbers already determine can be predicted well by a model that has learned nothing about the flow. Since every encounter in a case shares one $\mathbf{c}$, a latent that had merely memorised which case it was looking at would score just as well. The floor bounds that possibility: it is what the parameters alone can explain, and only closure above it is genuinely supplied by the state the encoder built. We compute it with a kernel-ridge regressor mapping $\mathbf{c}$ directly to each observable at $H = 16$, with no latent input, fit on training and evaluated on the held-out splits. On wake enstrophy the floor is low, so the wake closure reported in §4.1.2 is not something the parameters could have supplied, and that is what makes it the paper's central evidence that the latent carries wake state. At the impact frame the same floor is high, because the just-released gust nearly determines the instantaneous flow, and it falls as the wake evolves away from its initial condition.

[S3-10] The predictive encoder differs from the reconstructive baseline in three ways at once, its objective, its architecture and its auxiliary supervision, so a bare comparison of the two cannot say which difference produced the result. The controls exist to close that gap, each varying one factor at a time (table 12): a crossing of objective with architecture at matched auxiliary heads, an ablation of each auxiliary head in turn, and a conditioned variant in which the predictor is additionally given the gust parameters $\mathbf{c} = (G, D, Y)$ that the model otherwise never sees, which bounds how much of the forecast skill is available from the forcing alone (§4.4.1). All the controls share the one encoder of table 11 and differ only in what is trained on top of it. The objective-free supervised control isolates the objective most directly: it trains the shared encoder with the wake and lift heads and the anti-collapse term but with no predictive or reconstructive objective, no predictor and no decoder, so the supervision is held fixed while every objective is removed, leaving what the observable heads alone supply. The published reference recipes, a plain convolutional autoencoder in the Fukami–Taira lineage (Fukami & Taira 2023, 2025), the β -variational autoencoder of Solera-Rico *et al.* (2024), and a linear proper-orthogonal-decomposition

S3-08

Defines the latent-drift diagnostic used in table 8 and §4.4. Without it the near-null decomposition cannot be read.

KEEP



S3-09

Defines the parameter-only floor that makes the wake result non-trivial. §4.1 and §5.1 both depend on it.

KEEP



S3-10

The one-factor-at-a-time controls and the objective-free supervised control. This is the paragraph that answers the confound objection.

KEEP



basis, use their own native architectures but produce a pooled coefficient vector at the same latent dimension. Their forecast merit is refit under the legacy suited-operator protocol of §3.2.

3.5. Protocol, phases and uncertainty accounting

[S35-01] Every encounter is scored in three phases anchored to its impact frame: lead-in [$t_{\text{imp}} - 8, t_{\text{imp}}$), impact [$t_{\text{imp}}, t_{\text{imp}} + 16$) and relaxation [$t_{\text{imp}} + 16, t_{\text{imp}} + 48$), all half-open in frames and clamped to the encounter. Load metrics are reported in three complementary forms, because each corrects a failure mode of the others. The coefficient of determination $R^2 = 1 - \text{SSE}/\text{SST}$ is pooled over the phase frames, where SSE is the sum of squared errors of the estimate and SST the sum of squared deviations of the truth about its own mean, so R^2 measures skill against predicting that mean and is negative whenever the estimate is worse than the mean would have been. It is scale free, which is what makes it comparable across families, and that is also its failure mode: at a phase where the truth varies a great deal between encounters, a large SST flatters it. So we report alongside it the root-mean-square and mean absolute errors in lift-coefficient units, which are not scale free and expose exactly that case, and peak-relative quantities, which ask about the one instant an extreme-gust application cares about. The peak diagnostics locate the extremum of $|C_L|$ within the impact phase in the estimate and in the truth separately, and report the value difference, its magnitude as a percentage of the true peak, and the timing difference in convective units. Peak-region readability uses a pooled R^2 over windows of half-width 8 frames centred on each encounter's true peak, against a persistence floor that holds the mean of frames $[0, 25)$ constant; phase lag is the argmax of the normalised cross-correlation within ± 20 frames with parabolic sub-frame refinement, positive when the estimate trails the truth. Decoded fields are scored by the structural similarity index of Wang *et al.* (2004) on three masks, the feathered near-body band of §3.1, a wake window spanning $0 < x/c < 4.5$ and $|y/c| < 1.25$, and the full frame; its constants and its dataset-dependent range are given in §2.2. Linear-probe R^2 is used throughout as a readability metric and never as an information metric: a low value means the quantity is not linearly accessible in that latent, not that it is absent. The control that motivates the distinction, a multilayer probe that equalises what a linear read-out cannot reach, is reported alongside every dimension comparison (§4.2.2). An observable read at one instant rather than over a window uses the same probe, fit on the training window and evaluated on held-out encounters at the pre-impact, impact and peak-lift frames (± 2), and is reported both as R^2 against the mean observable at that instant and as a root-mean-square deviation in the observable's own units, for the reason given above.

[S35-02] Uncertainty enters the reported numbers from six places. Two are sampling: a bootstrap over held-out encounters, and a clustered bootstrap over whole cases for every paired comparison between families, since encounters of one case are not independent of each other and treating them as such would understate the interval. Three are training stochasticity: the encoder seed, the seed of an operator fitted on a frozen representation, and the member-perturbation seed of the filter, which are separated because they answer different questions, namely whether a representation is reproducible, whether a downstream operator is, and whether the estimator is. The sixth is the read-out, cross-validated over cases, which bounds how much a probe result depends on the particular fit rather than on the latent. Table 16 gives the resampling or training unit for each, its count, and which results carry it; that table is the authoritative accounting, and it is there because no single number carries all six and which apply depends on what was varied to produce it. Every stochastic number in the paper states its count. Single-seed cells appear only in the dimension grid and say so in their captions. The acceptance gates for the conditioning study and for this

S35-01

KEEP

Defines the three phases and every load metric used in the paper. Trim internally: the SSIM convention is already given in S2-06.



S35-02

KEEP

Enumerates the six uncertainty sources and points at the one authoritative accounting. The sign-convention clause is still an open REVIEW-CLAIM.



campaign’s follow-up runs were written and committed before any corresponding result was read, and both declarations ship with the data record. The gust-intensity axis is reported in the inventory sign convention, and no test-C quantity is used for any selection anywhere in the pipeline.

4. Results

[S4-01] We evaluate three families of reduced-order state at matched pooled dimension: a predictive coefficient state, a matched-supervision reconstruction autoencoder, and a linear proper orthogonal basis, with the published-recipe reconstruction and a β -variational autoencoder retained as reference recipes. The controlled representational comparisons (§4.1 to §4.5) are reported on the in-distribution test encounters, with the boundary test set held in reserve so that the $|G| = 4$ extrapolation enters only as a final reported number and never as a selection signal. The estimator and the operating envelope (§4.6) are characterised with a frozen filter across all encounters, training, validation, in-distribution test and boundary test sets alike, and the per-encounter paired tests on the in-distribution test and boundary test sets are given in Appendix A. Every quantity quoted below is a held-out value at the readout horizon unless stated otherwise. One convention holds throughout: the wake-supervised predictive state (conditioning-cube cell CLW) is the single predictive representative carried into the forecasting and estimation comparisons, and the lift-focused predictive state is the lift-sharpened variant whose accuracy ceiling the construction section quantifies.

4.1. A non-collapsed predictive state

4.1.1. Constructing the state: which supervision supplies which property

[S4A-01] The construction question is answered by a $2 \times 2 \times 2$ cube over the three supervision heads $\{L, W, N\}$, trained at three seeds per cell on the pooled pipeline of §3.1 with matched reconstructive anchors, under gates written and archived before any result was read; probes are fit on train and scored on the held-out in-distribution test encounters, the validation set steering nothing and the boundary test set staying in reserve per the policy above. Under the predictive objective the scalar lift head is necessary for latent health. Every cell without L collapses, with participation ratios of 2.50 (C0) and 2.55 (CW) against a floor of 9.6, flat from the first diagnostic; neither the wake observable (figure 7) nor the near-body observable (appendix A.1), nor both together, substitutes for the scalar anchor. The reconstruction objective anchors without it: the wake-only autoencoder keeps a healthy latent (participation ratio 21.94) yet reads the peak loads poorly ($R^2 = 0.47$), so a healthy latent and a lift-readable latent are different achievements. The collapse is the low-intrinsic-dimension failure mode anticipated for the characteristic-function regulariser at its default configuration; the automatic variance-covariance fallback is armed at an iteration threshold the pooled runs never reach, so the cube exposes the mechanism rather than masking it.

[S4A-02] Nothing replaces the lift anchor. Replacing L by W fails decisively, the wake-only cell trailing the lift-only cell by a paired difference of -35.22 peak-region R^2 points (paired R^2 differences throughout are in units of 10^{-2} , signed towards the modified cell, with case-clustered intervals in figure 18), and replacing L by N fails the same way (-21.06). On top of L , a near-body force-element head sharpens the peak load further and is the strongest cell in the cube (appendix A.1). We carry the wake-headed wake-supervised predictive state as the single predictive representative downstream: its marginal value on top of L is geometry rather than lift, matching the lift-only cell on the peaks (the paired difference straddles zero) while adding near-body structural fidelity (0.017 SSIM in the

S4-01

KEEP

Names the three families and fixes the reporting policy that keeps the boundary set out of every selection. The section is unreadable without it.

S4A-01

KEEP

The conditioning cube and the collapse without the lift head. The construction finding the abstract leads with.

S4A-02

KEEP

Nothing substitutes for the scalar anchor, with the paired differences. Also fixes the wake-headed state as the representative carried downstream.

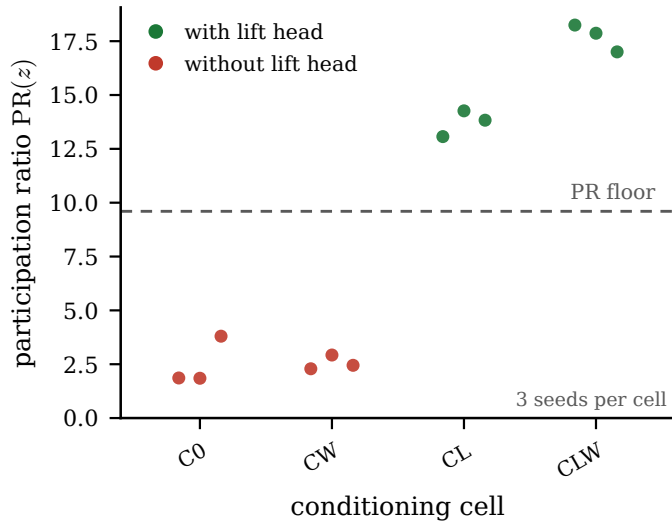


Figure 7. [S4A-C1] The conditioning cube on the lift and wake axes: final participation ratio per cell (in-distribution test set, three encoder seeds per cell) against the collapse floor $0.3 d = 9.6$, every no- L predictive cell (C0, CW) collapsing and every L cell (CL, CLW) healthy. The training history of the collapse, flat from the start of training while a healthy cell crosses the floor within the first quarter of the run, is figure 22 in appendix A; the near-body (N) cells and the paired peak-load increments are in appendix A.1.

S4A-C1

KEEP

The collapse against the participation-ratio floor, with the demoted panels named. Self-contained.

Wang convention of §3.5); the per-cell peak-region R^2 , seed spreads and intervals are in figure 18 and appendix A.1.

[S4A-03] The three heads therefore divide the labour: L anchors the latent, N sharpens the load, and W carries the wake state. The wake-entropy probe makes the last division quantitative: the wake-supervised predictive state reads the wake entropy E_w at $R^2 = 0.73$ linear and 0.84 through a multilayer probe (three seeds, in-distribution test set; this entropy-specific probe is distinct from the pooled wake-descriptor readability 0.751 quoted in table 2), where the lift-focused predictive state reaches only 0.37 and 0.63. The anchor comparison is required too: the reconstructive AE-LW carries the most linearly readable wake state of the anchors compared (0.77 linear, 0.90 MLP), as a reconstruction objective should; the predictive family's wake advantage is not probe readability but what survives forecasting and assimilation, quantified in §4.4 and §4.6.3. The full task-dependent readability matrix, five observables across every cell and anchor, is figure 20 in appendix A: the wake block lights up exactly where wake supervision or a reconstruction objective sits, and nowhere else. Because lift is included during training, lift accuracy alone cannot establish the role of the lift head. Its structural role is instead identified from the collapse of every predictive state trained without that head; the incremental effect of the near-body target is evaluated relative to an already lift-supervised state, under paired gates fixed in advance of evaluation. The two claims rest on different comparisons and fail independently.

S4A-03

KEEP

The division of labour between the three heads, with the entropy probe. The closing caveat about which comparison supports which claim is worth keeping.

4.1.2. What the coefficient state carries, and what supplies it

[S4-02] The comparison between reduced states is meaningful only if the wake is a discriminating endpoint, and it is. At the readout horizon the release parameters no longer fix the wake entropy, while they still fix the forces, so the wake is where a representation that has captured the gust-vortex physics parts company from one that has captured only its force signature. Table 2 reports what each pooled coefficient state carries, read by a fixed linear probe from a held-out in-distribution test field with no rollout. Within the controlled

S4-02

KEEP

Establishes that the wake is a discriminating endpoint and reads table 5. The central representational result.

block, the wake is read cleanly by every state that carries the wake-supervision head, with a held-out R^2 between 0.750 and 0.792, and it degrades in every state that does not, falling as far as 0.160 for the predictive latent trained without the head; the reference recipes of the bottom block, which vary more than one ingredient at a time, straddle rather than obey this split. Instantaneous readability of the wake is therefore supplied by the observable supervision, not by any training objective. The lift, read by the same windowed probe, is a different story: every anchored state carries it well, and the predictive state reads the lift at least as accurately as the wake (0.813 against 0.751), so the wake supervision sharpens the wake without costing the load.

[S4-03] The objective-free supervised encoder makes this concrete. Trained with the wake head but no predictive or reconstructive objective at all, it reads the wake at least as well as the full predictive latent (0.792 against 0.751): the paired readability gap is 0.041 in the wake with a confidence interval $[-0.047, 0.162]$ that includes zero, so the instantaneous readability is supplied by the supervision, not by the training objective. Under one shared forecast operator, the direct multi-horizon forecaster of §3.2.1 trained identically on every state, the three controlled wake-headed states are also the most forecastable and are statistically indistinguishable from one another (0.561 [0.349, 0.631], 0.574 [0.312, 0.634] and 0.443 [0.175, 0.548] at horizon sixteen, case-clustered intervals), so the forecastability of the coefficient state, like its readability, is supplied by the supervision rather than by the training objective: under the common forecaster the predictive objective does not dominate matched-supervision reconstruction in aggregate five-observable merit. What the predictive objective adds is the model it co-trains: the wake-supervised predictive state’s own vector predictor forecasts the five observables at 0.755 at horizon eight, above what the shared operator extracts from any latent there (0.680), and falls to 0.418 against the shared operator’s 0.561 at horizon sixteen, the error-accumulation contrast that §4.4 quantifies. We state the representational result as the supervision’s, and turn to what the predictive objective does add, which is dynamical.

[S4-04] One inversion in the record fixes the coordinate-level reading. Among the controlled states, the predictive latent trained without the wake head carries the strongest single-observable lift forecast (0.691) yet the weakest wake readability (0.160), because the forces are held redundantly across the latent while the wake is not, the distributed-code signature we return to in §4.3. Among the reference recipes the lift-augmented reconstructive autoencoder reads the wake only when it carries the wake head (0.432 against -0.094 without it), and the linear basis does not reach the wake at all (0.186).

4.2. Representation quality across dimension

4.2.1. How small the state can be

[S4B2-01] The dimension axis splits into two tiers, and the split is the paper’s compression statement (figure 8a). At $d = 4$ the predictive families read the peak loads through a linear probe at 0.903 for the wake-supervised predictive state lineage against 0.796 for the reconstructive baseline, three seeds each with non-overlapping ranges; the lift-focused lineage forecast through the direct multi-horizon forecaster rather than the autoregressive predictor is close to dimension-insensitive on the peaks, where the wake-supervised predictive state lineage declines with d (figure 8a). A four-coefficient lift-critical state is therefore viable. The wake state is not compressible to it: the decoded structural fidelity of the same lineage rises monotonically with dimension (figure 8c), so the wake-bearing state needs $d \geq 16$ while the lift saturates by $d \approx 4$ to 8. This decode-fidelity bar is distinct from, and stricter than, the wake-readability threshold of §4.2.3, where a wake R^2 of one half is reached from eight coefficients.

S4-03

KEEP

The objective-free control and the shared-operator tie, which together attribute readability and forecastability to supervision rather than to the objective. The paper’s most self-critical result.

S4-04

KEEP

The inversion that fixes the coordinate-level reading: best lift forecast, worst wake readability. Sets up the distributed-code argument.

S4B2-01

KEEP

The two-tier compression statement, which is the paper’s answer on how small the state can be. §6 quotes it directly.

Table 2. [TCLO-C1] What each pooled coefficient state carries, at matched dimension $d = 32$ on the 42 in-distribution test encounters. Held-out wake-ensrophy readability and, on the same windowed linear probe, lift readability (both fixed linear probes, no rollout); forecast merit under one shared operator, the direct multi-horizon forecaster of §3.2.1 trained identically on every state (three operator seeds; mean over the five observables of the pooled nonlinear-probe R^2 at horizon sixteen; seed means, with case-clustered bootstrap intervals in the text); field-reconstruction error (VRMSE at horizon eight); and decode similarity (SSIM). The controlled matrix (top block) moves one training ingredient at a time; the reference recipes (bottom block) use their own architectures (table 12). Wake readability separates on the wake-supervision head, not on the training objective, and under the shared operator the three wake-headed states are also the most forecastable, statistically indistinguishably from one another. The lift is read well by every anchored state and, for the predictive state, at least as well as the wake (0.813 against 0.751), so the wake supervision does not cost lift accuracy. The wake-supervised predictive state’s own co-trained predictor leads every fitted operator at short horizon (0.755 at horizon eight) and is overtaken by the direct multi-horizon forecaster at horizon sixteen (0.418 against 0.561), the error-accumulation contrast of §4.4.

State	Objective	Wake head	wake R^2	lift R^2	merit	field VRMSE	SSIM
predictive (wake)	predictive	yes	0.751	0.813	0.561	0.962	0.779
supervised only	none	yes	0.792	0.769	0.443	0.992	0.774
AE (wake)	reconstruction	yes	0.750	0.765	0.574	0.976	0.776
predictive (no wake)	predictive	no	0.160	0.797	−0.426	0.983	0.767
AE (no wake)	reconstruction	no	0.456	0.825	0.285	0.976	0.773
anti-collapse AE	reconstruction	no	0.333	0.206	0.068	0.949	0.782
β -VAE	reconstruction	yes	0.728	0.668	0.040	1.007	0.724
Fukami	reconstruction	no	−0.094	0.790	−0.195	1.016	0.708
Fukami (wake)	reconstruction	yes	0.432	0.727	0.162	1.026	0.705
POD	linear	no	0.186	0.610	0.036	0.930	0.775

[S4B2-02] The probe-dilution control disciplines how the first tier may be read (figure 8b). A multilayer probe nearly equalises every dimension (from 0.89 at $d = 4$ to 0.88 at $d = 32$ on the in-distribution test set): the lift information in the state is approximately unchanged over the tested dimensions, and small states do not know more. What changes with dimension is linear accessibility. The best four coordinates of a larger state recover only 0.66 to 0.55 through a linear read-out, so the code is distributed rather than diluted, and the $d = 4$ advantage is exactly that four coordinates are what a linear probe, a filter gain, or a controller reads directly. The precise form of the two-tier claim is therefore: compact and linear at $d = 4$, distributed and partly nonlinear above.

4.2.2. The decode floor

[S4B-01] Decoding closes the loop on the construction story: the same division of labour that the probes read appears in the decoded fields, scored on the in-distribution test set through identical-capacity frozen decoders with the Wang SSIM convention of §3.5. With the lift anchor present, the near-body head supplies the same near-body structural fidelity as the wake head (0.709 for the lift-focused predictive state against 0.709 for the wake-supervised predictive state), the collapsed no- L cells sit far below (from 0.470 for CW), and the reconstructive anchors reach parity (0.704 for AE-LW), as an objective trained on the field should (figures 25 and the per-cell bars in the data record). The decode panels make the same point qualitatively: all three healthy states recover the near-body vortex system and the coherent wake at matched fidelity, and the residual rows concentrate on the fine-grain shed structures no 32-coefficient state carries. The decode panels are figure 25 in appendix A.2, since field decode is a wash across the compact families and supports this claim rather than leading it.

TCLO-C1

KEEP

The representational closure table, the headline exhibit of §4.1. Reports readability and shared-operator merit side by side.

S4B2-02

KEEP

The probe-dilution control that disciplines the $d = 4$ claim. Without it the compression result would be over-read.

S4B-01

TRIM?

A whole subsection whose conclusion is that field decode is a wash, with its figure already in the appendix. S4-05 and S4-06 state the same result from the tables.

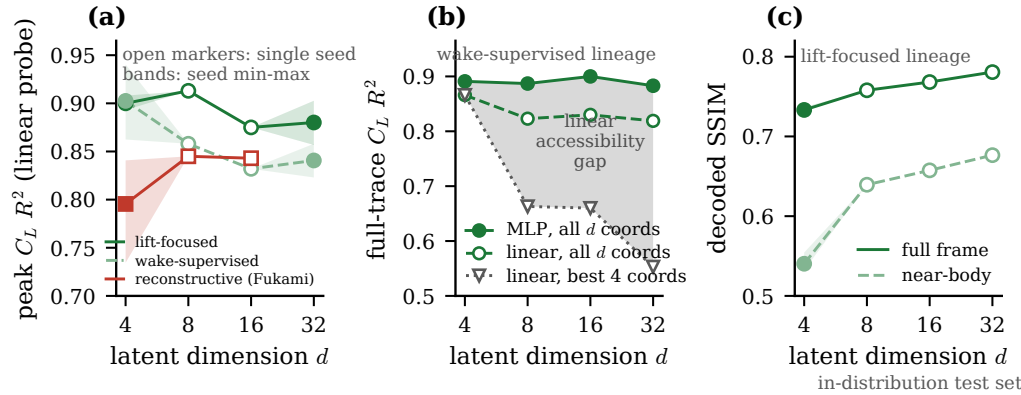


Figure 8. [S4B2-C1] The dimension axis on the in-distribution test set. (a) Peak-region lift R^2 (linear probe) against latent dimension for the three lineages; shaded bands are three-seed min-max ranges, open markers single seeds. (b) The probe-dilution control: a multilayer probe is flat in d while the best-four-coordinate linear probe falls, so the $d = 4$ advantage is linear accessibility of a distributed code, not information. (c) Decoded SSIM (Wang convention, §3.5) of the lift-focused lineage rises with d : the wake tier is not compressible.

S4B2-C1

KEEP

Three panels including the control, with the seed provenance stated. Self-contained.

S4-05

KEEP

Concedes the linear basis its field-reconstruction win before the wake result is read. Removing it would make the comparison look stacked.

S4-06

KEEP

Resolves the decode floor to the critical instants, which is where the linear readout actually breaks. Carries table 6.

S4-07

TRIM?

A single sentence pointing at an appendix figure to say, for the third time, that field decode does not separate the families. S4-05 and S4-06 have already made the point with a table.

S4-08

KEEP

The discriminating result the field pixels hide: supervision decides what a state carries at the critical instants. Carries table 7.

[S4-05] The linear basis is not disadvantaged in field reconstruction by this comparison, and it is worth saying why before the wake result is read. By field VRMSE the proper orthogonal basis is the most accurate of the ten families (0.930, against 0.962 for the predictive state), and its decoded similarity matches the nonlinear states (0.775 against 0.779 SSIM in the Wang convention of §3.5). VRMSE rewards an energy-optimal linear basis and penalises a nonlinear state that carries a high decode floor, so we route field-fidelity claims through SSIM and the decoded snapshots and reserve the wake-observable readability for the nonlinear-content claim. There the linear basis reaches only 0.186 at $d = 32$, and the POD coefficients do not recover the wake at any tested dimension.

[S4-06] The window-averaged decode floor can be resolved to the critical instants of the encounter, the pre-impact frame, the impact and the peak lift (table 3). Through the impact the compact nonlinear supervised states and the linear basis all render the field faithfully, at SSIM 0.81 for the predictive state, 0.81 for the matched-supervision reconstructive and 0.82 for the linear basis (near-body 0.63, 0.65 and 0.67), with only the published-recipe lineage a clear step behind (0.72 full, 0.45 near-body). Field decode at the critical instants is therefore not where the compact states separate: the energy-optimal linear basis reconstructs the instant as sharply as the nonlinear states, the same VRMSE advantage read above, and the families part company instead on the wake observables, which the POD coefficients do not reach, and on forecasting.

[S4-07] Across dimension the picture is the same (figure 26 in appendix A.2): decode fidelity at the impact instant rises and saturates for the compact nonlinear states and the energy-optimal linear basis together, the two indistinguishable at every dimension, while the published-recipe lineage stays lower.

[S4-08] The observables tell the discriminating story the field pixels hide (table 4). A linear probe reads the lift and the wake enstrophy off each state at the same three instants. Supervision decides what a state carries: the wake-supervised predictive state and the wake autoencoder carry both the lift and the wake at and beyond impact, whereas the lift-focused predictive state reads the lift as accurately as any state ($R^2 = 0.78$ at impact) but not the wake (0.05, against 0.56 for the wake-supervised predictive state), so the wake content is the wake head's contribution and not a by-product of the lift anchor. The linear basis fails differently again, losing the lift at the very impact instant ($R^2 = -0.05$, against 0.70 for the

Table 3. [TSSI-C1] Decode fidelity at the critical instants of the encounter, rather than averaged over the window: SSIM between each family’s decoded field and the DNS truth at the pre-impact frame, the impact frame and the peak-lift frame ($\arg\max |C_L|$), for the full field and the near-body band, averaged over the 42 in-distribution test encounters at $d = 32$. Decode floor: the true latent is decoded, with no forecast. Both nonlinear supervised predictive states, the wake-supervised predictive state and the lift-focused predictive state, together with the wake autoencoder and the energy-optimal linear basis, all render the field faithfully at every critical instant, including the near-body band; only the published-recipe lineage is a clear step behind. Field decode at the critical instants is thus not where the compact states separate, consistent with the linear basis’s field VRMSE advantage; they part on the wake observables (table 4) and on forecasting. The frozen decoders exist at a single encoder seed, so the decoded-field tables are single-seed.

State	full field			near-body band		
	pre-impact	impact	peak lift	pre-impact	impact	peak lift
predictive (wake-supervised predictive state)	0.83	0.81	0.82	0.72	0.63	0.65
predictive (lift-focused predictive state)	0.82	0.80	0.82	0.70	0.63	0.66
AE (wake)	0.82	0.81	0.81	0.73	0.65	0.64
Fukami (wake)	0.72	0.72	0.70	0.54	0.45	0.41
POD	0.82	0.82	0.81	0.74	0.67	0.67

Table 4. [TOBS-C1] Where the families separate at the critical instants: held-out readability of the lift C_L and the wake enstrophy E_w at the pre-impact, impact and peak-lift instants (± 2 frames), each cell the coefficient of determination R^2 and, in parentheses, the physical error (RMSE, in lift-coefficient units for C_L and in the pipeline’s enstrophy units for E_w , whose test-window standard deviation is 92). A linear probe fit on the training window reads each observable off the frozen latent; $d = 32$, on the 42 in-distribution test encounters. The learned predictive and AE states are means over three encoder seeds (encoder-seed standard deviation ≤ 0.09 in R^2 on every cell that reads the quantity); the published-recipe lineage has a single seed at this dimension and the linear basis is deterministic. Unlike the decoded field (table 3), the observables separate the states exactly where it matters. The wake-supervised predictive state and the wake autoencoder read both the lift and the wake enstrophy at and beyond impact (the autoencoder more weakly pre-impact, 0.39); the lift-focused predictive state reads the lift as accurately as any state ($R^2 = 0.78$ at impact) yet reads the wake enstrophy far worse (near zero at impact, 0.05), locating the wake content in the wake head rather than the lift anchor. The linear basis loses the lift at the impact instant ($R^2 = -0.05$) and carries the largest lift error at every instant, including where its R^2 looks healthy at the high-variance peak (0.74 but RMSE 1.80, the worst in the column), and it loses the wake enstrophy at and beyond impact (0.31 from 0.66 pre-impact). The energy-optimal linear basis renders the field but cannot read the load or the wake at the moments that decide the encounter.

State	lift C_L : R^2 (RMSE)			wake enstrophy E_w : R^2 (RMSE)		
	pre-impact	impact	peak lift	pre-impact	impact	peak lift
predictive (wake-supervised predictive state)	0.93 (1.00)	0.70 (0.84)	0.83 (1.44)	0.61 (59)	0.56 (59)	0.87 (46)
predictive (lift-focused predictive state)	0.93 (0.98)	0.78 (0.71)	0.84 (1.38)	-0.82 (127)	0.05 (85)	0.42 (96)
AE (wake)	0.89 (1.22)	0.76 (0.75)	0.81 (1.53)	0.39 (74)	0.70 (48)	0.83 (52)
Fukami (wake)	0.93 (0.98)	0.79 (0.69)	0.81 (1.55)	0.74 (48)	0.72 (48)	0.58 (81)
POD	0.77 (1.74)	-0.05 (1.57)	0.74 (1.80)	0.66 (56)	0.31 (74)	0.31 (105)

predictive state) and losing the wake enstrophy at and beyond impact (0.31 at the impact instant and at the load peak, down from 0.66 before impact). It is at the critical moments, not on the window average, that the linear readout breaks down: the energy-optimal basis renders the field but cannot read the load or the wake exactly when the encounter turns nonlinear.

4.2.3. The distributed wake code

[S4-09] The wake is a collective code rather than a labelled coordinate. The gap between the wake readability of the full predictive latent and that of its single most informative coordinate is 0.51, and 0.53 for the objective-free supervised state, against -0.03 and

TSSI-C1

KEEP

Decode fidelity at the critical instants rather than window-averaged. The exhibit that shows where the linear readout survives.

TOBS-C1

KEEP

Readability of lift and wake enstrophy at the three critical instants. The discriminating exhibit of §4.2.

S4-09

KEEP

The distributed wake code and the readability plateau at $d = 32$. Both are cited in §5 and §6.

0.04 for the published-recipe reconstruction and the linear basis, so no single predictive coordinate carries the wake and the information is spread across many. The dimension at which each family first reaches a wake R^2 of one half makes the same point from the other side: the predictive and the wake-headed reconstructive states need eight latent dimensions (8 each), while the linear basis never reaches it at any dimension we tested. The headline dimension of thirty-two sits on a plateau rather than a cliff, the wake readability moving only 0.053 across sixteen, thirty-two and sixty-four (0.751 at thirty-two), so the choice of dimension is a robustness statement and not a tuned operating point.

[S4-10] The decodability-versus-estimability trade this raises, and why the coefficient state rather than a field-resolved latent is the object estimated here, is taken up in §5.3.

4.3. The physics the latent state holds

[S4-11] Before the state is forecast or estimated, we read what the representation holds, in measures that do not depend on any filter. The predictive latent carries the shedding clock. Fitting a linear operator to the encoded limit-cycle trajectory by dynamic mode decomposition places a marginally stable oscillatory mode at Strouhal number 0.666, against the measured shedding line marked in figure 9(d), with modulus 0.993 just below unity. The Fukami-lineage autoencoders lose that clock: their leading latent mode is heavily damped and off-frequency (0.182 at modulus 0.590 without the wake head, 0.147 at 0.294 with it). The structured states keep it, whether their structure comes from prediction, from supervision, from anti-collapse, or from a linear basis: the objective-free supervised latent, the anti-collapse reconstruction, and the proper orthogonal basis all place the shedding mode near the measured frequency on a near-unit-modulus orbit (0.654, 0.676 and 0.673 at moduli 0.989, 0.995 and 0.996). It is the Fukami-lineage reconstruction, not reconstruction in general, that fails to hold the shedding spectrum.

[S4-12] The coordinate spectra say the same in the frequency domain and locate the reconstruction failure precisely. Averaging a per-coordinate Welch spectrum over the training encounters, the published-recipe reconstruction spreads its latent energy across frequency, at a median spectral flatness of 0.068 against 0.012 for the predictive latent, 0.024 for the matched-supervision reconstruction and 0.030 for the linear basis (figure 9f); the published recipe exceeds the predictive latent by 0.056 in flatness with a case-clustered interval [0.048, 0.065] that excludes zero. The predictive coordinates are the most sharply peaked at the shedding line, and it is the published-recipe reconstruction alone, not the matched-supervision autoencoder, whose coordinates are broadband: the broadband, phase-scrambling latent that a reconstruction objective without matched supervision produces is the frequency-domain signature of the rollout collapse the same latents show under the shared forecast operator. The same contrast appears in the phase portraits of the latent trajectories (figure 19): the predictive, linear and matched-supervision states trace coherent cycles that the gust perturbs, while the published-recipe reconstruction wanders without a clean orbit.

[S4-13] The parametric manifold coheres with the release parameters where the training data resolve them. An impact-frame probe of the predictive latent recovers the gust strength and the core diameter well (0.88 and 0.73 on the in-distribution test set), which is the atlas of figure 9 (top) coloured by parameter. The wall-normal offset, unreadable at the linear-probe level in earlier work, becomes weakly readable at the linear level here (0.29; 0.79 under a nonlinear probe), plausibly because the additional cases sample it more densely near the training centre; we note this rather than lean on it, since the offset remains the least resolved parameter (§2.2).

[S4-14] A state that reads the wake at a single instant need not survive a rollout, because a probe attached to a drifted latent is queried outside its support. We ask where each

S4-10

DELETE

A one-line forward pointer to §5.3 and nothing else. The reader reaches §5.3 without it.

S4-11

KEEP

The shedding clock read by dynamic mode decomposition, and the scoping to the Fukami lineage rather than reconstruction in general. A precision the paper needs.

S4-12

KEEP

The coordinate spectra that locate the reconstruction failure in the frequency domain. The flatness gate behind the broadband wording passed here.

S4-13

KEEP

The parametric manifold, honestly scoped on the weakly readable offset. §5.4 cites it as a limitation.

S4-14

KEEP

The near-null decomposition showing supervision, not anti-collapse, builds the protected subspace. The mechanism claim of the paper.

rolled state goes relative to the training distribution by decomposing its departure onto the principal and the near-null directions of the encoded covariance (table 14 and figure 11a). The reconstruction carrying only an anti-collapse term is near-isotropic, its covariance condition number (11) more than an order of magnitude below the supervised states', and its rollout leaks a tenth of its departure into the near-null directions (0.100). The supervised states, predictive and objective-free alike, are strongly anisotropic, their variance concentrated in a low-rank subspace (condition numbers 451 and 1041), and their rollouts keep their departures inside it (0.014 and 0.029, against the 0.25 of an isotropic null). Supervision, not the anti-collapse term alone, builds the protected subspace: it pins the encoded variance onto the observable-aligned directions, and the forward model, having only ever seen states there, moves within it. The anti-collapse term on its own drives the latent toward isotropy, as an independent adaptation of the same regulariser to biosignals also reports (Broustail *et al.* 2026), and it is the supervision that supplies the geometry the rollout needs.

4.4. Forecastability

[S4C-01] The family ordering survives an operator-robust protocol, which resolves the operator confound the suited-operator comparison of §3.2 left open: one direct multi-horizon forecaster (§3.2.1), trained identically on each family's frozen latents, is stable on every latent including the reconstructive ones. Under it the predictive states carry the more forecastable lift, decoded lift $R^2 = 0.638$ for the wake-supervised predictive state and 0.685 for the lift-focused predictive state against 0.539 for the reconstructive anchor, three operator-training seeds per family on the in-distribution test set, with the lift-focused predictive state latents the most lift-forecastable at the seed mean (figure 10a). On the five-observable merit of table 2, recomputed under the same shared operator across all ten families, the wake-headed states are statistically tied (§4.1), so the operator-robust statement is two-fold: supervision, not the training objective, decides which states are forecastable at all, and the lift-focused predictive state leads specifically on the lift readout.

[S4C-02] The forecasting mechanism is the direct-versus-autoregressive contrast. Rolled autoregressively through the impact, the matched transformer compounds its own errors: forty steps from the pre-impact context its decoded lift closes at -0.62 on the wake-supervised predictive state latents and -0.23 on the lift-focused predictive state latents. The direct multi-horizon forecaster, emitting all forty steps in one shot from the same context, closes at 0.701 on the identical protocol (figure 10b): the large improvement of the direct multi-horizon forecast indicates that recursive error accumulation is the dominant limitation of the autoregressive implementation over this horizon. The direct forecast is a conditional median, and the correct reading is that of a median: it contracts toward the climatology of the shedding cycle where the encounter is unresolved, which is precisely the model-error regime the assimilation of §4.6.3 must correct.

[S4C-03] Knowing the gust does not help. Conditioning the forecaster on the true episode parameters (G, D, Y), an oracle no deployment has, degrades the forecast in every one of three operator seeds, 0.539 against 0.699 unconditioned, with non-overlapping seed bands: at the present case count the oracle covariates overfit held-out parameter combinations. A deployable phase covariate is a wash within seed noise (0.710). This closes the estimation thesis from the third side: forecast alone contracts, parameter oracles hurt, and what remains is the model-plus-pressure route of part D.

[S4-15] The forecast is not uniformly hard across the encounter, and resolving it by phase locates where the families separate (figure 27). Splitting every shared-operator forecast by whether its horizon stays in the pre-impact shedding or crosses the impact transient, merit is highest before impact and drops through it for every family, the wake-supervised predictive

S4C-01

KEEP

The operator-robust protocol that removes the operator confound, and the lift-forecastability result scoped to the lift readout. Carefully reconciled with the five-observable tie and should stay that way.

S4C-02

KEEP

Direct against autoregressive: the mechanism behind the forecasting result. Also flags the median-contraction behaviour the filter must correct.

S4C-03

KEEP

The oracle-conditioning negative, which closes the deployment argument from the third side. A genuine negative result worth its space.

S4-15

ANNEX

A number-dense narration of a figure that already lives in appendix A. Paragraph and figure should sit together, wherever they end up.

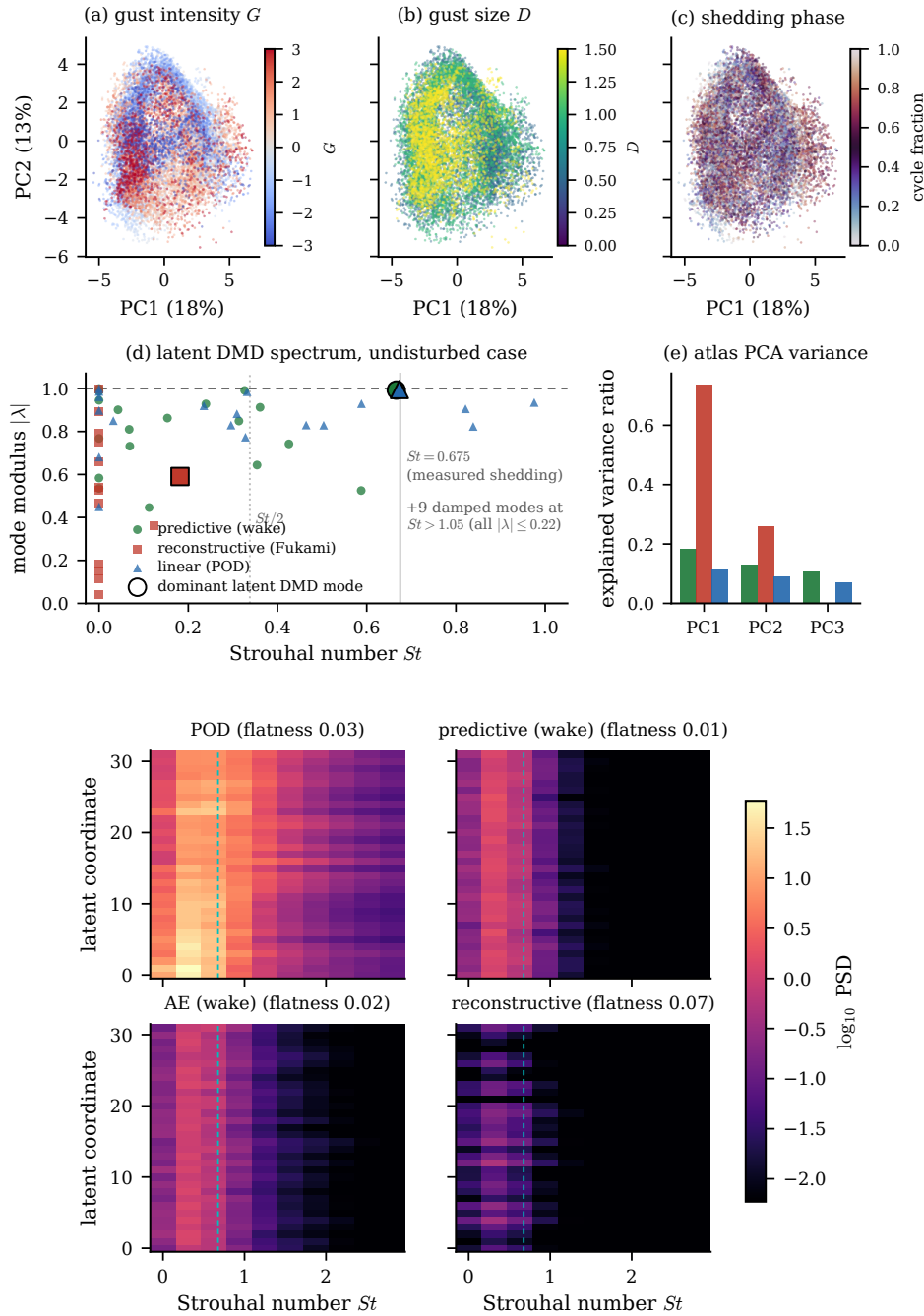


Figure 9. [S4-C1] The physics the latent state holds. (a-e) Two-dimensional projections of the predictive coefficient latent coloured by gust strength, core diameter and shedding phase, showing a parametric manifold that coheres with the release parameters, with the latent dynamic-mode-decomposition spectrum (modulus against Strouhal number, by family; the predictive and linear latents place a marginally stable mode at the measured shedding frequency while the Fukami-lineage reconstruction is damped and off-frequency), and the explained-variance ratio of the leading atlas principal components per family (e; colours as in d). (f) Per-coordinate latent spectra by family (training encounters, $d = 32$; mean Welch power spectral density against Strouhal number, coordinates ordered by peak frequency, dashed line the shedding Strouhal): the predictive coordinates concentrate at the shedding line while the published-recipe reconstruction spreads energy across frequency and the matched-supervision autoencoder does not.

S4-C1

KEEP

Covers the atlas, the DMD spectrum and the per-coordinate spectra in one caption. Panel (f) is lettered at float level.

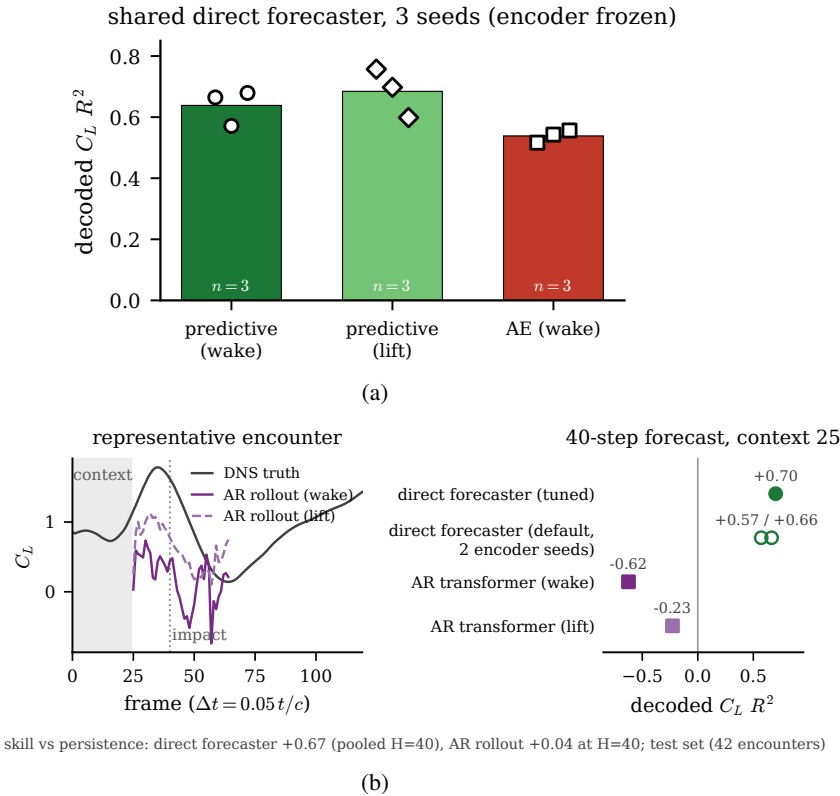


Figure 10. [S4C-C1] Forecasting the state on the in-distribution test set. (a) Decoded lift R^2 of the shared direct multi-horizon forecaster per family (three operator seeds each, the forecaster retrained on a frozen representation, bars at the seed mean): the predictive states forecast the lift best. (b) Direct versus autoregressive over forty steps through the impact from a 25-frame context: the one-shot forecast stays usable where the rollout compounds. The conditioning null, where the (G, D, Y) oracle degrades the forecast and the phase covariate is a wash, is figure 21 in appendix A.

S4C-C1

KEEP

Both forecast panels with the seed provenance corrected to operator seeds. Self-contained.

state from 0.78 pre-impact to 0.63 through impact at horizon eight. The separation is in the transient: through the impact the wake-supervised states carry the encounter (0.63 for the wake-supervised predictive state and 0.70 for the reconstructive AE-LW at horizon eight, 0.57 and 0.61 at horizon sixteen), while the published-recipe reconstruction collapses across it (near zero at horizon eight, -0.29 at horizon sixteen) and the linear basis holds the shedding but fades into the relaxation (through-impact 0.36 against post-impact -0.16 at horizon sixteen). The through-impact window, not the periodic shedding, is what a forecast model must survive, and it is where the anchored, wake-supervised states hold while the reconstruction fails (figure 27 in appendix A.2).

[S4-16] When the forecast is decoded to the field rather than read in the latent, the families no longer separate. Decoding each family's forecast through its frozen decoder and scoring SSIM against the DNS truth, all three states forecast the field to comparable fidelity through the impact at horizon eight, SSIM 0.78 for the predictive state, 0.76 for the reconstructive and 0.76 for the linear, the predictive state only modestly ahead on the near-body band (0.54 against 0.52 and 0.49). The decoded field, like the decode floor of §4.2, is not where the states separate. What separates them is the observable content the field-averaged SSIM washes out, the wake-observable merit above, where the wake-supervised and predictive states lead and the linear basis does not.

S4-16

TRIM?

The fourth statement that field decode does not separate the families, this time for the forecast. One sentence in S4-05 would carry it.

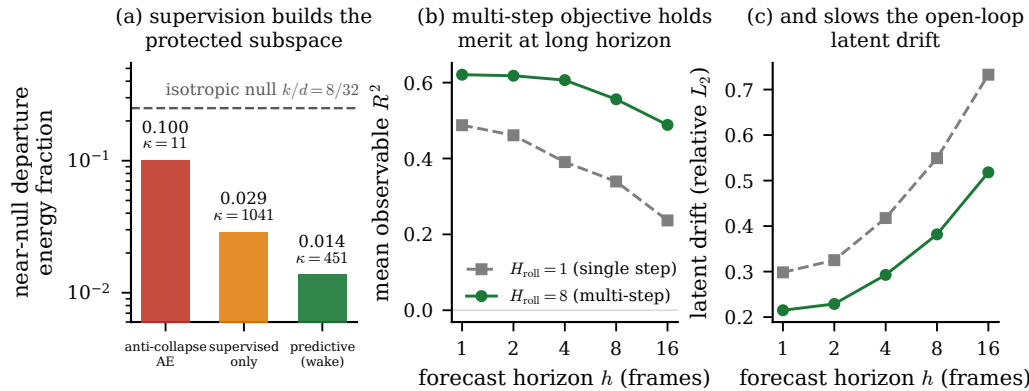


Figure 11. [S4-C2] The rollout mechanism and what the multi-step objective adds. (a) Near-null departure fraction by family against the isotropic null, annotated with each latent covariance condition number: supervision builds the protected subspace. (b) Forecast merit and (c) rollout drift against horizon for the single-step and multi-step objectives, tied at one step and separated by sixteen.

S4-C2

KEEP

The rollout mechanism figure: protected subspace, merit and drift against horizon. Self-contained.

S4-17

KEEP

What the multi-step objective adds: long-horizon stability and lower drift. This is the only place the predictive objective earns its own credit.

S4-18

KEEP

The physics-versus-variance ordering, the wall observability result. §5.1 and §6 both rest on it.

4.4.1. Why the state stays usable under rollout

[S4-17] What the predictive objective adds is long-horizon stability, and it is the multi-step form of the objective that adds it. Trained with a single-step rather than a multi-step rollout, the same predictive latent forecasts the five observables less well at every horizon, and the margin widens with the rollout (figure 11b,c): by sixteen steps the single-step latent's merit has fallen to 0.237 while the multi-step latent holds 0.488, with markedly lower on-manifold drift (0.518 against 0.733 at horizon sixteen). The single-step latent is marginally better on the load alone a few steps out (0.685 against 0.664 in C_L closure at eight steps), a single-observable exception to a forecast advantage that otherwise grows with horizon. The objective-free supervised state, which has the readability and the protected subspace but no predictive training, is the weakest field forecaster of the supervised group. The contribution of the predictive objective is therefore forward rather than instantaneous: it keeps the rolled state both in distribution and near the true trajectory, which is exactly what a sequential estimator queries.

4.5. What wall pressure observes

[S4-18] At deployment the state must be read from the wall, so we ask how much of each latent a sparse set of pressure sensors recovers over a window ending at the readout instant (table 5). Two findings organise the answer. First, raw-variance recoverability and physics recoverability disagree: the reconstruction latent is the most recoverable in state variance (0.921) yet carries the least wall-recoverable physics (0.470 in observable closure), while the predictive latent carries the most physics (0.589) from a less recoverable state (0.659 ± 0.009 over three seeds). The wall aligns with many of the reconstruction latent's directions, but they are the low-variance ones that do not carry the observables; it aligns with fewer of the predictive latent's directions, but those are the ones that hold the state. Second, perturbing the predictive state along its principal directions and measuring the signal that reaches the pressure head shows the wall reading the force and wake-entropy directions and blind to the highest-variance wake-circulation direction, a ranking unchanged under a fixed-amplitude perturbation and so a property of the learned dynamics rather than of the normalisation. The wall, in short, reads the force well and the negative-signed wake circulation not at all, which is why the state must be estimated sequentially rather than inverted from one frame.

S4-19**KEEP**

Concedes that the margin over a field latent is modest and declines the strong-form claim. Exactly the honesty a referee rewards.

S4-20**KEEP**

The delay window recovers the coordinate the instantaneous pressure misses. Confirms the delay-embedding reading empirically.

S4-21**KEEP**

The tap-count against window-length trade, which is the deployment knob §5.2 names. Carries figure 10.

S4-22**TRIM?**

Reconciliation arithmetic plus a single-tap filter control, and the reconciliation number is quoted a third time in APB-02. The filter-cannot-spare-taps finding is the part worth keeping.

S4-23**KEEP**

Runs the estimator and reads the hero figure. Also states the control-literature pathway this closes.

[S4-19] The margin of the coefficient state over a raw high-dimensional field latent is real but modest. Its recoverability advantage is 0.120 in R^2 , with a confidence interval [0.096, 0.145] that clears zero. We therefore do not claim a strong-form estimability gate; the defensible statement is the clean physics-versus-variance ordering above.

4.5.1. Sensors traded for delays

[S4-20] That the wall is blind to the negative-signed wake circulation at a single instant is a statement about one frame, not about the sensor count, and the delay-embedding reading of §3.3 makes a testable prediction: the circulation the instantaneous pressure misses should become recoverable as the pressure is read over a longer window. It does. Holding the sensor count at eight and lengthening the window, the recovery of the wall-blind wake circulation rises from near its single-frame value to 0.33 at a thirty-frame window, a paired case-mean gain of 0.224 with a case-clustered interval [0.136, 0.343] that excludes zero, while the force, already visible at one instant, barely moves. The delays recover the coordinate the sensors cannot.

[S4-21] Varying the sensor count and the window together traces a spatial-for-temporal trade (figure 12). A single sensor read over a thirty-frame window approaches the recovery obtained from eight sensors read at one instant, and intermediate combinations of sensor count and window meet in between, because the encounter is low-dimensional and a state that filled its ambient dimension would not admit the trade. The eight-sensor, thirty-frame recovery reconciles closely with the static recovery above (0.656), the delay stride sits at the cache cadence, and the thirty-frame window spans the first mutual-information minimum of the sensor signals at 29 frames.

[S4-22] The grid uses a shared target-blind sensor array so that the trade is free of any model-conditioned placement; rerunning its eight-sensor, thirty-frame cell on the model-conditioned sensors of table 5 closely reproduces that table's recovery (0.656 against 0.659), so the two analyses meet where their sensing coincides. This trade belongs to the static delay-coordinate reconstruction, and it does not transfer to the frozen sequential filter, which needs its eight sensors. Rerunning the filter with only its own leading sensors, and changing nothing else, drives the analysis lift below the skill of the encounter mean at every gust strength the eight-sensor filter tracks: at $|G| = 1$ the single-sensor analysis is worse than the eight-sensor filter by a paired margin whose interval $[-107.96, -4.78]$ excludes zero, and the loss stays significant at $|G| = 1.5$ and $|G| = 2$. The rank of the innovation, with the process and observation noise held at their calibrated values, is what the filter cannot spare: its sensor budget is a calibration constraint, not a delay-embedding one, which points at the noise model of §5.4 rather than at fewer sensors.

4.6. Estimating the encounter from the wall

4.6.1. Tracking the encounter from the wall

[S4-23] We now run the estimator. It is initialised from wall pressure alone before impact and corrected at every frame through the encounter, and it never sees the vorticity field. Figure 13 follows representative held-out encounters, the true lift and wake enstrophy against the open-loop rollout with no correction, the sliding-window pressure-only regression with no dynamics, and the filter analysis. The open-loop rollout tracks the pre-impact baseline and then loses the encounter, as a forecast with no measurement must; the pressure-only regression follows the force while its window is representative and goes stale through the transient; the filter follows the load through impact and into relaxation. This realises, on the wake-supervised coefficient state, the online-estimation pathway the extreme-aerodynamic control literature has named but not closed (Fukami *et al.* 2024; Mousavi & Eldredge 2025;

Table 5. [TREC-C1] What eight wall-pressure sensors recover of each family’s coefficient state, on the in-distribution test set, with the recovery estimator selected by cross-validation per family and the observables read through the frozen probes. Raw-variance recoverability and physics recoverability disagree: the reconstruction latent is the most recoverable in state variance but the least in physics, while the predictive latent carries the most wall-recoverable physics from a less recoverable state. The absolute state recovery is estimator-limited: a per-family tuned LSTM (hidden width validation-selected up to 512) recovers more from the same sensors, raising the predictive state to 0.833 and the reconstruction to 0.937, and it had not saturated at the largest network tested, so these are a lower bound on the wall-recoverable state; the family ordering, which is the claim, is unchanged.

State	recovery estimator	state R^2	observable R^2
predictive	LSTM	0.659	0.589
reconstructive (Fukami)	LSTM	0.921	0.470
linear (POD)	MLP	0.561	0.534

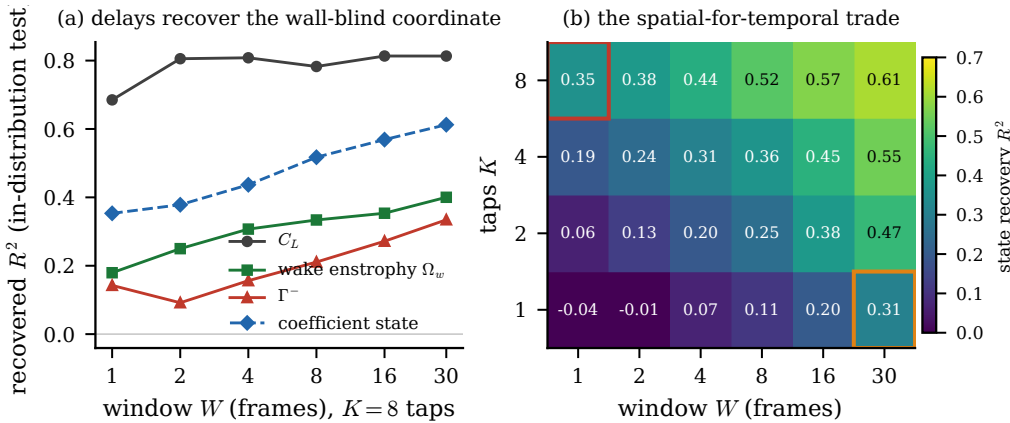


Figure 12. [S4-C3] Sensors traded for delays. (a) Recovery of each observable from eight wall-pressure sensors against the length of the pressure window (in-distribution test set); the wall-blind wake circulation rises with the window while the force, visible at one instant, does not. (b) Coefficient-state recovery over the sensor-count by window-length grid, with the single-sensor, thirty-frame cell and the reference eight-sensor, single-frame cell outlined: a long window with few sensors matches a short window with many.

TREC-C1 **KEEP**
What eight taps recover of each family’s state, with the per-family estimator selection disclosed. The exhibit behind the physics-against-variance ordering.

S4-C3 **KEEP**
Defines both panels of the trade figure and states which placement each uses. Self-contained.

Eldredge & Mousavi 2025): the state a real-time controller would need is recovered from sparse wall pressure and advanced by the learned dynamics, without ever measuring the field.

[S4-24] Two boundaries frame what this demonstration does and does not establish, and we state them before the envelope. First, tracking the load inside the training range is not unique to this state. Running the identical frozen filter on every family (table 6), the nonlinear latents, the wake-headed published-recipe variant excepted, track the lift for $|G|$ between one and two, and the wake-supervised reconstructive control is the best of them at the mildest gusts (0.86 at $|G| = 1$ against 0.74 for the predictive state); only the linear basis fails outright. What no static approach achieves is the strong-gust range: a windowed regression from the same sensors straight to the lift, with no latent at all, fades to 0.08 by a gust ratio of three, and even the full wall array recovers only 0.14 at the boundary, so the sequential accumulation of pressure through a learned nonlinear state, rather than any particular training objective, is what carries the load into the strong-gust regime. Where the objectives separate is at that boundary and in the filter’s statistical health. The predictive state combines a boundary closure among the highest (0.84, against 0.66 to 0.85 for the

S4-24 **KEEP**
The two boundaries: in-range tracking is not unique to this state, and no filter tracks the wake. Scoping without which the estimation claim would be oversold.

Table 6. [TFAM-C1] The frozen filter run identically on every family: median analysis lift closure by gust strength, with the consistency-failure rate (§3.3.2) at the $|G| = 4$ boundary, and the direct no-latent baseline (windowed regression from the same eight sensors straight to the lift) beneath. Load-tracking inside the training range is a property of the nonlinear latents generally, not of any one objective; the families separate at the strongest gusts, in calibration, and in what else the tracked state carries (§4.6.1). The linear basis and the no-latent regression both lose most of their skill beyond $|G| = 2$.

state	$ G = 1$	1.5	2	3	4	cons. fail. at 4
predictive (wake)	0.74	0.83	0.73	0.63	0.84	0.72
AE (wake)	0.86	0.86	0.81	0.49	0.66	0.93
Fukami	0.79	0.87	0.78	0.68	0.76	0.97
Fukami (wake)	−0.39	0.72	0.68	0.77	0.85	0.95
POD	0.53	0.52	−0.33	−0.27	0.25	0.93
direct regression (no latent)	0.41	0.29	0.21	0.08	0.08	−

alternatives), by far the lowest consistency-failure rate there (0.72 against 0.93 to 0.97), and consistency across gust strengths where the wake-headed reconstructive baseline posts −0.39 at $|G| = 1$. Second, no family’s filter tracks the wake enstrophy: the analysis wake closure is below the mean predictor for every state, so the wake content of the encounter is established at the representational tier (§4.1.2) and is not yet recoverable through the assimilation. The filter is a load estimator whose state carries wake information; it is not a wake estimator, and we say so plainly.

4.6.2. The gust-intensity operating envelope

[S4-25] This subsection diagnoses the reach of the base frozen filter across the envelope; the production estimator whose performance we report is the two-stage filter of §4.6.3, and the two are kept distinct throughout. A reduced state earns its place at deployment only if it can be followed from the sensors that are present, and the wall pressure a small vehicle can carry does not, at a single instant, fix the wake it leaves behind (§4.5). The question is therefore not whether the state is readable but how far into the gust an estimator that sees only the wall can track it. The sequential filter reaches much further than the single-frame inverse. While the encounter stays mild, the static single-frame recovery of the lift holds, with an R^2 of 0.73 at $|G| = 1$ and 0.65 at $|G| = 1.5$ pooled across the strata of table 7; as the gust strengthens it collapses, falling below the skill of predicting the encounter mean by a gust ratio of three (−0.46) and remaining there at the $|G| = 4$ boundary (−0.33), because a single pressure frame cannot see the reorganised wake a strong gust deposits. Reading the same sensors through the learned dynamics, the filter follows the load further, but the accuracy that matters at deployment is the error in real lift units, not the variance-normalised score, and that error grows with the gust. The median analysis C_L error rises from 0.31 at $|G| = 1$ to 0.68 at a ratio of three and 0.72 at the boundary (table 7), even as its R^2 rises to 0.84: the strong gust inflates the lift excursion faster than the error grows, so the normalised score flatters the extreme-gust tracking while the absolute error roughly doubles. The open-loop forecast never tracks the load at any gust strength, so it is the correction step, not the forward model, that recovers the state, and it does so from pressure the visibility analysis found blind to the wake circulation.

[S4-26] Figure 13b shows the two-stage filter of §3.3.3 tracking one representative encounter per gust intensity, including the $|G| = 4$ boundary, where the same models run open loop lose the transient.

[S4-27] Run identically on each family, each under its own validation-tuned inflation, the comparison is a division of degradation rather than a clean win (table 8). At moderate

TFAM-C1 **KEEP**
The frozen filter run identically on every family. The exhibit behind the in-range tracking is not unique claim.

S4-25 **KEEP**
Separates the base diagnostic filter from the production estimator, and gives the R^2 -flatters-the-boundary warning the abstract now respects. The most important honesty paragraph in §4.

S4-26 **TRIM?**
A one-sentence pointer to a panel whose caption already says the same thing. Nothing depends on it.

S4-27 **KEEP**
The per-family filter comparison at the boundary, framed as a division of degradation rather than a win. Carries table 12.

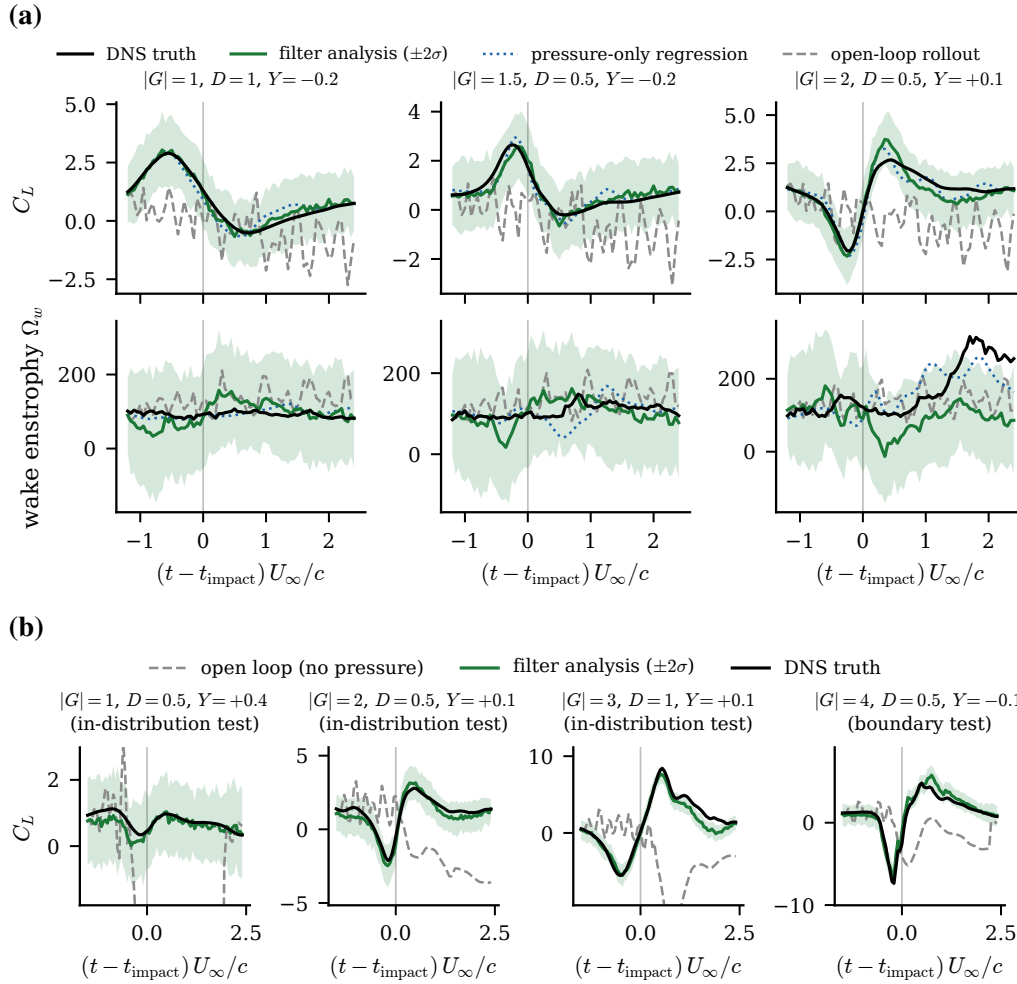


Figure 13. [S4-C4] The estimator tracks the encounter from wall pressure alone. (a) Representative held-out encounters at $|G| = 1, 1.5$ and 2 (columns), lift coefficient and wake enstrophy (rows); each panel shows the DNS truth, the filter analysis with its $\pm 2\sigma$ ensemble band, the pressure-only regression, and the open-loop rollout, the vertical line marking impact. The filter follows the load through impact and relaxation where the open-loop forecast loses it and the pressure-only regression goes stale. (b) C_L estimation across the gust-intensity envelope on the wake-supervised predictive state ($d = 32$) under the two-stage filter of §3.3.3 (one representative encounter per $|G|$ stratum, in-distribution test set for $|G| \leq 3$ and boundary test set at $|G| = 4$, where this filter posts no large-error encounter); inside the impact window the members are advanced by the direct multi-horizon forecaster’s median with its quantile band as state-dependent process noise (production scale $c = 1.77$, validation-calibrated) and corrected by the encoded pressure observation, outside it by the matched transformer with the sensor-space update. Eight wall-pressure sensors, $N = 64$ members; the lift probe producing the read-out never enters the innovation; grey is the same forecast run open loop.

S4-C4

KEEP

Covers both the hero traces and the envelope traces, with the production filter named. Panels lettered at float level.

gusts every filter tracks the load, the reconstruction filter marginally best. At the $|G| = 4$ boundary every filter is stressed, its innovations flagged inconsistent on 0.72 to 0.95 of encounters; the predictive filter degrades least, with the lowest median error (0.72 against 0.79 for the reconstruction and 1.33 for the linear basis) and the lowest consistency-failure rate, and the linear filter degrades worst. The predictive state’s advantage at the boundary is one of robustness, a degraded regime rather than a solved one.

Table 7. [TENV-C1] The operating envelope, by gust strength, for the predictive coefficient state under the base frozen filter, used here as a diagnostic. Median analysis lift closure, median analysis lift error in lift units, and consistency-failure rate of the filter (the fixed innovation-consistency criterion of §3.3.2, not a load-tracking failure) against the static single-frame recovery, over all encounters (train, validation, in-distribution test, and the $|G| = 4$ boundary test sets), stratified by $|G|$. The $|G| \leq 0.5$ closure is low because the true lift is near-constant there and the R^2 denominator vanishes, not because the filter fails; the per-encounter paired tests on the in-distribution test and boundary test sets are in Appendix A.

$ G $	filter $C_L R^2$	filter C_L RMSE	consistency-failure rate	static recovery $C_L R^2$
≤ 0.5	0.25	0.23	0.00	0.34
1	0.74	0.31	0.00	0.73
1.5	0.83	0.38	0.09	0.65
2	0.73	0.63	0.33	0.24
3	0.63	0.68	0.68	−0.46
4	0.84	0.72	0.72	−0.33

Table 8. [TFIL-C1] What the wall-pressure filter costs in real lift units. Median analysis C_L error (RMSE, dimensionless) and filter consistency-failure rate at the impact window, by gust strength, each family run with its own validation-selected inflation ($\rho = 1.00$ for the predictive filter, 1.05 for Fukami, 1.00 for POD). The consistency-failure rate is the fraction of encounters flagged by the fixed innovation-consistency criterion of §3.3.2 (sustained normalised innovation squared or an analysis Mahalanobis ratio above the threshold); the ensemble mean can still track the load where the consistency diagnostic fails. The variance-normalised R^2 rises with gust amplitude and hides the cost: at $|G| = 4$ the predictive filter scores $R^2 = 0.84$ while its absolute error is 0.72. Two readings follow. At moderate gusts every filter tracks the load, the reconstruction filter marginally best. At the strongest gusts every filter degrades, with consistency-failure rates of 0.72–0.95; the predictive filter degrades least, in both error and consistency, and the linear filter worst. The strongest gusts remain a degraded regime for every filter, not a solved one.

Filter	median C_L RMSE			consistency-failure rate		
	$ G =1$	$ G =2$	$ G =4$	$ G =1$	$ G =2$	$ G =4$
predictive	0.31	0.63	0.72	0.00	0.33	0.72
reconstructive (Fukami)	0.26	0.61	0.79	0.02	0.53	0.95
linear (POD)	0.36	1.14	1.33	0.01	0.49	0.93

[S4-28] What weakens with gust strength is not the tracking, whose median holds through $|G| = 4$ (table 7), but the filter’s own account of its uncertainty, which degrades from a gust ratio of about three. The consistency-failure rate climbs from none at $|G| = 1$ to a fraction 0.72 of encounters at the boundary, and the mean normalised innovation squared from 0.6 in the undisturbed limit to 15.2 at $|G| = 4$, crossing the half-consistency-failure mark at a gust ratio of 3.0 for the intermediate and the widest cores alike, at the resolution of the tested $|G|$ grid; the widest core already reaches a rate of one half at $|G| = 2$, and the most compact does not cross the mark inside the envelope. The ensemble collapses while the analysis mean still follows the load, so the filter is under-dispersed rather than mistracking. We read this as the signature of a missing noise model: the innovations are correlated in time rather than white, and the remedy is a calibrated process and observation covariance rather than the covariance inflation we deliberately withheld (§5.4).

[S4-29] The envelope is not symmetric in the sign of the gust (figure 14(b)). The physically positive gusts, which roll up the larger and longer-lived leading-edge vortex (§4.3), leave the wider usable envelope. Sampling the $|G| = 4$ boundary at both signs (§2.2) is what lets this asymmetry be measured rather than assumed.

TENV-C1

KEEP

The operating envelope in both R^2 and lift units, under the diagnostic filter as the caption states. The RMSE column is what keeps the boundary claim honest.

TFIL-C1

KEEP

What the filter costs in real lift units, per family and per gust strength. Decision D252’s exhibit.

S4-28

KEEP

Where the filter degrades: under-dispersion from a gust ratio of about three, with the innovation statistic. The missing-noise-model diagnosis follows from it.

S4-29

KEEP

The sign asymmetry of the envelope, measurable only because the boundary was sampled at both signs. Pays off S2-05.

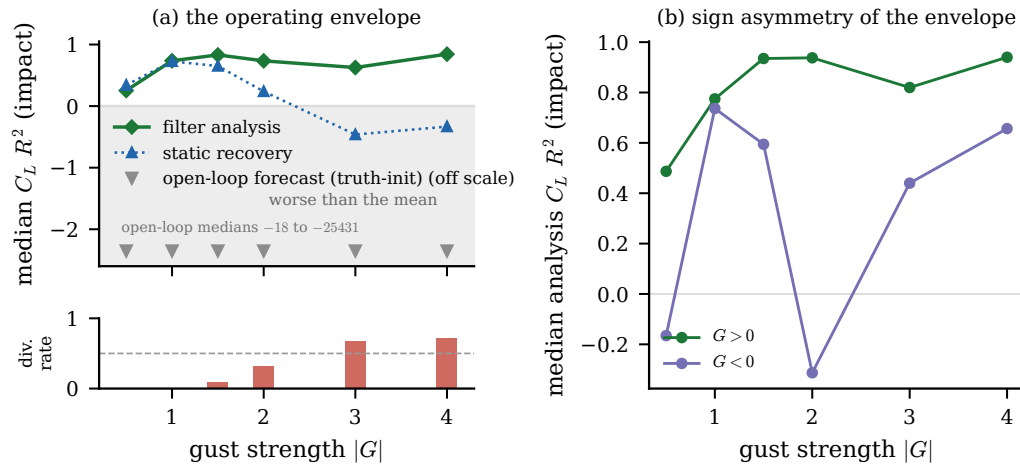


Figure 14. [S4-C5] The gust-intensity operating envelope, diagnosed with the base frozen filter (figure 15 carries the forecast-noise filter’s phase-resolved errors). (a) Median analysis lift closure (R^2 , impact window) against gust strength for the base filter, the static single-frame recovery, and the truth-initialised open-loop forecast, with the filter consistency-failure rate as the strip beneath; the shaded band marks closure worse than predicting the encounter mean, and the open-loop forecast lies off the scale below it. The base filter tracks the median throughout but becomes under-dispersed from a gust ratio of about three, where the consistency-failure rate climbs. (b) The same base-filter closure resolved by the physical sign of the gust, showing the wider usable envelope of the positive gusts.

S4-C5

KEEP

Names the diagnostic filter explicitly and points to the production tracking figure. The disambiguation lives in the caption as well as the text.

S4-30

KEEP

Places the limit at $|G| = 3$ and splits the envelope into two claims with different owners. The paragraph that keeps the estimation result correctly attributed.

[S4-30] The boundary sits at a gust ratio of three, inside the training range, so it marks a limit of what the mid-plane pressure can observe and the filter assimilate, not a gap in training coverage. Read through delay coordinates the two limits are one. The filter follows the load where the single frame fails because it reconstructs the state from a window of pressure rather than an instant (§3.3), and it loses its calibration where the strengthening gust drives the interaction three-dimensional and the mid-plane pressure history captures a smaller fraction of the state (§4.5). Only at $|G| = 4$ does the mid-plane representation meet the substantially larger three-dimensional content the field itself carries (§2.1), now seen from the filter. The envelope is therefore two claims with different owners: its extension beyond every static delay-embedded inverse belongs to the sequential estimator on a nonlinear latent, a property the family comparison of §4.6.1 shows is shared; its width at the strongest gusts, its calibration margin, and the wake content of the state being tracked belong to the wake-supervised predictive state. This envelope, not the single-instant readability, is what a wall-estimable reduced state is for.

4.6.3. The estimator ladder, its calibration and its deployment envelope

[S4D-01] Physical units reorganise the phase story. Read in R^2 the relaxation phase looks like a failure; read in lift units it is the best phase the estimator has, because the relaxation variance is small and the ratio, not the error, collapses (figure 15, which carries the per-phase errors). Assimilation is decisive at impact: the filter cuts the open-loop drift about five-fold, to 0.30 in lift units, with the median peak error at 11.7 per cent of the true peak and peak timing within one frame. Relative accuracy is approximately scale-invariant across the training envelope, near 12 to 14 per cent for the forecast-derived filter at all but the $|G| = 1$ stratum, which sits somewhat higher, while the linear filter is non-uniform (figure 30); the absolute error grows with the peak it tracks, which is what a scale-invariant estimator must do.

S4D-01

KEEP

Physical units reorganise the phase story and expose the relaxation R^2 as a variance artefact. Decision D252 in one paragraph.

S4D-02 **KEEP**

The estimator ladder with the static inverse on top, the qualifier that scopes the entire estimation result. The abstract depends on this paragraph.

S4D-03 **ANNEX**

The full calibration narrative, including the excluded test-selected variant, told at length in the body. Appendix C already exists for exactly this audit trail; the body needs the one-sentence outcome.

S4D-04 **KEEP**

The smoother result and the deployment rule it changes. Short, concrete and cited in appendix B guidance.

S4D-05 **KEEP**

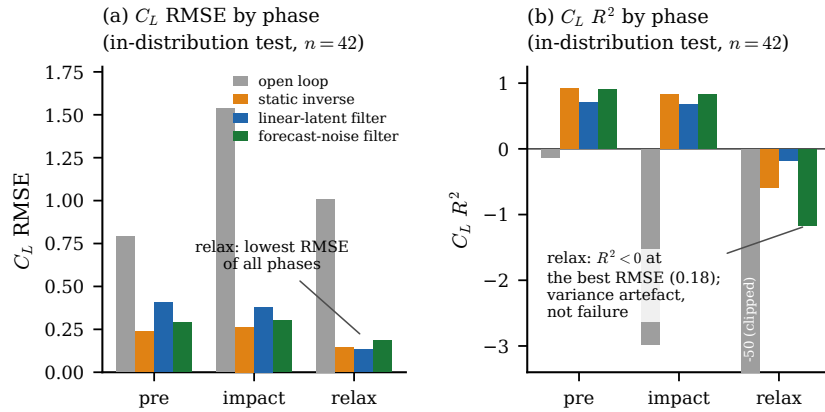
The per-family end-to-end comparison, which is the fair version of the estimator claim. Also the sensor-budget and noise-robustness separation.

1170 [S4D-02] The estimator ladder orders the suite at impact (appendix C, figure 32), and its
 1171 top rung is unexpected: the static delay-embedded inverse, no dynamics at all, matches
 1172 the best in-distribution test impact medians on the ladder (0.83; figure 32a). The filters'
 1173 case does not rest on the in-distribution median. It rests on the tails, where the static delay-
 1174 embedded inverse posts a large-error encounter, its impact-phase R^2 falling below -1 ;
 1175 on the extrapolation boundary, where the two-stage filter reaches 0.837 median R^2 on the
 1176 $|G| = 4$ boundary test encounters with no large-error case and a positive relaxation (0.120);
 1177 and on state tracking beyond the single read-out the inverse regresses. The linear filter with
 1178 encoded observations adds the robustness column: zero consistency failures under the fixed
 1179 criterion of §3.3.2 and graceful degradation as the assimilation rate drops, where the base
 1180 filter collapses already at every-second-frame updates (supplementary material).

1181 [S4D-03] Calibrating the forecast-derived noise did not go the way the calibration hypothesis
 1182 expected, and the failure is informative. On the held-out training encounters the pooled
 1183 impact-phase innovation statistic sits below one at every band scale in the grid fixed in
 1184 advance (0.53 at the selected edge $c^* = 1.00$) while the same split's tracking improves
 1185 monotonically with the band (figure 32): innovation consistency and tracking optimality
 1186 point in opposite directions, so the frozen NIS-selected run, the single-stage forecast-
 1187 noise filter at the NIS-selected band c^* , closes at 0.638 at impact on the in-distribution
 1188 test set, below the production two-stage rung, and the consistency route to a larger band
 1189 is not supported by the held-out training encounters. The mechanism is model error,
 1190 not miscalibration: inflating the state-dependent noise compensates the forecast median's
 1191 contraction, buying gain at the price of consistency, and the test-selected variant excluded
 1192 by the freeze rule is disclosed, with its number, only in appendix C. The five-seed member-
 1193 noise mean of the forecast-noise filter at this band is 0.764 at impact. The legitimate gain
 1194 comes from structure instead of noise: the two-stage filter of §3.3.3, at the production
 1195 coverage band (a single validation-calibrated noise scale, no test-set tuning), reaches 0.794
 1196 at impact on the in-distribution test set (root-mean-square error 0.286) and carries the
 1197 boundary test set as above, the best filter in the ladder among those selected without
 1198 reference to the test set.

1199 [S4D-04] When a quarter-convective-time latency is acceptable, the smoother changes the
 1200 deployment rule. The fixed-lag recursion of the fixed-lag smoother of appendix C on
 1201 the linear stack cuts the impact error to 0.286 from the filter's 0.375 and leaves the
 1202 relaxation essentially unchanged (0.149 against 0.141), a full-interval reanalysis at lag five;
 1203 the ensemble smoother on the forecast-noise filter degrades it (supplementary material).
 1204 Online estimation therefore takes the nonlinear filter and a $0.25 t/c$ latency budget takes
 1205 the closed-form linear smoother.

1206 [S4D-05] What the training objective confers survives to the assimilation stage, and per-
 1207 family end-to-end evaluation makes the comparison fair: each family senses at its own
 1208 sensors, encodes its own observations, forecasts with its own operator and decodes
 1209 through its own decoder. The wake-supervised predictive state cuts the reconstructive
 1210 anchor's assimilated load error by more than a third at impact and by more than half in
 1211 relaxation (impact 0.26 against 0.41, relaxation 0.14 against 0.37; the lift-focused predictive
 1212 state between at 0.31), while decoded-field fidelity is family-insensitive (figure 31a). The
 1213 sensor axis separates the families further: the predictive state exploits every added sensor
 1214 monotonically where the reconstructive one saturates at $K = 8$, and it degrades more
 1215 slowly with sensor noise, still beating the clean reconstructive stack at twenty per cent
 1216 noise (figure 31b,c). Streaming multi-encounter assimilation under a deployment-realism
 1217 protocol, and a filter on the four-dimensional lift-critical state of §4.2.1, are reported in
 1218 appendix B.



$K = 8$ wall-pressure taps, delay-10, $N = 64$ members; phase evaluation at the val-calibrated band $c = 1.77$.

Figure 15. [S4D-C1] Phase-resolved lift tracking with the forecast-noise filter at the production coverage band on the in-distribution test set ($n = 42$ encounters, $K = 8$ sensors, delay 10, $N = 64$ members, every-frame pressure). Lift error per phase in physical units (a) and in R^2 (b) at the val-calibrated band: the relaxation R^2 collapse is a variance artefact, the physical error the honest phase metric. The estimator ladder, its noise calibration and the fixed-lag smoother are in appendix C (figure 32).

S4D-C1

KEEP

The phase-resolved tracking figure with the production band stated. Self-contained.

S4D-06

KEEP

The dimension-by-family grid and the reconstructive seed fragility, with the observation-side mechanism verified. This is the deployment-robustness argument of the paper.

S4D-07

TRIM?

Two loose ends that restate §4.6.2's under-dispersion and §4.5's sensing geometry. Its consistency-failures-removed claim is also the one number in the section with no bound macro.

[S4D-06] The closing study runs the whole design axis through one identical per-family end-to-end protocol: three families at four dimensions (figure 16). The predictive family is uniform, impact error between 0.265 and 0.359 across $d \in \{4, 8, 16, 32\}$ with peak errors of ten to fifteen per cent everywhere; even its $d = 4$ cell (0.298) beats the linear floor's best (0.346 at $d = 32$). The linear basis is stable and a step worse everywhere. The reconstructive baseline is non-monotonic in dimension and irreproducible across seeds: it fails outright at $d = 4$ and $d = 32$ (1.620 and 2.247, peak errors above one hundred per cent, assimilation at $d = 32$ making the open loop worse), and its $d = 16$ cell, the best single cell in the grid at 0.180, does not survive seed replication: across three encoder seeds the same cell spans 2.252 impact error on average with individual seeds at 0.18, 0.65 and 5.9, the worst of them failing exactly as the neighbouring dimensions do. The mechanism is verified and lives on the observation side: linear probes on the true reconstructive latents are healthy at every dimension and every seed, while the pressure-to-latent map fails to recover the lift-carrying directions, insensitive to its regularisation, under a protocol that succeeds for the other two families in every cell. Wall-pressure estimatability of the reconstructive geometry is unpredictable across both dimension and training seed; the anchored families are uniformly estimatable, which is the deployment-robustness argument of this paper.

[S4D-07] Two loose ends close the section. The under-dispersion of the base filter has two regimes, not one: at impact it was calibration, and the state-dependent noise removes the consistency failures entirely; in relaxation it is model error, the forecast median's contraction, which no noise model fixes and the smoother does. And the wake-ensrophy null of the frozen filter, no family's filter tracks E_w , now has its mechanism: the wall constrains the lift-carrying near-body directions, the ones the Chang decomposition of §3.1 weights, while the wake ensrophy lives in shed structure the sensors reach only through the delay window, so the null is the sensing geometry, not the state; the delay-for-sensors trade of §4.5 bounds what any wall-limited estimator of this flow can see.

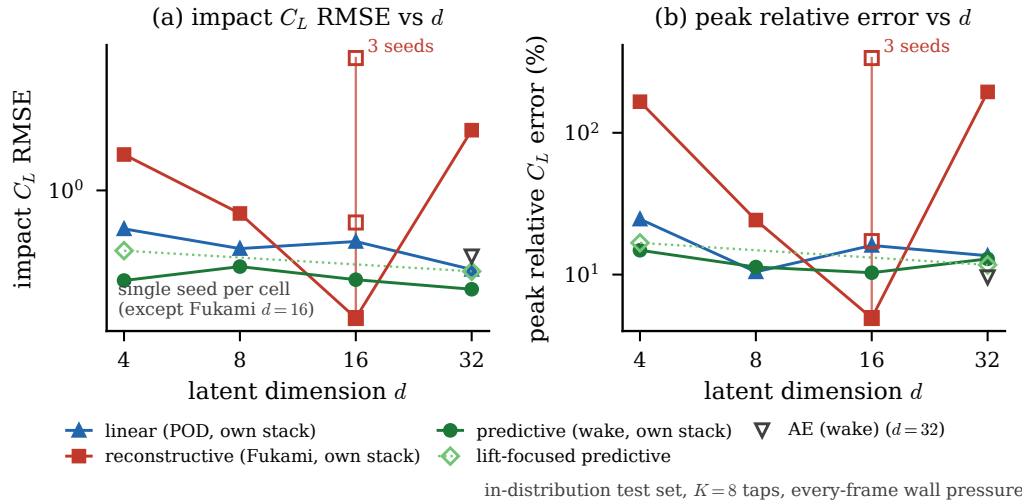


Figure 16. [S4D-C2] The assimilation-versus-dimension grid (in-distribution test set, per-family end-to-end stacks, $K=8$, every-frame pressure; single seed per cell except the reconstructive $d=16$ cell, which carries three): impact error and relative peak error against dimension for the three families. The predictive family is uniform, the linear basis stable, the reconstructive baseline non-monotonic in dimension and irreproducible across seeds.

S4D-C2

KEEP

The dimension grid, with the single-seed disclosure per cell. Self-contained.

5. Discussion

5.1. What a wall-estimable state must contain

[S5-01] The comparison has a direction, read off the chain of tasks a wall-borne reduced state must perform. On the decoded field the three families are indistinguishable, the energy-optimal linear basis as sharp as the nonlinear states, so field fidelity does not separate them. The separation opens at the first task that needs the wake, the observable read-out, where the linear basis loses the load at the impact instant and neither it nor the published-recipe reconstruction reads the wake; and it widens at every step toward deployment. Through the impact the published-recipe reconstruction's forecast collapses and the linear basis fades; recovering the state from sparse wall pressure, the predictive latent carries the most physics from a less recoverable state; and under assimilation at the strong-gust boundary every filter is stressed: all of them fail innovation consistency on most encounters, least often the predictive filter (0.72 of encounters against 0.93 to 0.97 for the others). The predictive state holds furthest down the chain, and that is the design rule stated operationally: a state earns its place at deployment not by rendering the field but by remaining readable, forecastable, wall-recoverable and assimilable at once.

[S5-02] The central result is a design rule for a reduced state that a wall-pressure estimator can track. Such a state must carry the wake observables that govern the post-impact transient, must be organised so that a forward model can advance it without leaving its own support, and must be recoverable from the surface pressure that is the only measurement a vehicle carries. These three requirements are supplied by different ingredients, and separating them is the paper's contribution. Observable supervision on the wake supplies the first, the instantaneous readability, and, less obviously, much of the second: it concentrates the encoded variance into a low-rank subspace aligned with the observables, and the forward rollout, trained and living in that subspace, stays in distribution, where an anti-collapse regulariser applied to a bare reconstruction produces a near-isotropic latent with no protected subspace and a rollout that leaks into the directions the training

S5-01

KEEP

Reads the comparison as a chain of tasks and says where each family drops out. The clearest single statement of the paper's result.

S5-02

KEEP

The design rule and the geometric mechanism, with the earlier anti-collapse reading explicitly corrected. The contribution paragraph.

data never constrained (§4.4.1). The mechanism is geometric, and it refines our earlier reading, which credited the anti-collapse term: a reconstruction objective fixes the latent only up to a diffeomorphism (Kelshaw & Magri 2024), the anti-collapse term on its own drives the latent toward isotropy (Broustail *et al.* 2026), and it is the supervision that pins the variance onto the observable-aligned directions the rollout respects. The multi-step predictive objective supplies the rest of the second requirement, the long-horizon stability that a single-step objective and an objective-free supervised encoder both lack, and, across retrains, the reliability of the forecast itself (§5.4). The third requirement, wall recoverability, is where energy and information part company: the reconstruction latent is the most recoverable in raw variance but the least in physics, while the predictive latent carries the most wall-recoverable physics (§4.5), so the coefficient state a controller would want is not the one a reconstruction objective produces.

[S5-03] The wake, and not the forces, is where these requirements separate the families, and the coordinate-level reading of §4.2.3 explains why. The forces are carried redundantly in every family, so an energy or reconstruction objective retains the force signature of an encounter without retaining the distributed combination of coordinates that carries the vorticity redistribution producing it. The gust parameters alone predict the forces well and fail on the wake enstrophy at the readout horizon, so a state can look accurate on the load while carrying none of the physics the load is built from, which is exactly the failure mode the estimation results expose. The linear basis, for its part, remains genuinely useful: it is competitive on the body forces, amplitude-accurate in reconstruction, and it holds the shedding clock as well as the predictive latent (§4.3); the result is not that POD is obsolete but that the POD coefficients do not achieve comparable linear readability of the wake-enstrophy target at any dimension we tested.

[S5-04] One caveat is inherited from the compactness-versus-forecast comparison that motivates this study (Solera-Rico *et al.* 2026): a more expressive predictor might narrow the forecast gaps between the families. Three controls argue that the ordering is a property of the representations rather than of an underpowered predictor. The shared forecaster was capacity-selected on the validation set, a sequence-LSTM variant included; the matched autoregressive transformer diverges on the published-recipe latents at equal recipe and budget, so its failure is not a capacity artefact; and a per-family tuned recovery estimator raises every family without changing their order (table 5; supplementary material). The orderings reported here are therefore floors set by the representation, not ceilings set by the predictor.

5.2. The estimation demonstration and its limits

[S5-05] The filter turns the design rule into a working estimator: from eight wall-pressure sensors and no field measurement it tracks the load through the encounter and extends the usable gust-intensity range beyond every static inverse, whose in-distribution impact median it matches rather than beats (§4.6.3), realising the online-estimation pathway the extreme-aerodynamic control literature has named (Fukami *et al.* 2024; Mousavi & Eldredge 2025; Eldredge & Mousavi 2025). We are deliberate about its limits, and the first is one the family comparison forces: the in-range load-tracking is a property of sequential estimation on a nonlinear latent generally, not of this state, so the paper does not rest the representational claim on it. What the predictive state contributes to the estimator is the strong-gust boundary, the calibration margin, and a tracked state whose coordinates carry the wake; what no state yet contributes is a filtered wake estimate, since the analysis wake closure is below the mean predictor for every family. Its recoverability advantage over a raw high-dimensional field latent is real but modest (§4.5), so the case for the coefficient state rests on the physics-recovery ordering and the envelope, not on a large recoverability

S5-03

KEEP

Why the wake and not the forces separates the families, and what the linear basis remains good for. Keeps the POD comparison fair.

S5-04

KEEP

The capacity caveat and the three controls that answer it. Pre-empt the most likely referee objection.

S5-05

KEEP

The estimation demonstration with its limits stated in the same breath. The feasibility-demonstration framing lives here.

margin. Its point tracking is good but its uncertainty is under-dispersed at the strongest and widest gusts, where the innovations are temporally correlated and the ensemble collapses. We conjecture, beyond what the exhibits establish, that the forward model carries gain along data-unconstrained directions it never learned to damp, and that those inject spread-consuming noise during assimilation; what the record does establish is that the innovations are temporally correlated rather than white (§4.6.2). Either reading points to a calibrated process and observation noise model, rather than covariance inflation, as the outstanding requirement, and we report the filter as a feasibility demonstration with a characterised soft spot, not as a deployed estimator.

[S5-06] The delay-embedding view names the deployment knob and its limit. Because the encounter is low-dimensional, the state can be tracked from few sensors read over a short window, which is the sensing budget a small vehicle could carry; the trade between sensor count and window length is explicit for the static reconstruction (§4.5.1), so a sensor-poor platform can in principle compensate with a longer window within the encounter’s own timescale. The frozen filter does not inherit that trade, because its innovation rank, not the embedding, is what binds at reduced sensor count, and the two statements together sharpen the same conclusion: the noise model, not the sensor count, is the binding constraint. The limit that no budget escapes is the growth of the encounter’s effective dimension with gust strength. Read through delay coordinates the encounter is an intermittently forced low-dimensional system (Brunton *et al.* 2017), the baseline shedding carrying the delay-coordinate dynamics and the vortex impact acting as a forcing whose amplitude is the gust ratio, which is consistent with the envelope being organised by $|G|$: no fixed sensing budget embeds the strongest, most three-dimensional encounters, which is why the envelope closes where it does (§4.6.2).

5.3. Decodability versus estimability

[S5-07] A reduced state can be optimised for two deployments that pull apart. A spatially resolved latent decodes the vorticity field more sharply than the coefficient state, but it is of far higher dimension than the wall measurement and so is not recoverable from a sparse set of pressure sensors. The coefficient state decodes the field less sharply and is the object a wall-pressure filter can track. We therefore separate the two roles: the coefficient state is the estimable, forward-usable object this paper is about, and the field-resolved latent is a decodability instrument rather than an estimation target. Pooling the encoder onto the coefficient state costs field fidelity across every family, by -0.093 in decode similarity and -0.095 in fitted-operator forecast merit for the predictive latent (supplementary material), and it gains nothing in wall recoverability that the coefficient state does not already have. The state a controller would carry is therefore not the one a field-reconstruction objective produces.

5.4. Limitations and outlook

[S5-08] Five limitations bound the claims. First, the attribution of the instantaneous results is unglamorous: the wake readability is supplied by the observable supervision and its tie between the predictive and reconstructive wake-headed states is seed-robust (0.727 ± 0.021 against 0.760 ± 0.019). The shared-operator forecast merit of §4.1.2 shows the same picture dynamically: over three operator seeds the three wake-headed states are statistically indistinguishable, with overlapping case-clustered intervals. An earlier seed study of the fitted merit, run under the residual U-Net operator as a stability check, cautions the same way: the supervised control is seed-fragile (spread 0.163 , with one healthy retrain in three collapsing), the predictive latent is tighter (± 0.098), and the family means overlap. No fitted-operator margin between the wake-headed states is therefore decisive on its own.

S5-06

KEEP

The delay-embedding view of the deployment knob and the reason the envelope closes where it does. Ties the sensing and envelope results together.

S5-07

TRIM?

A one-paragraph subsection whose two numbers are both in the supplementary material. It could become two sentences inside §5.1 without loss.

S5-08

KEEP

Five limitations, including the seed-fragility disclosure and the calibration priority. The rehearsal of the superseded U-Net seed study inside it is the removable part.

The defensible dynamical claim rests instead on the co-trained model and its rollout: the wake-supervised predictive state’s own predictor forecasts better than the shared operator extracts from any latent at short horizon (§4.1.2), and the multi-step objective confers long-horizon stability and lower drift, not a larger instantaneous readability margin. Second, the wall-normal offset Y remains the least resolved release parameter because the training mass concentrates near $Y \approx 0$; the additional cases make it weakly readable where it was unreadable before (§4.3), and we note this without leaning on it. Third, the $|G| = 4$ extrapolation degrades for a physical reason: the encoder observes a single mid-plane vorticity slice that captures a substantially smaller fraction of the three-dimensional content there (§2.1), so the boundary test set marks where that representation becomes incomplete, not a parametric-generalisation claim, and the filter meets the same limit from the sensing side (§4.6.2). Fourth, the visualisation decoder is a diagnostic on the frozen encoder, never part of the predictive loss, and the pooled coefficient state trades decode sharpness for estimability by construction (§5.3). Fifth, and foremost for what comes next, the filter’s calibration is the priority open problem: its uncertainty is under-dispersed at the strongest gusts, and the natural next step is a learned or physically motivated process and observation covariance, ahead of any closed-loop use.

[S5-09] These limitations come into sharpest focus against the use the architecture is meant for. The predictor is, by construction, a latent dynamics function of the kind an estimator or controller requires, the latent is orders of magnitude cheaper to advance than the field, and the filter demonstrates the estimation half of that loop from the wall alone. Closed-loop actuation is outside the present evidence: the predictor is trained to forecast observed encounters within the training envelope, and predicting how an encounter would differ under intervention is a task its training does not address. The contribution is narrow by design, a controlled comparison and a frozen-filter characterisation on a single parametric flow, and what it establishes is bounded accordingly: a wall-estimable, forward-usable coefficient state, the ingredients that produce it, and the gust-intensity envelope over which a fixed sensing budget can follow it. The natural next steps are the noise model above and the same matched-estimator protocol on other parametric flows, to test how general the estimability ordering reported here actually is.

[S5-10] A further direction concerns the wake target itself. The present descriptor fixes where the wake vortices sit and at which scales their energy resides, but its coordinates are generic. Weak-form dominant-balance analysis (Ahnert *et al.* 2026) suggests a physically grounded alternative: inner products of the conservation-form vorticity-transport terms against compactly supported test functions, so that each supervised coordinate is the local magnitude of a governing mechanism, evaluated without differentiating the data. Such a target is capacity-matched to the present descriptor by construction and would align the protected subspace of §4.4.1 with the mechanisms themselves; the controlled substitution is left to future work.

6. Concluding remarks

[S6-01] We asked whether a reduced state of an extreme vortex-gust encounter can be constructed to carry the wake dynamics, advanced in time, and recovered from wall pressure alone. On construction, under the predictive objective a scalar lift target is necessary for a non-collapsed latent representation and a wake target is what makes the state carry the wake observables; the near-body refinement that further sharpens the peak load is reported in §4.1. The collapse result concerns latent health and the peak-load result concerns a paired increment over an already lift-supervised baseline; the two rest on different comparisons.

S5-09

KEEP

Scopes the work against closed-loop control and states what the evidence does not cover. The narrowness is deliberate and should be visible.

S5-10

TRIM?

A speculative alternative wake target with a citation and no test. Interesting, but it is future work stated at paragraph length.

S6-01

KEEP

Answers the construction question and separates the collapse claim from the peak-load claim. Both rest on different comparisons and the text says so.

S6-02

KEEP

The compression and forecasting statements, both two-tier and both scoped. Matches §4.2 and §4.4 exactly.

S6-03

KEEP

The estimation statement, now leading with the in-distribution pair rather than the flattered boundary score. This is the F01 repair.

S6-04

KEEP

The design principle and the three results that bound it. The static-inverse qualifier here is what appendix C must support.

[S6-02] On compression, the statement is two-tier: load information is approximately unchanged over the tested dimensions under a nonlinear probe and is linearly accessible at four coefficients, whereas moderate linear readability of the wake diagnostics appears from eight coefficients and stable reconstruction of the spatial wake structure requires approximately sixteen or more over the tested dimensions. On forecasting, one shared direct multi-horizon forecaster makes the wake-supervised states the most forecastable and the predictive states the most forecastable on the lift; a one-shot forecast remains usable over forty steps where autoregressive rollout degrades through error accumulation, and supplying the true gust parameters did not improve the direct forecaster for this dataset and model class, so the deployable route to the encounter is a model corrected by pressure rather than the release parameters.

[S6-03] On estimation, a two-stage ensemble filter sensing eight wall-pressure sensors tracks the lift through the encounter, phase-resolved and in physical units, with an impact-phase median $R^2 = 0.794$ at a root-mean-square lift error of 0.286 on the in-distribution test set, and holds its median tracking at the $|G| = 4$ boundary. Across latent dimension and training seed the predictive states remain uniformly estimable from the wall, whereas the reconstructive geometry does not, for a verified observation-side reason; this robustness, rather than any single configuration, is the deployment argument.

[S6-04] The design principle is that a nonlinear latent can be made forecastable and wall-estimable by training it predictively under observable anchoring, at no cost in compactness. Three results bound the claims. A static delay-embedded inverse matches the filter's in-distribution impact median, and the filter's advantage lies in the tails, at the boundary set and in continuous state tracking. The wake enstrophy is not reliably recovered by the sensor counts, delay windows and estimators considered here, although the encoded state retains it, which points to the present observation configuration rather than to the state. And the mid-plane representation omits a substantially larger fraction of the three-dimensional vorticity content at the strongest gusts (four cases), where the filter's uncertainty consistency degrades even as the median tracking holds. The natural next steps are a calibrated process and observation noise model for the filter, and the same matched-estimator protocol on other parametric flows, to test how general the estimability ordering reported here is.

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Declaration of interests. The authors report no conflict of interest.

Data availability statement. The reduced-order-model and evaluation code, the configuration files, the v2.2 data-split manifest, and the analysis outputs that regenerate the tables and figures are deposited at <https://doi.org/10.xxxx/zenodo.PLACEHOLDER> under the MIT License (DOI to be finalised on acceptance). The processed per-encounter cache, or a representative subset sufficient to rerun the heavier steps, is available from the corresponding author. The raw direct numerical simulations were computed with the SOD2D solver (Gasparino *et al.* 2024) and are owned by the simulation collaborators.

Author contributions. [Authors to insert contributions, for example in CRediT format.]

Use of generative-AI tools. Generative-AI assistance (Anthropic Claude, accessed through the Claude Code command-line interface) was used during 2026 to help with data analysis, figure generation, and manuscript drafting. All numerical results were computed by the authors' own code and verified against the underlying data, and the authors take full responsibility for the content of the article.

Appendix A. Architecture, anti-collapse regularisation, and uncertainty quantification

Architecture detail

[APA-01] The encoder is a hybrid convolutional stem followed by a small vision transformer. Three downsampling convolutional stages map the single-channel 192×96 vorticity field to a 24×12 feature map at 256 channels, that

APA-01

TRIM?

Opens by restating the architecture already given in §3.1 and table 14, then spends twenty lines narrating a suited-operator protocol the paper no longer uses. The superseded narrative belongs with the suited table in the supplementary material.

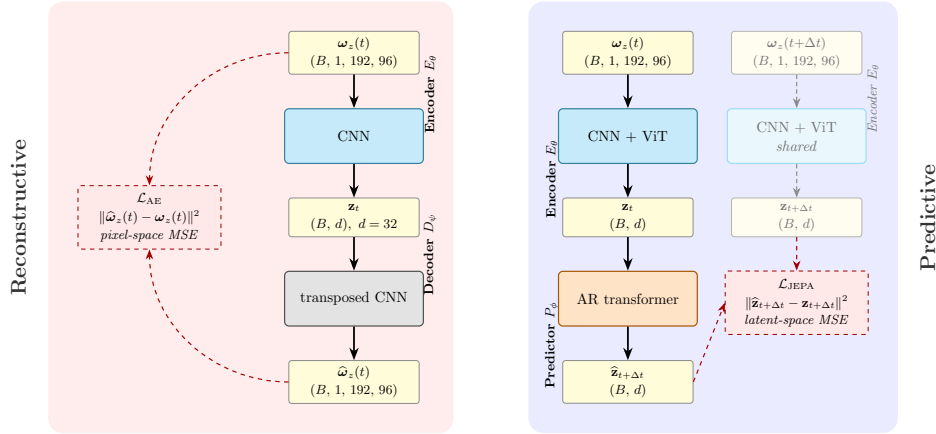


Figure 17. [APA-C1] Predictive versus reconstructive training. Left: the reconstructive autoencoder encodes the field at t , decodes it, and incurs a field-space error. Right: the predictive model encodes the fields at t and $t + \Delta t$ with shared weights and incurs a latent-space error between the predicted and observed target embeddings, with no field-space loss and no decoder during training. This is the contrast the matched-predictor protocol of §3.2 evaluates; the visualisation decoder is trained only afterwards, on the frozen encoder.

APA-C1

KEEP

The predictive-against-reconstructive schematic, which is the clearest statement of what the objective changes. Referenced from §3.1.

is 288 image tokens, which a six-layer pre-norm transformer of width 256 with eight heads processes; the latent is read from a prepended class token through a one-layer projection whose output passes a BatchNorm, the layer the characteristic-function regulariser below requires. The encoder is unconditional and carries no information about the gust parameters. The training predictor advances the pooled 32-vector directly: its function class is a six-layer autoregressive transformer of width 384 with sixteen heads, rotary temporal positions and a causal mask, mapping a short history of 32-vectors to the next 32-vector, with no spatial map or tiling anywhere in the loop. Training minimises an open-loop rollout loss over eight steps, with no teacher forcing, against online encoder targets that are detached (no exponential-moving-average target network), plus the regulariser below, with AdamW and bf16 mixed precision. The predictor advances the latent from its own history alone, with no gust input, so the gust parameters $\mathbf{c} = (G, D, Y)$ enter nowhere in the model. The forecast merit of table 2 is computed under one shared operator, the direct multi-horizon forecaster of §3.2.1 trained identically on every state, and under it the three controlled wake-headed states are statistically tied (§4.1). The superseded protocol that fitted a different operator to each family is documented in the supplementary material. The ensemble filter of §3.3 uses the same matched transformer on the wake-supervised predictive state's own pooled trajectories, where it is stable, and the wake-supervised predictive state's own co-trained predictor, the co-trained reduced-order model, is reported alongside.

1479 Anti-collapse regularisation

[APA-02] Because the encoder and predictor are trained end to end on latent-space prediction with no decoder, the latent is held away from a collapsed solution by an explicit regulariser rather than by a reconstruction term. We use SIGReg, the characteristic-function (Epps–Pulley) regulariser of the from-pixels joint-embedding line (Maes *et al.* 2026; Balestrieri & LeCun 2025), with M random one-dimensional projections of the latent and a fixed weight $\lambda_S = 0.02$ in the loss, the value used in every run reported here. The same regulariser has since been adopted independently for biosignal time series, where it is likewise reported to drive the embedding toward isotropy so that a structuring objective is needed alongside it (Broustail *et al.* 2026), consistent with the supervision-anisotropy mechanism of §4.4.1. As a safeguard the participation ratio of the latent is monitored every thousand iterations, with an automatic fall-back to a variance-covariance objective (Bardes *et al.* 2022) if it indicates collapse while the parameter probe is still uninformative; on the runs reported here the participation ratio stays healthy and the fall-back does not trigger.

1491 Uncertainty quantification

[APA-03] The six sources of uncertainty, their units and their counts are collected in table 16; this paragraph records what that table cannot, which is how they compare at the present sample size. The encounter bootstrap dominates the scatter: with $n = 42$ held-out in-distribution test encounters and $n = 40$ boundary test encounters

APA-02

KEEP

The anti-collapse regulariser, its weight and its safeguard. Needed to reproduce the training.

APA-03

KEEP

What the accounting table cannot say; which source dominates at this sample size, and why the plain encounter bootstrap overstates it. No longer re-enumerates the protocol.

Table 9. [APA-C2] Paired per-encounter tests for the estimation endpoints, on the frozen filter records. The family was fixed before any statistic was computed: the primary family F1 compares the filter analysis against the static single-frame recovery and F2 against the field-free open-loop forecast, on the endpoints $\{C_L, E_w\}$ in the impact and relaxation windows, one-sided with the filter declared in advance as better; ΔR^2 is the median per-case mean improvement, the p is a case-level Wilcoxon signed-rank, and Holm corrects within each family. On the pooled in-distribution test encounters the filter beats the open-loop forecast decisively on the load but not the static recovery, which is accurate at the moderate gust strengths the in-distribution test set samples; the filter’s advantage over the static delay-embedded inverse concentrates at strong gusts, where the boundary test annex (characterisation only, $n = 8$ cases) is consistent on the load in both windows at the smallest p the sample size resolves (uncorrected). The wake enstrophy through the filter does not beat the static recovery at the boundary, a null result.

Family	Endpoint	Window	median ΔR^2	Wilcoxon p	Holm p
F1 (vs static)	C_L	impact	0.16	0.161	0.483
F1 (vs static)	C_L	relax.	0.53	0.116	0.465
F1 (vs static)	E_w	impact	−25.15	0.986	1.000
F1 (vs static)	E_w	relax.	−2.47	0.839	1.000
F2 (vs open loop)	C_L	impact	9.43	0.001	0.003
F2 (vs open loop)	C_L	relax.	44.84	0.001	0.003
F2 (vs open loop)	E_w	impact	17.39	0.246	0.246
boundary test annex	C_L	impact	1.11	0.0078	—
boundary test annex	C_L	relax.	6.41	0.0039	—

its interval is wide for the individual wake observables, as noted in §4.1.2, and wider than the seed spread of any cell reported here. Because encounters of one case share a shedding history, the plain encounter bootstrap overstates the effective sample size, which is why every paired family comparison is additionally read through the case-clustered resampling of the same table. A single split with bootstrap intervals, seed retrains and probe-level cross-validation is the community standard for this class of study.

Paired tests on the estimation endpoints

[APA-04] Table 9 reports the per-encounter paired family behind the tracking and envelope claims of §4.6.1 and §4.6.2. Pairing cancels the encounter-to-encounter difficulty that widens marginal intervals, and clustering by case respects the shared shedding history of a case’s encounters.

Topology of the encoded encounter

[APA-05] A coordinate-free check on the geometric mechanism of §4.4.1: Vietoris–Rips persistence of the encoded trajectories, per family, with a generator declared significant when its persistence exceeds a noise floor set at a fraction of the cloud diameter, after per-family standardisation so that coordinate scale cannot carry the result. On the gusted held-out encounters the supervised states keep a single clean H_1 cycle in about half the encounters (0.64 for the predictive and 0.52 for the objective-free supervised state) where the anti-collapse reconstruction fragments (0.10); on the single-period no-gust control the same separation holds (0.62 against 0.00), and it survives whitening, so the fragmentation is a property of the reconstructive encoding rather than of any coordinate scale. The linear basis keeps the cleanest no-gust loop of all (0.62), consistent with a linear projection of a smooth field; the claim is that the anti-collapse reconstruction fragments, not that only the predictive latent carries a loop.

Preprocessing robustness

[APA-06] The wake enstrophy squares the vorticity, so the per-encounter high-percentile amplitude clip of the ω_z preprocessing could in principle shape the primary endpoint. We test this on the frozen encoders, with no retraining, by recomputing the readout-frame wake-enstrophy target under three training-only clip treatments of the raw field and re-fitting the frozen-latent-to-wake ridge probe per family: no amplitude clip, the production per-encounter $p_{99.99}$ clip, and a single leak-free $p_{99.99}$ threshold pooled over the training encounters. The predictive latent leads the wake on the in-distribution test set under all three (table 10), with the largest shift in the predictive-minus-best-baseline advantage relative to the production clip being 0.07 in R^2 . The leak-free training-global clip gives the largest predictive advantage, so the per-encounter clip is not the source of the wake result.

APA-C2 **KEEP**
The paired-test table, including the null on the wake enstrophy. The pre-declared family structure is stated in the caption.

APA-04 **KEEP**
Explains why the estimation endpoints are tested paired and case-clustered. Two sentences that justify table 15.

APA-05 **SUPPL.**
A coordinate-free topology check that no claim in the body depends on. Genuine supporting evidence, which is what the supplementary material is for.

APA-06 **SUPPL.**
A robustness check run on the previous-generation $d = 64$ encoders, honestly disclosed as such. The body sentence it supports can point at the supplementary material instead.

Table 10. [APA-C3] Preprocessing robustness of the representational wake closure, established on the previous-generation spatial encoders ($d = 64$, the 85-case v2.1 split) and retained as evidence that the amplitude clip does not shape the endpoint; the pipeline is unchanged in the present dataset revision, and the probe protocol is identical. Held-out in-distribution test wake-ensrophy R^2 from a frozen-latent ridge probe (no retraining), recomputed under three training-only amplitude-clip treatments of the raw ω_z field. The predictive latent leads the wake under every treatment; the advantage keeps its sign and approximate magnitude.

Family	no clip	per-encounter	training-global
predictive	0.75	0.80	0.84
reconstructive (Fukami)	0.12	0.10	0.08
linear (POD)	-0.27	-0.05	0.00

A.1. The near-body lift-element head

[APA-07] The auxiliary potential ϕ_L behind the lift-element density of §3.1 is defined by the exterior boundary-value problem

$$\nabla^2 \phi_L = 0 \text{ outside the body,} \quad -\mathbf{n} \cdot \nabla \phi_L = \mathbf{n} \cdot \mathbf{e}_L \text{ on the body,} \quad \phi_L \rightarrow 0 \text{ far away,} \quad (\text{A } 1)$$

with $\mathbf{e}_L = (-\sin \alpha, \cos \alpha)$ the lift direction at $\alpha = 14^\circ$. Equation (A 1) is discretised once on the 192×96 cache grid with a staircase immersed Neumann condition on the solid-adjacent cell faces and a homogeneous Dirichlet far field, and solved directly (figure 24); the interior Laplacian residual of the stored solution is 6.4×10^{-13} in the maximum norm. The lift-element density is then the pointwise field

$$e(\mathbf{x}, t) = \omega_z (-v \partial_x \phi_L + u \partial_y \phi_L), \quad (\text{A } 2)$$

evaluated from the raw mid-plane velocity and vorticity with the stored sign convention verified empirically (the cache stores $\partial u / \partial y - \partial v / \partial x$, so the standard ω_z is its negation). Because the cache window truncates the domain, the full-domain integral of (A 2) does not track the lift; the band-restricted field does, which is why the head supervises e inside the feathered band $w(\mathbf{x}) = \max\{0, 1 - \delta(\mathbf{x})/0.3c\}$, with δ the Euclidean distance to the body and its adjacent cells. The banded field $w e$, scaled by a fixed constant chosen so its 99th percentile magnitude is order one, is compressed to its 80-dimensional target by exactly the wake-observable construction of §3.1: 64 sign-preserving patch energies (an 8×4 grid of $\log(1 + \text{relu}(\pm x)^2)$ pooled energies) concatenated with a 16-bin Hann-windowed radial power spectrum. A precompute quality gate requires the median lagged correlation between the band-integrated element and the stored lift over the gust-bearing training encounters (lag search $|\ell| \leq 25$ frames) to exceed 0.4; the achieved value is 0.74. The principled weighting is not interchangeable with a plain vorticity-magnitude proxy: the patch-block cosine between the two targets is 0.68 on the comparison sample. The potential and its supervision band are shown in figure 24 in appendix A.2.

[APA-08] On top of the lift anchor the near-body force-element (Chang) head confers real peak-load accuracy, the strongest cell of the $2 \times 2 \times 2$ conditioning cube of §4.1: the lift-focused predictive state (cube code CLN) beats the lift-only cell CL by 2.15 peak-region R^2 points with a case-clustered interval that excludes zero ($[0.03, 5.59]$), at a pooled peak-region $R^2 = 0.86 \pm 0.003$ over three seeds, the tightest seed band anywhere in the cube, and a decoded phase lag of $-0.011 t/c$ (figure 18). Downstream the wake-headed wake-supervised predictive state is the single predictive representative; the near-body head is a lift-sharpening refinement, not the primary state.

A.2. Field-decode and phase-resolved figures

[APA-09] The figures collected here support the decode-floor, physics and forecastability claims of §4.2–§4.4: the Chang auxiliary potential behind the near-body head, the decode galleries and their dimension curve, and the phase-resolved cut of the shared-operator forecast merit.

A.3. Methods and comparison tables

[APA-10] The tables collected here give the full configuration record: the encoder and training-predictor architecture, the compared families, the ensemble-filter configuration (table 13), the rollout-departure decomposition, the per-estimator settings of the ladder, and the uncertainty accounting (table 16).

APA-C3

SUPPL.

The preprocessing-robustness table, on the same superseded encoders as APA-06. Moves with it.

APA-07

KEEP

The full construction of the near-body head, moved out of Methods so the three heads sit at one depth in the body. Two equations, the discretisation and the pre-committed gate.

APA-08

KEEP

The near-body conditioning increment, demoted here from §4.1 and referenced from it. The strongest cell in the cube deserves this much.

APA-09

TRIM?

A list of what the following figures show, immediately before figures whose captions each say it. Pure connective tissue.

APA-10

TRIM?

A one-sentence list of the tables that follow, each of which has a caption. Pure connective tissue.

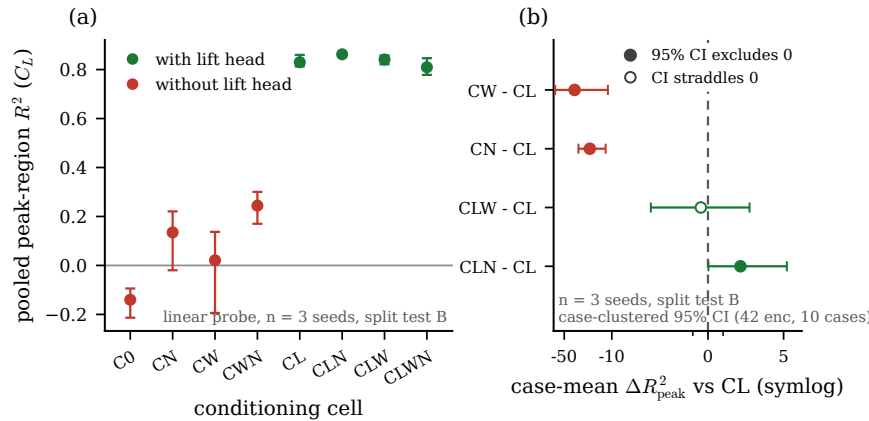


Figure 18. [APA-C4] Paired peak-region lift R^2 per conditioning-cube cell (left) and case-mean deltas against the lift-only cell CL (right, symmetric-log axis, 95 % case-clustered bootstrap intervals over the 42 in-distribution test encounters of 10 cases); filled markers mark intervals that exclude zero. The near-body lift-focused predictive state cell (CLN) is the unique positive increment; the no- L cells sit one to two orders of magnitude lower.

APA-C4 **KEEP**
The paired cube deltas with case-clustered intervals. Self-contained.

Table 11. [APA-C14] Encoder and training-predictor configuration, counted from the released $d = 32$ checkpoint. The model is unconditional: no gust parameters $\mathbf{c} = (G, D, Y)$ enter the encoder or the predictor. Two small auxiliary heads read observables off the encoder output during training (a lift head and an 80-dimensional wake-spectrum head; §3.1). The estimation-stage prediction model of §3.3 is a predictor of the same class refitted on the frozen pooled trajectories.

	Encoder E_θ	Predictor P_ϕ
role	field $\rightarrow$ pooled state (unconditional)	pooled-state rollout (unconditional)
input	$\omega_z \in \mathbb{R}^{192 \times 96}$	pooled states $\mathbf{z} \in \mathbb{R}^{32}$
backbone	3 conv. stages + 6-layer ViT	6-layer causal transformer (RoPE)
width / heads	256 / 8	384 / 16
image tokens / context	$24 \times 12 = 288$	2 frames, rolled 8 steps open loop
conditioning	none	none
state read-out	class token $\rightarrow$ linear $\rightarrow$ BatchNorm	next pooled state
parameters	6.7M	10.7M

APA-C14 **KEEP**
The encoder and predictor configuration, counted from the released checkpoint. The reproducibility record.

Appendix B. Sparse wall-pressure state estimation

[APB-01] At deployment the vorticity field is not available; only the wall pressure is. This appendix records the sensing conventions behind §4.5 and §4.6.1 and the placement-robustness checks that the main text summarises. The input throughout is the span-averaged pressure on the 192 surface sensors; the recovery estimator is selected per family by encounter-grouped cross-validation among a kernel-ridge regressor on the flattened window, a multilayer perceptron, and a recurrent network over the window sequence, and the recovered state is read by the same frozen probes as everywhere else.

APB-01 **KEEP**
States the sensing conventions and how the recovery estimator is selected. Needed before any recovery number can be read.

Placement, and why it does not drive the ordering

[APB-02] Two placements are used, deliberately. The filter and the recovery of table 5 sense at each family’s own model-conditioned sensors, the greedy selection that best exposes that family’s latent, because the estimator comparison should give every family its best sensing. The delay-trade grid of §4.5.1 instead uses a single target-blind array, the QR-pivot optimum of the wall-pressure modes (Manohar *et al.* 2018), nested so that every sensor count is a prefix of the next, because a sensor-budget claim must not depend on a placement tuned to any model. The two meet where their sensing coincides: the grid’s eight-sensor, thirty-frame cell rerun on the model-conditioned sensors closely reproduces the table’s recovery (0.656). The energy-versus-information split of §4.5 is also placement-robust: under the shared target-blind array with the kernel-ridge estimator alone, the reconstructive state remains the most recoverable in variance (0.775 against 0.667 for the predictive and

APB-02 **KEEP**
Why the ordering is not an artefact of model-conditioned placement, with the target-blind rerun. Answers a real objection.

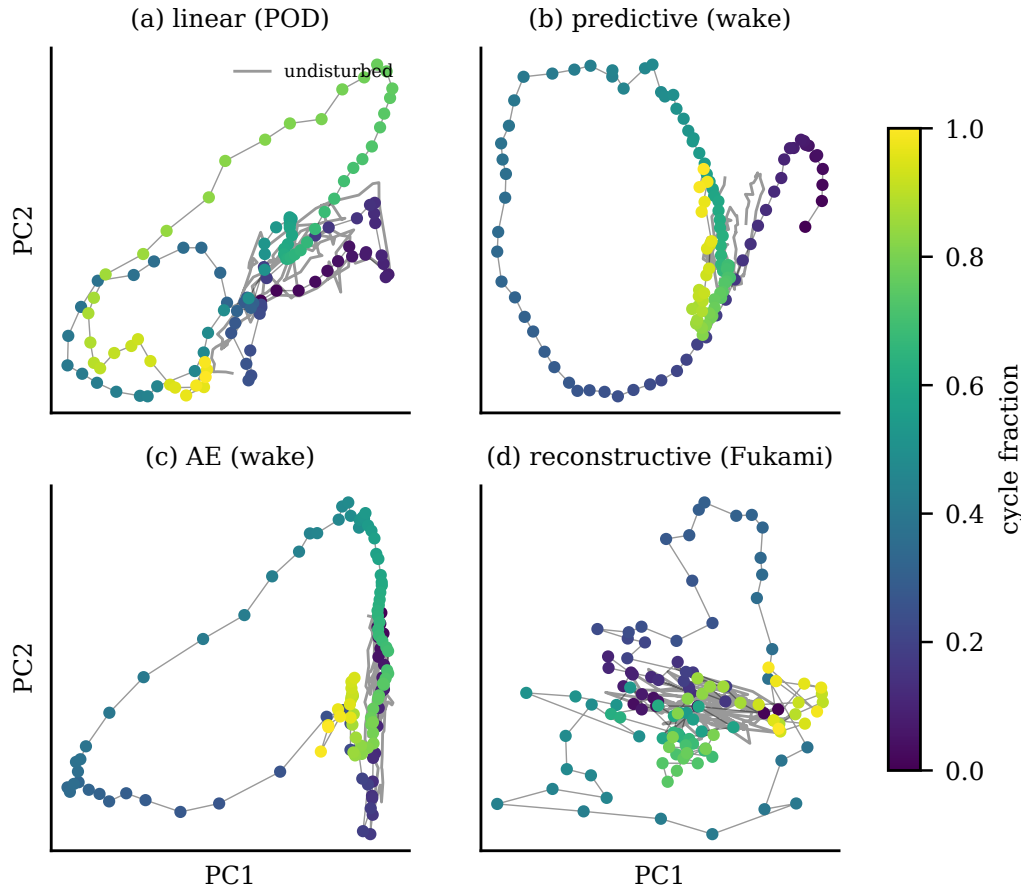


Figure 19. [APA-C5] Latent trajectory portraits by family (leading two principal components of the $d = 32$ state; a representative gusted encounter coloured by cycle fraction, undisturbed limit cycle in grey). The predictive, linear and matched-supervision states trace coherent cycles perturbed by the gust; the published-recipe reconstruction does not.

APA-C5

KEEP

The latent portraits that corroborate the spectral result qualitatively. Referenced from §4.3.

1578 0.500 for the linear basis) and the predictive state retains the most recoverable physics (0.523 against 0.363 and
1579 0.391), so neither ordering is an artefact of the model-conditioned placement.

1580 *The greedy placement criterion*

1581 [APB-03] The model-conditioned placement above is built one sensor at a time. Starting from an empty set,
1582 each step adds the sensor that most improves a ridge prediction of the leading principal component of that
1583 family's own impact-frame states, made from the 30-frame causal pressure windows of the sensors selected
1584 so far together with the candidate. Writing r_j for the ridge residual mean square of the extended feature set
1585 and $g_j = \max\{0, r_\emptyset - r_j\}$ for the reduction it achieves over the target variance, the sensor added is the one
1586 maximising

$$1587 \quad J(j) = n g_j - \frac{1}{2} g_j - \frac{1}{2} n r_j + \frac{1}{2} \frac{n g_j}{g_j + 10^{-6}}, \quad (\text{B } 1)$$

1588 with n the number of training encounters. At this configuration the first and third terms dominate the other two,
1589 so in practice the criterion is explained-variance gain penalised by the residual that remains; the smaller terms
1590 are a complexity penalty that does not change the ordering here. Selection is nested by construction, each budget
1591 warm-starting from the previous one, so the $K = 16$ array contains the $K = 8$ array and a sensor-count sweep
1592 varies only the budget. The shared target-blind reference array is instead the column-pivoted QR (qDEIM)
1593 selection of Drmač & Gugercin (2016) on the leading pressure modes, which uses no model at all. Which
1594 placement a given figure or table uses is stated in its caption.

APB-03

KEEP

The greedy placement criterion, moved out of §3.4. It sits beside APB-02, which argues that placement does not drive the ordering.

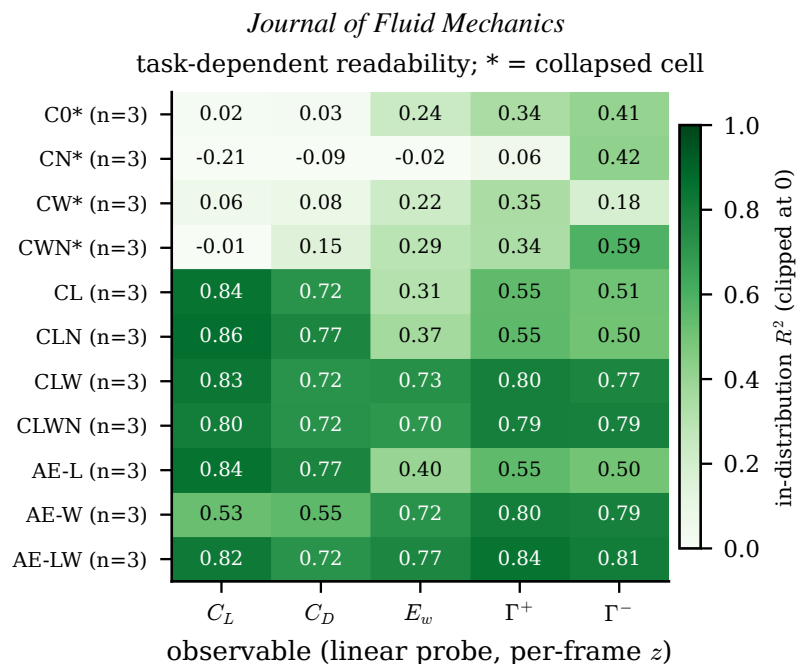


Figure 20. [APA-C6] Task-dependent readability (in-distribution test set, linear probes on the per-frame state, seed mean over three encoder seeds per cell): lift and drag follow the L head, the wake block (E_w , $\Gamma^\pm$) follows the W head or the reconstruction objective, and collapsed cells (asterisks) read nothing linearly.

APA-C6 **KEEP**
 The full task-dependent readability matrix behind the division-of-labour claim.
 Referenced from §4.1.

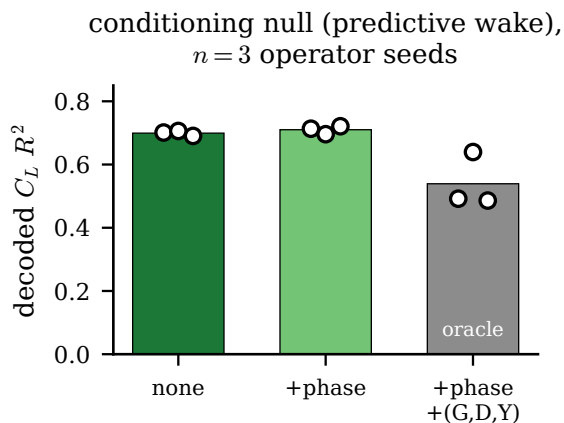


Figure 21. [APA-C7] The conditioning null on the shared direct multi-horizon forecaster (demoted from the forecast figure 10): decoded lift R^2 of the tuned forecaster on the wake-supervised predictive state latents under no covariate, an added phase covariate, and an added (G, D, Y) gust-parameter oracle (three operator seeds per arm). The deployable phase covariate is a wash with the unconditioned forecaster, while the gust-parameter oracle degrades it.

APA-C7 **KEEP**
 The conditioning null, demoted from the forecast figure and referenced from §4.4.
 Self-contained.

1595 *What the window recovers that the instant cannot*

1596 [APB-04] At a single frame the wall is blind to the wake-circulation direction of the predictive state (§4.5); over
 1597 the thirty-frame window the blind coordinate becomes partially recoverable (0.33 held-out, from near zero at
 1598 one frame), and the wake entrophy read through the recovered state reaches 0.40. The window, not the sensor
 1599 count, is what carries this: the same quantities barely move as sensors are added at a fixed single frame. This
 1600 is the delay-coordinate content of the sensing result, quantified in the grid of §4.5.1; it is a statement about the

APB-04 **TRIM?**
 Restates the window-against-instant result already given in S4-20 and S4-21, with the same macros. The appendix adds only the entrophy figure.

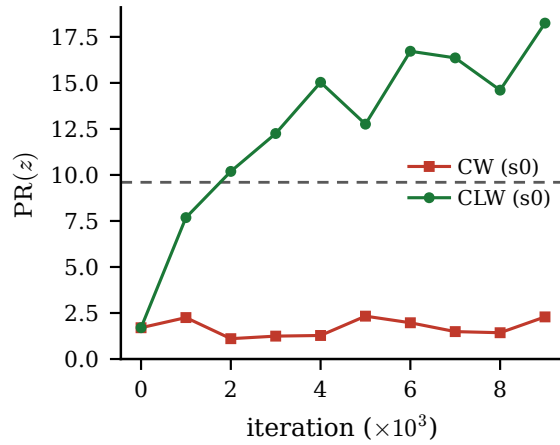
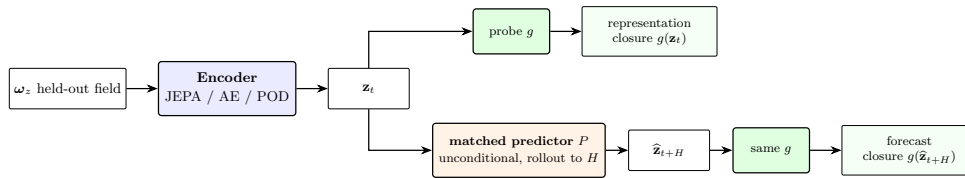


Figure 22. [APA-C8] Training history of the conditioning-cube collapse (demoted from figure 7): participation ratio against iteration for one representative collapsed cell (CW, seed 0) and one healthy cell (CLW, seed 0), with the collapse flat from the first diagnostic while the healthy cell crosses the floor within the first quarter of the run.

APA-C8

KEEP

The collapse training history, demoted from the cube figure. Shows the collapse is flat from the first diagnostic.



The predictor (unconditional, no gust parameters) and probe family g use the same architecture and protocol, trained separately for each encoder family.

Figure 23. [APA-C9] The evaluation protocol, applied identically to every encoder family (demoted from the architecture figure 4). The same matched forecast operator is refit and the same probe family g is applied to every encoder: a probe on the simulation-encoded latent measures representational closure, the headline, and the same probe on the rolled-out latent $\hat{z}_{t+H}$ measures forecast closure, reported as secondary.

APA-C9

KEEP

The evaluation protocol applied identically to every family. Referenced from §3.2 and the framework caption.

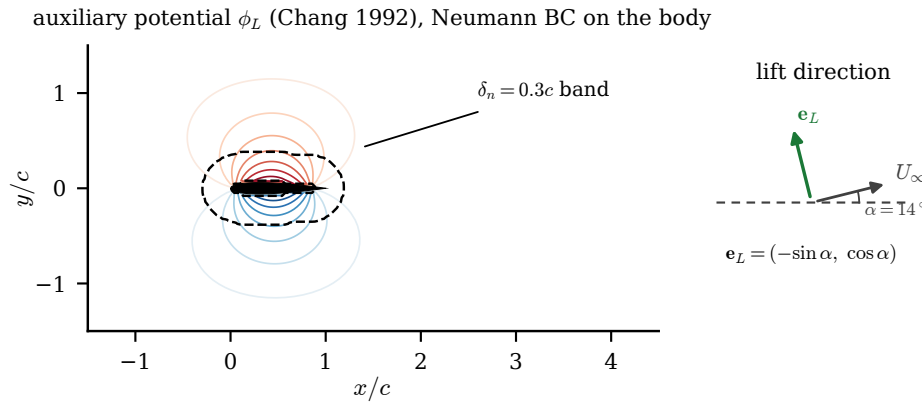
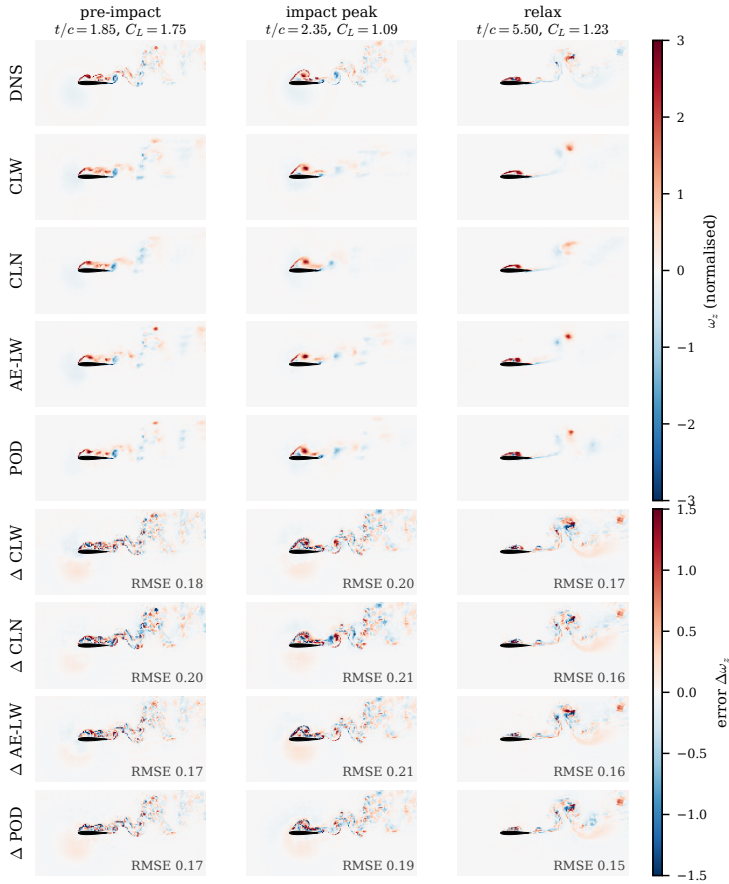


Figure 24. [APA-C10] The Chang auxiliary potential ϕ_L of equation (A 1) on the cache grid (contours), the feathered $\delta_n = 0.3c$ supervision band (dashed) and the airfoil footprint; the inset shows the lift direction $\mathbf{e}_L$ against the inclined freestream at $\alpha = 14^\circ$.

APA-C10

KEEP

The Chang auxiliary potential and its supervision band. Referenced from §3.1.2.



case G-0.50_D1.00_Y-0.40 (in-distribution test), encounter 0; seed s0 per model; pooled z tiled to the 24 x 12 grid, frozen-encoder decode-floor decoders

Figure 25. [APA-C11] Decode floor on a representative held-out in-distribution test encounter (one decoder seed per model): DNS, the wake-supervised predictive state, the lift-focused predictive state, the wake autoencoder and the linear basis, and the error rows at three phases. Identical-capacity frozen decoders; normalised vorticity at the fixed ± 3 range. All four states render the field faithfully, including the linear basis: field decode is not where the states separate.

APA-C11 **KEEP**
The decode gallery behind the decode-floor claim, with the linear basis included. Self-contained.

static reconstruction, and the filter inherits the sequential version of it through its assimilation window rather than through any explicit delay embedding.

The recovered field

[APB-05] Decoding the pressure-recovered state through each family’s diagnostic decoder makes the recovery visible in physical space (figure 28). The comparison within each row, recovered field against that family’s own decode ceiling, isolates the pressure step from the decoder’s reconstruction limit: the loss from ceiling to recovery is the part attributable to the eight-sensor window, and the panels show it concentrating in the wake structures rather than in the near-body field, consistent with the wall reading the force-carrying directions best (§4.5).

APB-05 **KEEP**
Reads the recovered-field panels and localises the loss to the wake structures. Connects the sensing result to physical space.

The out-of-distribution boundary from the sensing side

[APB-06] At the $|G| = 4$ boundary every static approach fails on the load, the via-latent recovery at -0.33 and the direct no-latent regression at 0.08 from the same sensors (0.14 with the full array), while the nonlinear filters still track it (table 6). The boundary is the same larger three-dimensional content the field carries there (§2.1): the mid-plane state captures less of the dynamics, so a static delay-embedded inverse recovers little from a single instant, and only the accumulated pressure history carries the load through.

APB-06 **TRIM?**
Restates the boundary failure of every static approach, already carried by S4-25 and table 12. The appendix repeats the numbers without adding evidence.

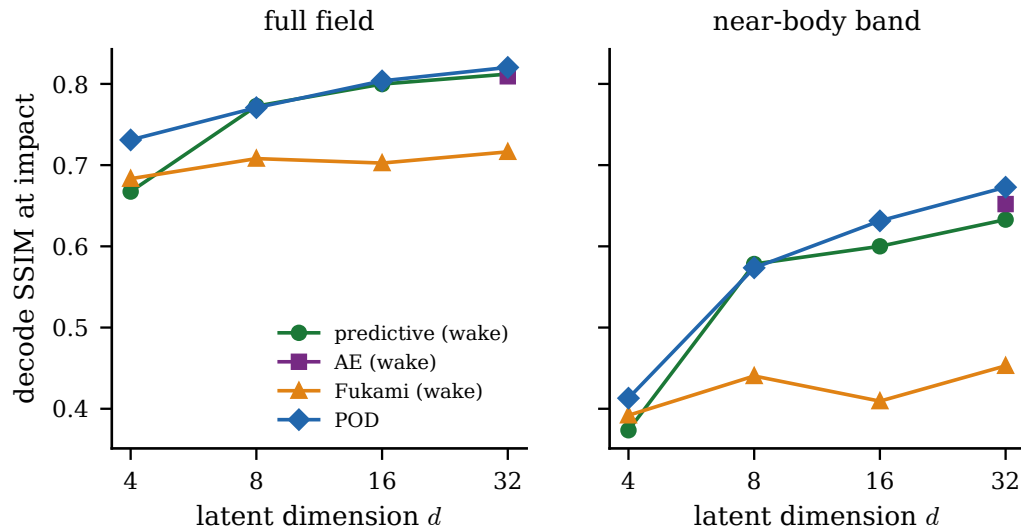


Figure 26. [APA-C12] Decode SSIM at the impact instant against latent dimension d (full field and near-body band, in-distribution test set, decode floor). The compact nonlinear states and the linear basis improve with d and saturate together; only the published-recipe lineage plateaus lower. Field decode is not the discriminator at any dimension. Single encoder/decoder seed.

APA-C12

KEEP

Decode fidelity against dimension, the exhibit S4-07 points at. Single-seed provenance disclosed.

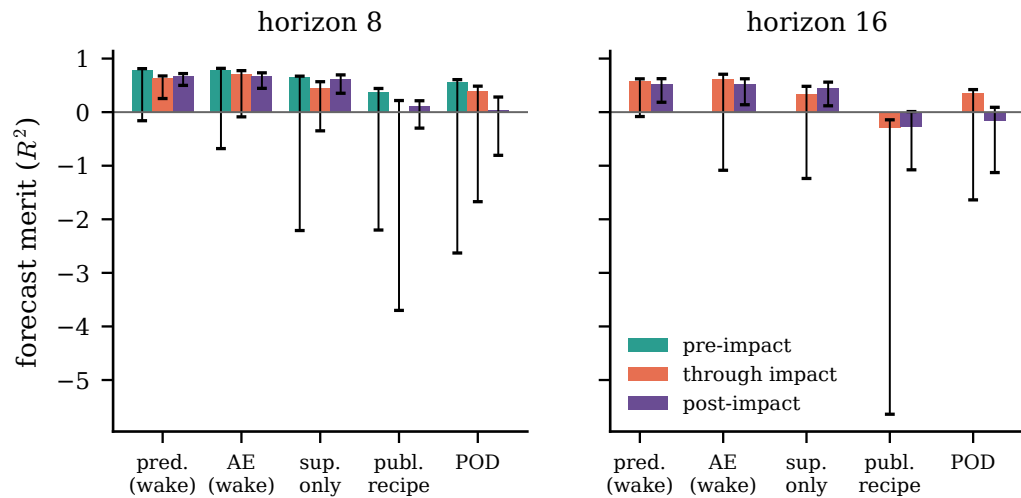


Figure 27. [APA-C13] Shared-operator forecast merit split by phase (pre-impact, through-impact, post-impact), per family at horizons eight and sixteen (in-distribution test set, five observables, three operator seeds; bars at the seed mean, case-clustered 95 % intervals). Through-impact merit separates the families: the wake-supervised states hold, the published-recipe reconstruction collapses across the transient, and the linear basis fades into the relaxation.

APA-C13

KEEP

The phase-split forecast merit, the exhibit S4-15 narrates. Self-contained.

1616 Delay-coordinate knobs

1617 [APB-07] The window sweeps of §4.5.1 keep the delay stride at the cache cadence ($\Delta t = 0.05 c/u_\infty$), and this
 1618 choice is a convention rather than a tuning: the first minimum of the time-lagged mutual information of the
 1619 sensor signals (Fraser & Swinney 1986) sits near 29 frames, about one shedding period, so the thirty-frame
 1620 window spans roughly one decorrelation time of the wall pressure at the cadence used. The window length
 1621 itself is selected by the measured recovery trade rather than by a false-nearest-neighbours criterion (Kennel
 1622 *et al.* 1992), which we regard as a cross-check and not a design input: the delay-embedding guarantees are

APB-07

KEEP

Justifies the delay stride and window as convention rather than tuning, and declines the false-nearest-neighbours criterion honestly. Methodologically careful.

Table 12. [TBAS-C1] The compared families: encoder, latent, and training objective. Every family produces a pooled $d = 32$ coefficient vector at the same dimension. The top block is the controlled matrix, one shared encoder (the hybrid convolutional-stem plus vision transformer of table 11) with the training ingredient varied one at a time; the objective-free supervised control trains that encoder with the observable heads and the anti-collapse term alone, with no predictor and no decoder, so it isolates what the supervision by itself supplies. The bottom block is the published reference recipes: the Fukami autoencoder and the β -VAE are plain convolutional autoencoders, a convolutional stack down to a fully-connected bottleneck with no skip connections, and POD is a linear basis. The forecast merit of table 2 reads every latent through one shared operator, the direct multi-horizon forecaster trained identically on each; the superseded suited-operator protocol (an autoregressive transformer and a residual U-Net, a different one stable on the reference latents) is described in Appendix A and its table retained in the supplementary material.

Family	Encoder	Latent	Training objective	Source
predictive	CNN + ViT (shared)	pooled 32-vec	latent-space prediction + heads + SIGReg	this work
supervised only	CNN + ViT (shared)	pooled 32-vec	heads + SIGReg only (no objective)	this work
reconstructive AE	CNN + ViT (shared)	pooled 32-vec	field reconstruction + heads + SIGReg	this work
anti-collapse AE	CNN + ViT (shared)	pooled 32-vec	field reconstruction + SIGReg (no heads)	this work
Fukami AE	conv. AE, no skips	pooled 32-vec	reconstruction + observable head	Fukami & Taira (2023, 2025)
β -VAE	conv. VAE, no skips	32-vec (mean)	reconstruction + KL + heads	Solera-Rico <i>et al.</i> (2024)
POD	linear projection	32 leading modes	none (SVD basis)	classical

TBAS-C1

KEEP

Defines every compared family in one place. The matched-dimension claim is checkable only against this table.

Table 13. [TENK-C1] Configuration of the pressure-only ensemble Kalman estimator in the latent space (§3.3, figure 6). Every setting is frozen before the held-out evaluation. The four configurations share one predict-correct loop and differ only in how members are propagated and what is assimilated; the two-stage configuration is the production estimator and switches between the other two regimes at the frozen impact-window boundary, which is not tuned. The multiplicative inflation is selected per family on the validation split, since the families differ in dispersion: the predictive and linear filters select no inflation ($\rho = 1.00$), being under-dispersed, while the reconstruction filter selects $\rho = 1.05$.

<i>Shared by every configuration</i>		
Filter	stochastic ensemble Kalman filter, latent space	
Ensemble size N	64 members	
Sensor count K	8 sensors, per-model optimal sensor placement (OSP)	
Observation noise $\mathbf{R}$	calibrated from held-out pressure residuals	
Inflation ρ	validation-selected per family (1.00 predictive/linear, 1.05 reconstruction)	
Initialisation	field-free: windowed pressure-to-latent regressor + its residual covariance	
Assimilation window	[impact – 24, impact + 48] frames	
Leakage guard	lift and wake probes never enter the innovation (enforced, unit-tested)	
Configuration	Propagation F , $\mathbf{Q}$	Assimilated observation
Base (§3.3.1)	matched autoregressive transformer, state-independent $\mathbf{Q}$ from its one-step residual covariance	K sensors through a ridge read-out; innovation in sensor space
linear latent filter (§3.3.2)	linear $F(\mathbf{z}) = \mathbf{A}\mathbf{z}$, singular values projected onto $[0, 1]$; $\mathbf{Q}$ the covariance of its fit residuals	encoded E_{obs} , eq. (3.7): delay-embedded sensor history, $h = \text{Id}$; innovation in state space
forecast-noise filter (§3.3.3)	direct multi-horizon forecaster median, eq. (3.8); state-dependent $\mathbf{Q}$ read off its predicted inter-quantile width	encoded E_{obs} , as above
two-stage filter (production)	forecast-noise filter inside the impact window, matched transformer outside	encoded inside the window, sensor space outside

TENK-C1

KEEP

The four estimator configurations side by side, which no other exhibit gives. Recently widened from a single-filter list.

1623 generic, wall pressure is a physical and non-generic observable strongly coupled to the near-body vorticity,
1624 and the operative evidence is the measured recovery, with the theorem as the organising principle rather than a
1625 proof that a given sensor count suffices.

Table 14. [TMEC-C1] Where each rolled state goes relative to the training distribution, on the in-distribution test set at rollout horizon sixteen. The latent covariance condition number measures anisotropy, and the near-null departure fraction is the share of the rollout departure that falls into the bottom-quartile directions of the encoded covariance. An isotropic null would leak 0.25. The anti-collapse reconstruction is near-isotropic and leaks; the supervised states are anisotropic and stay in their low-rank subspace.

State	covariance condition number	near-null departure fraction
anti-collapse AE	11	0.100
predictive (wake)	451	0.014
supervised only	1041	0.029

Table 15. [APA-C15] Per-estimator configuration (Gupta *et al.* 2026, table 1 analogue). All fits use the training split only; ρ and the two c calibrations use validation-family splits as stated in the text and nothing downstream of them. Ensemble size $N = 64$, sensors $K = 8$ at the family’s own placement, delay $m = 10$, every-frame assimilation, and the consistency-failure criterion of §3.3.2 are common to all rows. Every ridge fit in the suite, the linear dynamics A , the sensor read-out and the delay-embedding map W , uses unit penalty; the value is stated here rather than three times in §3.3.

estimator	dynamics F	observation h	process noise Q_t	noise scale
static delay-embedded inverse	none	E_{obs} , eq. (3.7)	none	none
base EnKF (frozen)	matched transformer	linear $\mathbf{z} \rightarrow \mathbf{p}\mathbf{K}$	teacher-forced residual cov.	$\rho = 1.00 / 1.05$
linear latent filter	linear A (SV-clipped)	$E_{\text{obs}}, h = \text{Id}$	A -fit residual cov.	$\rho = 1.0$
forecast-noise filter	forecaster median, eq. (3.8)	$E_{\text{obs}}, h = \text{Id}$	quantile band, eq. (3.8)	$c = 1.77$ (coverage)
two-stage filter	median forecast inside impact, transformer outside	E_{obs} inside, sensor space outside	as per stage	c as the forecast-noise filter
fixed-lag RTS	linear A	$E_{\text{obs}}, h = \text{Id}$	A -fit residual cov.	lag $L = 5$

Table 16. [APA-C16] The six sources of uncertainty, with the unit each one varies or resamples, its count, and the results that carry it. No single number carries all six: which apply depends on what was varied to produce it. Seed counts are the number of independent trainings; single-seed cells occur only in the dimension grid and say so in their captions. All resampling is on held-out data, and no interval is computed on the boundary test set for any purpose other than reporting.

Source	Unit varied or resampled	Count	Carried by
Sampling	held-out encounter, with replacement	2000	every held-out R^2 and mean absolute error
Sampling, clustered	whole case, with replacement	$2000 / 10^4$	every paired family comparison, Holm-corrected
Encoder training	random seed	3	conditioning cube, $d = 4$ bands, baseline bands
Operator training	random seed, frozen representation	3	shared-forecaster comparisons (table 2)
Filter member noise	random seed, perturbed observations	5	filter bands and the estimator ladder
Read-out	five-fold cross-validation over cases	5	every probe R^2

Streaming deployment realism and the four-coefficient filter

[APB-08] Under a deployment-realism protocol, streaming multi-encounter assimilation with a pressure-only initialisation and no oracle anywhere, the filter on the wake-supervised predictive state holds an impact-phase lift closure of 0.789 on clean sensors and 0.780, 0.749 and 0.646 at five, ten and twenty per cent sensor noise (three member-noise seeds each, figure 29). The four-dimensional lift-critical state of §4.2.1 supports a filter of its own at 0.782 impact median over five seeds, so a four-coefficient state is estimable from the wall as well as forecastable.

Choosing among the estimators

[APB-09] The ladder of §4.6.3 supports explicit selection guidance (Gupta *et al.* 2026). When only the in-distribution median matters and latency is irrelevant, the static delay-embedded inverse is the strongest and simplest tool; the moment the tails, the extrapolation boundary or continuous state tracking matter, the filters take over, the two-stage filter online and the closed-form fixed-lag smoother on the linear stack within a quarter-convective-time latency budget, while sparse or intermittent pressure favours the linear filter with encoded observations (the only member with zero consistency failures under subsampled updates). Across model families

TMEC-C1 **KEEP**
The rollout-departure decomposition behind the protected-subspace mechanism. Referenced from §4.3.

APA-C15 **KEEP**
The per-estimator configuration table, the Gupta table-1 analogue. What makes the estimator suite reproducible.

APA-C16 **KEEP**
The single authoritative uncertainty accounting, six sources with their units and counts. Replaces three partial statements that disagreed with each other.

APB-08 **KEEP**
The streaming realism protocol and the four-coefficient filter, both demoted from §4.6. Real results, correctly placed.

APB-09 **KEEP**
Explicit estimator-selection guidance in the Gupta idiom, ending in one sentence a practitioner can act on. The most useful paragraph in the appendices.

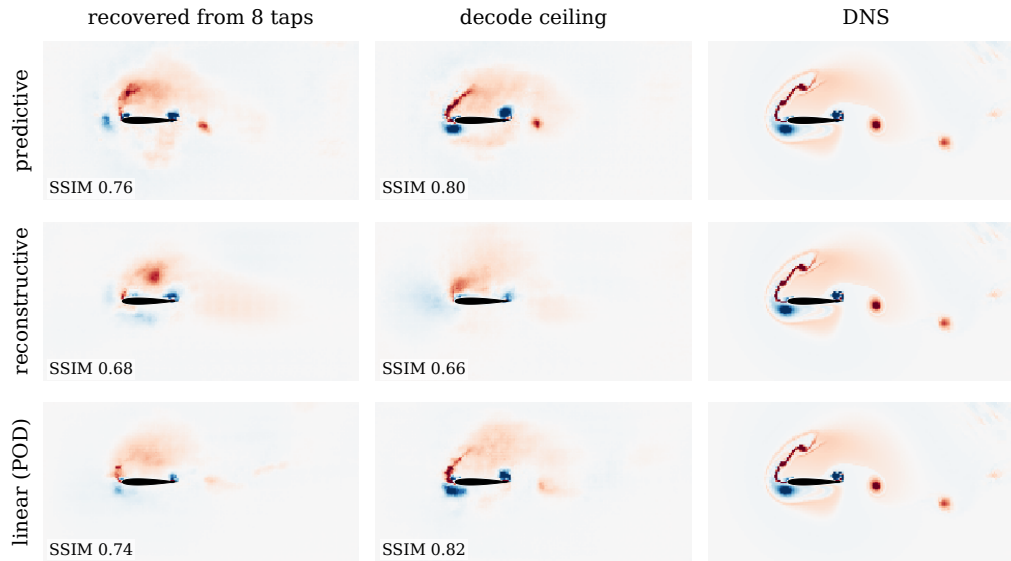


Figure 28. [APB-C1] Flow recovered from eight wall-pressure sensors on the representative held-out encounter of figure 13, at the impact-side frame, one row per family. Columns: the field decoded from the pressure-recovered coefficient state, the decode ceiling (the same decoder applied to the encoded state, no pressure step), and the simulation; per-panel structural similarity annotated. The comparison is within each row, since the decoders' ceilings differ (§5.3); the ceiling-to-recovered gap is the cost of the pressure step.

APB-C1

KEEP

The recovered field against each family's own decode ceiling, which is the only fair comparison. The caption says so.

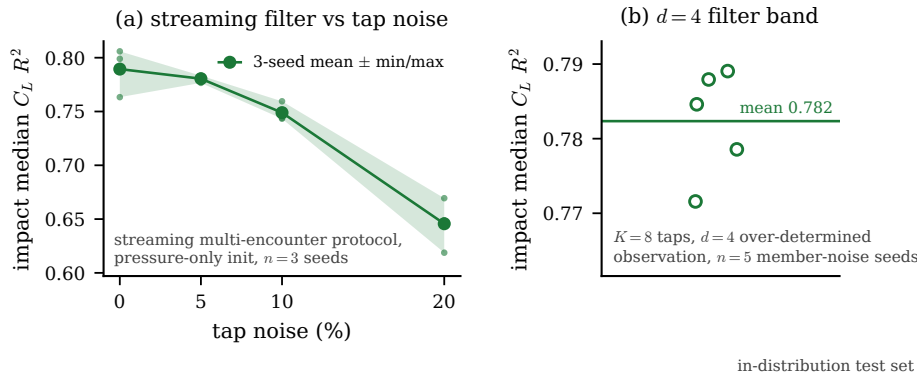


Figure 29. [APB-C2] Deployment realism (in-distribution test set): (a) streaming multi-encounter filter against sensor noise, three member-noise seeds, validation-selected band (no test-set reference); (b) the $d = 4$ filter band over five member-noise seeds.

APB-C2

KEEP

The deployment-realism panels with the seed counts. Self-contained.

the guidance is one sentence: anchored latent geometries, predictive or linear, can be selected on dimension and budget alone, while unanchored reconstructive geometries require a per-configuration estimatability check that their probes do not predict.

Appendix C. Estimator selection and calibration disclosure

[APC-01] The forecast-derived filter's band scale carries a full audit trail. The production value $c = 1.77$ is the coverage calibration of §3.3.3, selected on validation before any test encounter was seen. During exploration a band of $c = 4$ was evaluated against the in-distribution test set and reached 0.840 impact median there; because that value was selected with the test split in view, it is excluded from every table and headline of this paper and quoted only here. The consistency calibration (grid and rule committed before any value was seen) subsequently showed that the innovation statistic rejects large bands on held-out training encounters even

APC-01

KEEP

The full audit trail for the band scale, including the excluded test-selected value quoted only here. Integrity material that must stay in the paper.

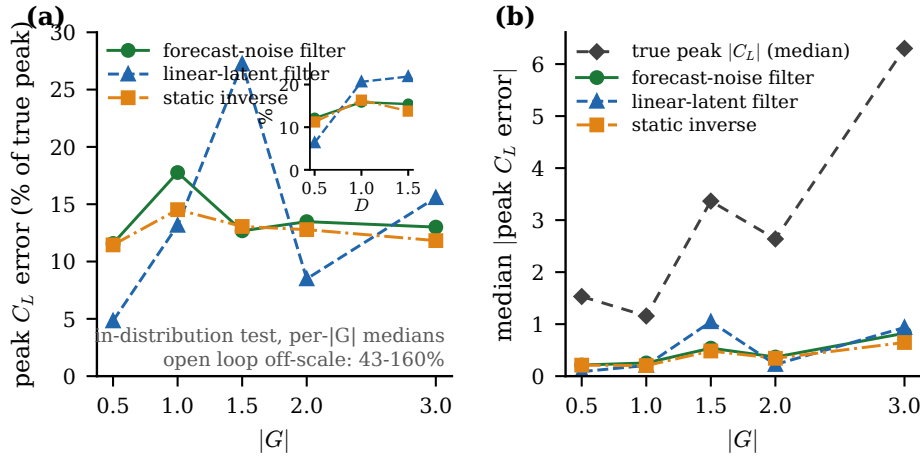


Figure 30. [APB-C3] Relative peak error across the gust envelope (in-distribution test set, medians per $|G|$): the forecast-derived filter stays near 12 to 14 per cent at all but the $|G| = 1$ stratum while the linear filter is non-uniform; the absolute peak error (right) grows with the peak itself.

APB-C3

KEEP

Relative peak error across the envelope, the scale-invariance exhibit §4.6.3 points at. Self-contained.

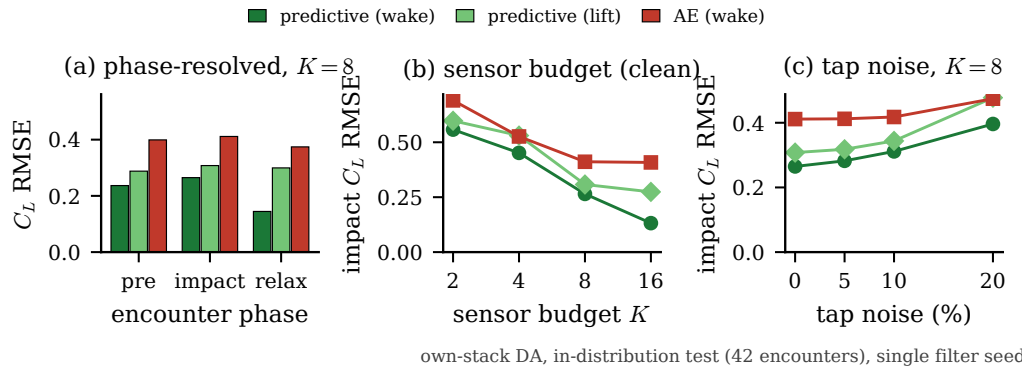


Figure 31. [APB-C4] Per-family end-to-end comparison (in-distribution test set, 42 encounters, single filter seed): (a) phase-resolved lift error per family; (b) sensor-budget sweep on each family’s own nested placement; (c) sensor-noise sweep at $K = 8$.

APB-C4

KEEP

The per-family end-to-end sweeps behind the own-stack claim. Single-seed provenance disclosed.

though their tracking is better (§4.6.3): the large-band gain is real and reproducible on encounters never used for selection, but it is model-error compensation, not calibration, and the paper’s route, selected without reference to the test set, to comparable performance is the two-stage filter instead. The two-stage filter run at the coverage band was itself declared in the pre-registration record after the consistency-selected run was seen and before any run at the declared configuration existed; both frozen runs are reported in §4.6.3 regardless of outcome, and the declaration order ships with the data record.

The fixed-lag smoother

[APC-02] The reanalysis reference of §3.3.3 is a fixed-lag Rauch–Tung–Striebel smoother running on the linear stack of §3.3.2. A filter reports the best estimate available from pressure up to the present frame; a smoother goes back and revises that estimate once a few later frames have arrived, which is legitimate whenever the application can tolerate the delay. Writing $(\mathbf{z}_k^f, P_k^f)$ and $(\mathbf{z}_k^a, P_k^a)$ for the forecast and analysis moments from the forward Kalman pass, the backward recursion over the lag window is

$$C_k = P_k^a A^\top \left(P_{k+1}^f \right)^{-1}, \quad \mathbf{z}_k^s = \mathbf{z}_k^a + C_k \left(\mathbf{z}_{k+1}^s - \mathbf{z}_{k+1}^f \right), \quad (\text{C } 1)$$

in which C_k weights how much of the later smoothed estimate to propagate back, large where the forward analysis was uncertain relative to the forecast it led to. We apply it over a lag of $L = 5$ frames, a fixed reporting

APC-02

KEEP

The fixed-lag recursion, moved out of §3.4. Its result is three sentences in §4.6.3, and the figure below already carries the smoother comparison.

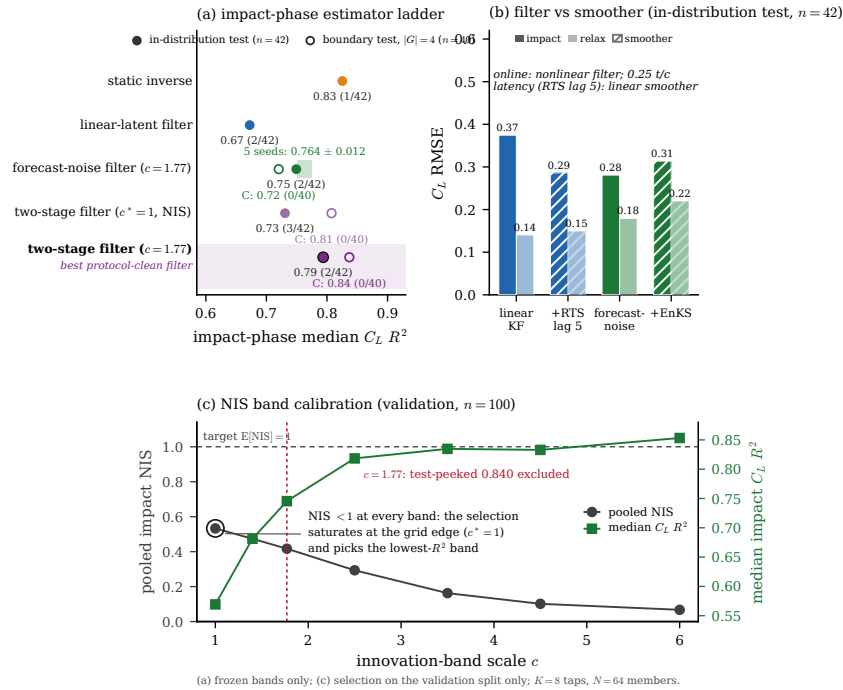


Figure 32. [APC-C1] Estimator calibration and selection (in-distribution test, validation and, through the open markers, boundary test sets; parenthetical counts on the ladder are large-error encounters, impact-phase $R^2 < -1$, per set). (a) The impact-phase estimator ladder with the five-seed member-noise band, open markers the frozen boundary test values; (b) the filter against the fixed-lag smoother; (c) the band-scale calibration on the held-out training encounters, where innovation consistency and tracking disagree and the selection saturates at the grid edge.

APC-C1

KEEP

The estimator ladder, the smoother comparison and the band calibration in one figure. Carries the static-inverse claim the abstract makes.

latency of $0.25 c/u_\infty$: the smoothed estimate at t uses pressure up to $t + L$ and nothing later, so it remains causal at that delay. The ensemble smoother variant on the forecast-noise filter, a lagged cross-covariance update of the stored member history, degrades the filter and is reported as a negative in the supplementary material.

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